Final Lecture

Geometric Algorithms

CAS CS 132

Objectives

» Course evaluations

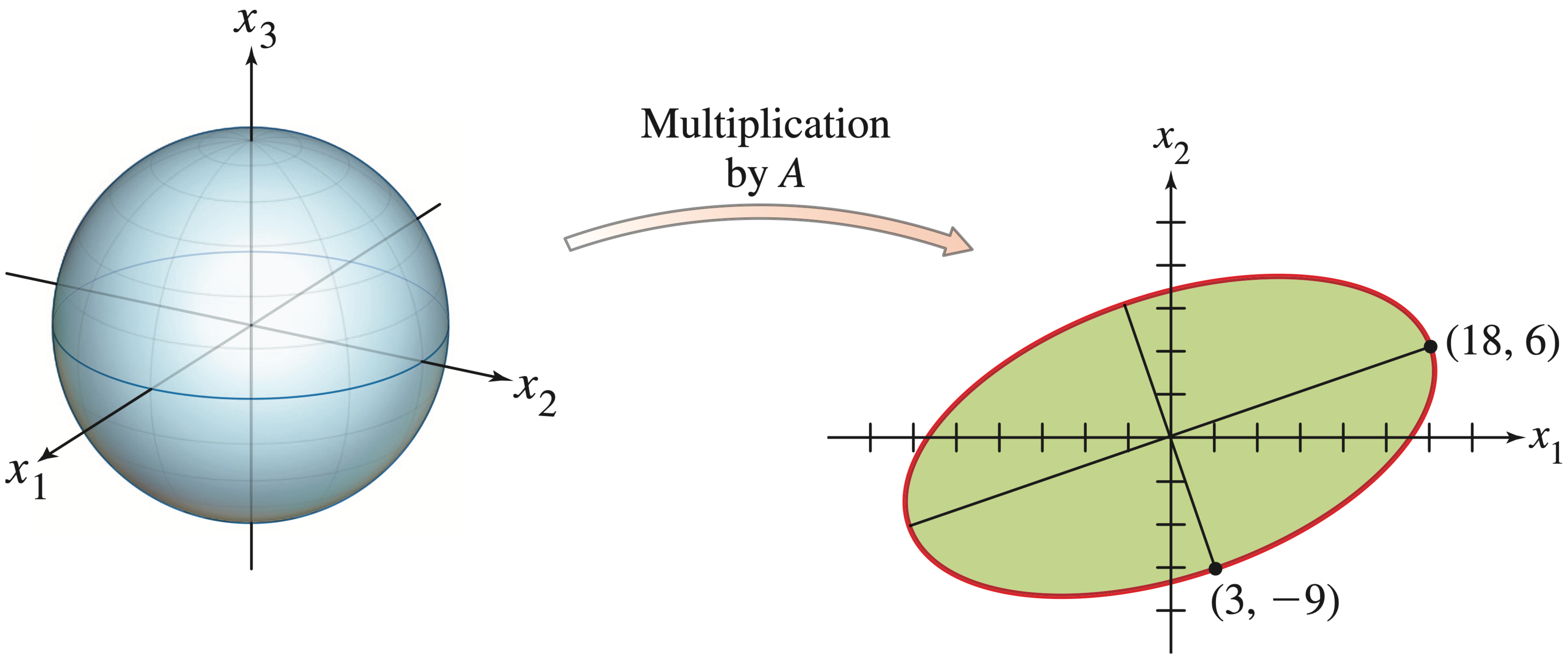
» Discuss applications of the SVD

» Final exam review

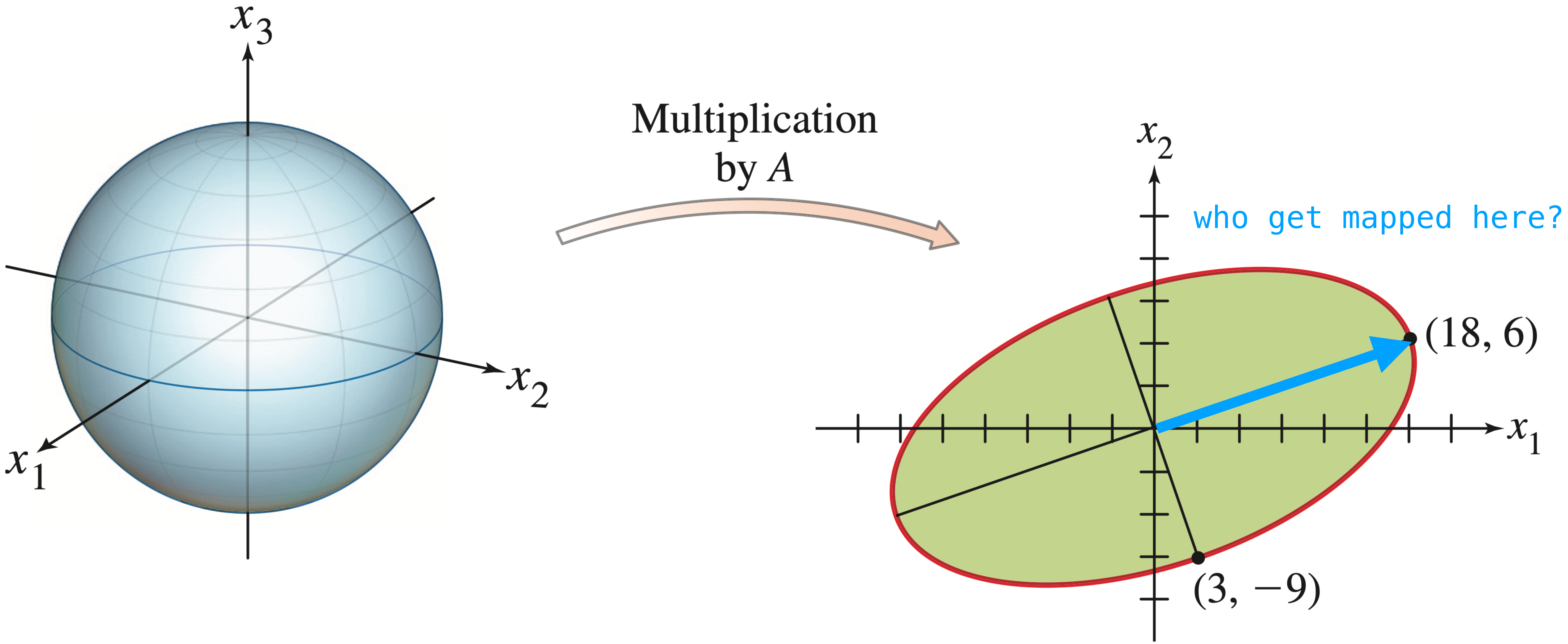
course evals

Recap

Recall: The Picture

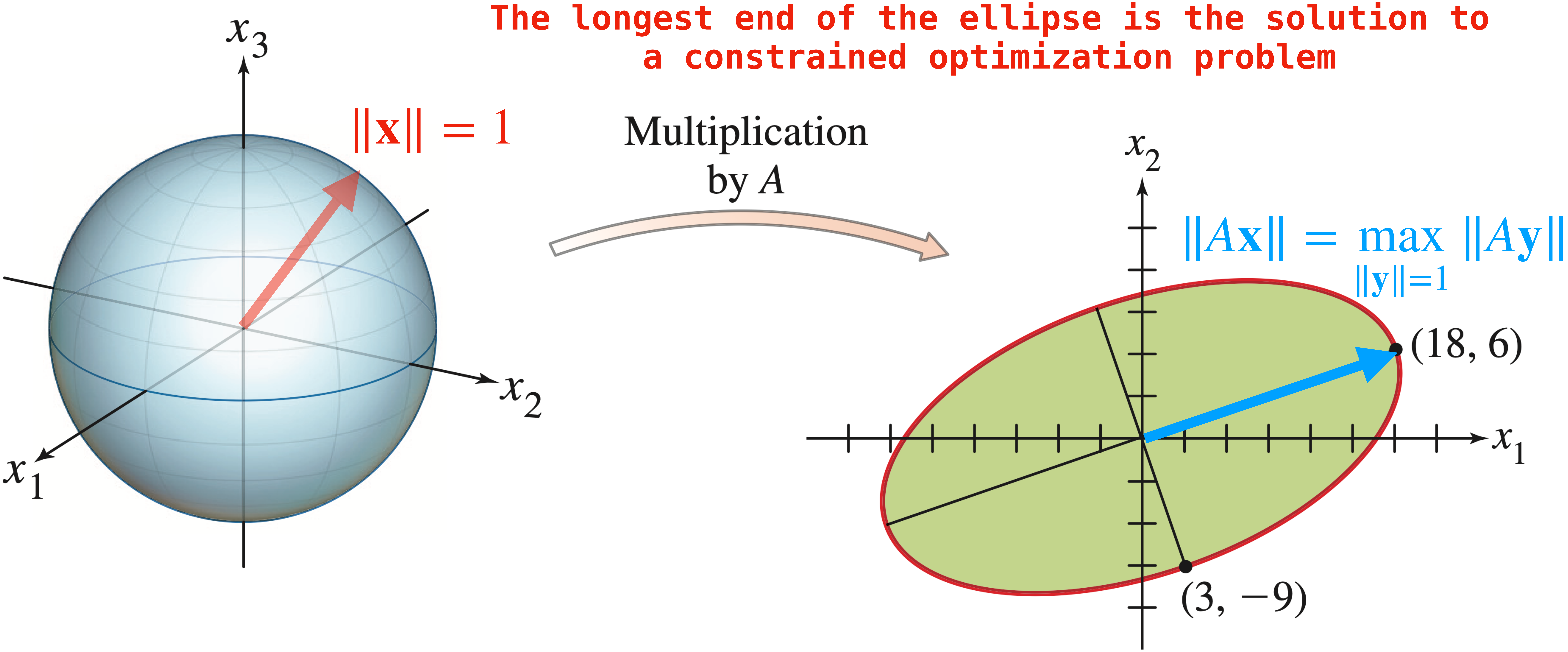


Recall: The Picture

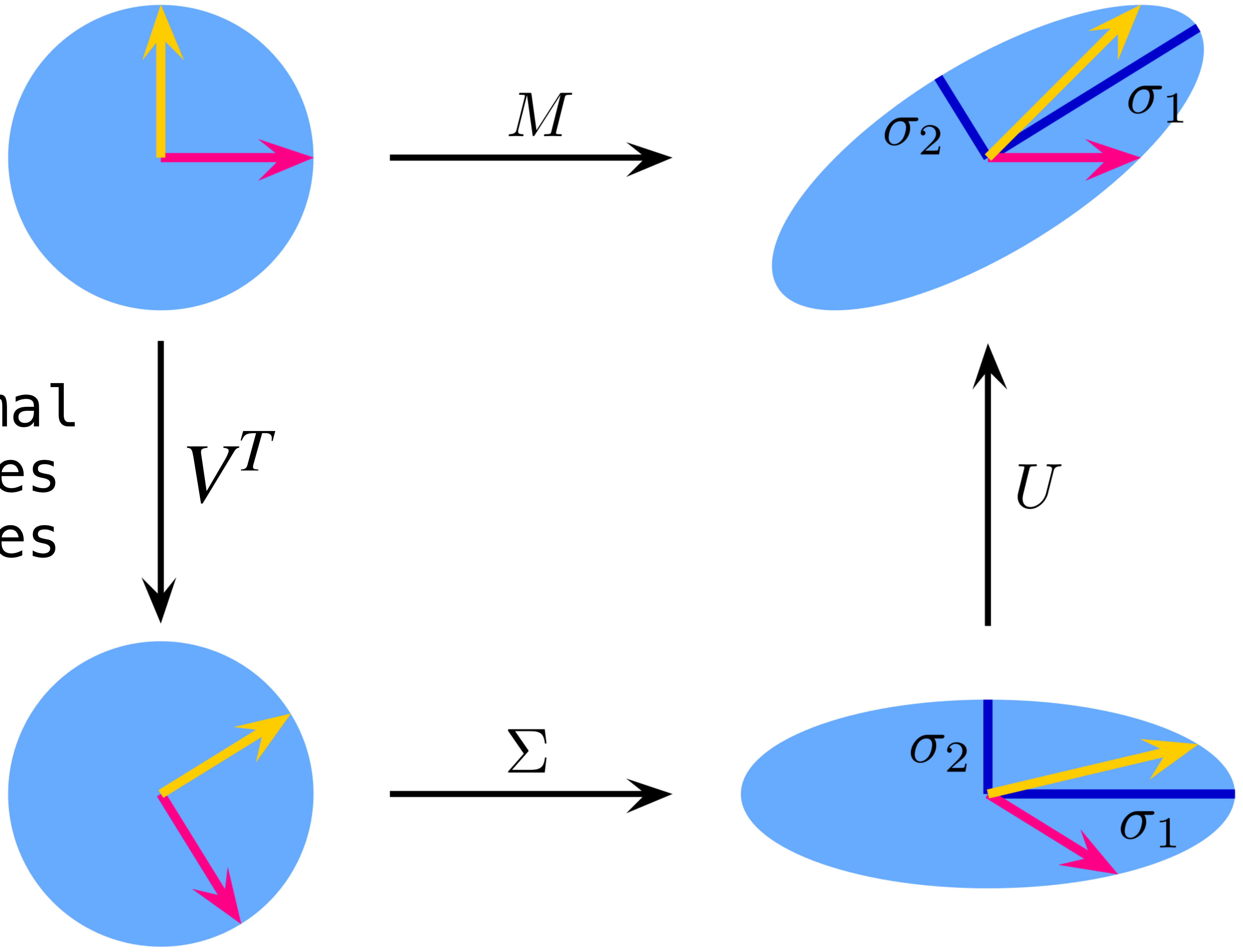


Recall: The Picture

The longest end of the ellipse is the solution to a constrained optimization problem



Recall: Another Picture



<u>recall</u>: Orthonormal matrices preserves lengths and angles

$$M = U \cdot \Sigma \cdot V^T$$

https://commons.wikimedia.org/wiki/File:Singular-Value-Decomposition.svg

Recall: Important Theorem

Let $\mathbf{v}_1, \ldots, \mathbf{v}_n$ be an **orthogonal** eigenbasis of $\mathbb{R}^n$ for A^TA and suppose A has r *nonzero* singular values

Thm. $A\mathbf{v}_1, \ldots, A\mathbf{v}_r$ is an orthogonal basis of $\text{Col}(A)$

Recall: All Together

$$\mathbf{u}_i = \frac{A\mathbf{v}_i}{\|A\mathbf{v}_i\|}$$

$$A\mathbf{v}_i = \|A\mathbf{v}_i\|\mathbf{u}_i = \sigma_i\mathbf{u}_i$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A\mathbf{v}_i\|$

The Important Equality

$$A[\mathbf{v}_1 \ \ldots \ \mathbf{v}_n] = [\sigma_1\mathbf{u}_1 \ \ldots\sigma_n\mathbf{u}_n]$$

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Let's take $V = [\mathbf{v}_1 \ \ldots \ \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \ \ldots \ \mathbf{u}_m]$ and

The Important Equality

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Let's take $V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \ \dots \ \mathbf{u}_m]$ and

$m > n$

$$\Sigma = \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \\ 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix}$$

$m < n$

or $\Sigma = \begin{bmatrix} \sigma_1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_m & 0 & \dots & 0 \end{bmatrix}$

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remember: U is orthonormal

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$m = n$

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The Important Equality

$$AV = U\Sigma$$

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Let's take $V = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \ \dots \ \mathbf{u}_m]$ and

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$$\overset{m > n}{\Sigma = \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \\ 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix}} \text{ or } \overset{m < n}{\Sigma = \begin{bmatrix} \sigma_1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_m & 0 & \dots & 0 \end{bmatrix}} \text{ or } \overset{m = n}{\Sigma = \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \end{bmatrix}}$$

The Important Equality

$$\underset{m\times n}{A}\;\underset{n\times n}{V} = \underset{m\times m}{U}\;\underset{m\times n}{\Sigma}$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value,
which is the length $\|A\mathbf{v}_i\|$

Let's take $V = [\mathbf{v}_1 \;\; \dots \;\; \mathbf{v}_n]$ and $U = [\mathbf{u}_1 \;\; \dots \;\; \mathbf{u}_m]$ and

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$$\Sigma = \overset{m>n}{\begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \\ 0 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 0 \end{bmatrix}} \text{ or } \Sigma = \overset{m<n}{\begin{bmatrix} \sigma_1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_m & 0 & \dots & 0 \end{bmatrix}} \text{ or } \Sigma = \overset{m=n}{\begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \end{bmatrix}}$$

The Important Equality

$$AVV^T = U\Sigma V^T$$

Remember that $\sigma_i = \sqrt{\lambda_i}$ is the singular value, which is the length $\|A\mathbf{v}_i\|$

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The Important Equality

$$A = U\Sigma V^T$$

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The Important Equality

singular value decomposition

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$$\Sigma = \begin{bmatrix} \sigma_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n \end{bmatrix}$$

Recall: Singular Value Decomposition

Thm. For $A \in \mathbb{R}^{m \times n}$, there are *orthogonal* matrices $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ such that

$$A = \overset{m \times m}{\boxed{U}} \underset{m \times n}{\Sigma} \overset{n \times n}{\boxed{V^T}}$$

where diagonal entries* of Σ are $\sigma_1, \ldots, \sigma_n$ the singular values of A

* these are diagonal entries in a *non-square* matrix

Recall: Singular Value Decomposition

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left singular   right singular

$$A = \underbrace{U}_{m\times m} \underbrace{\Sigma}_{m\times n} \underbrace{V^T}_{n\times n}$$

where diagonal entries* of Σ are $\sigma_1, \ldots, \sigma_n$ the singular values of A

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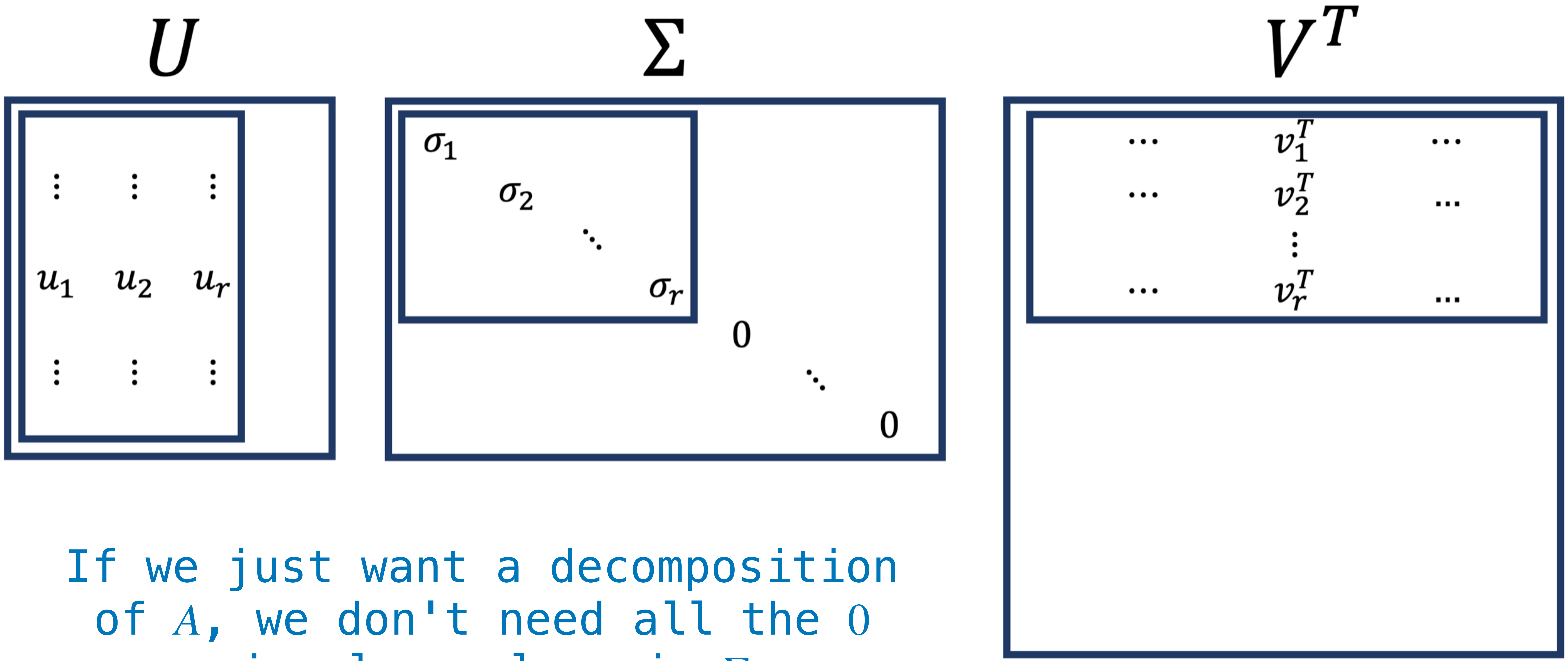
Recall: Reduced SVD



If we just want a decomposition of A, we don't need all the 0 singular values in Σ

Recall: The Reduced SVD

Thm. For every matrix A of rank r, there is an orthonormal matrix $U \in \mathbb{R}^{m \times r}$, a diagonal matrix $\Sigma \in \mathbb{R}^{r \times r}$ with **positive** entries on the diagonal, and an orthonormal matrix $V \in \mathbb{R}^{n \times r}$ such that

$$A = U \Sigma V^T$$

Recall: The Pseudoinverse

Def. Given a reduced SVD $A = U\Sigma V^T$, the **pseudoinverse** of A is $A^+ = V\Sigma^{-1}U^T$

Thm. $A^+\mathbf{b}$ is the *minimum length* **least squares solution** of $A\mathbf{x} = \mathbf{b}$

(in Python we have ***numpy.linalg.pinv****)*

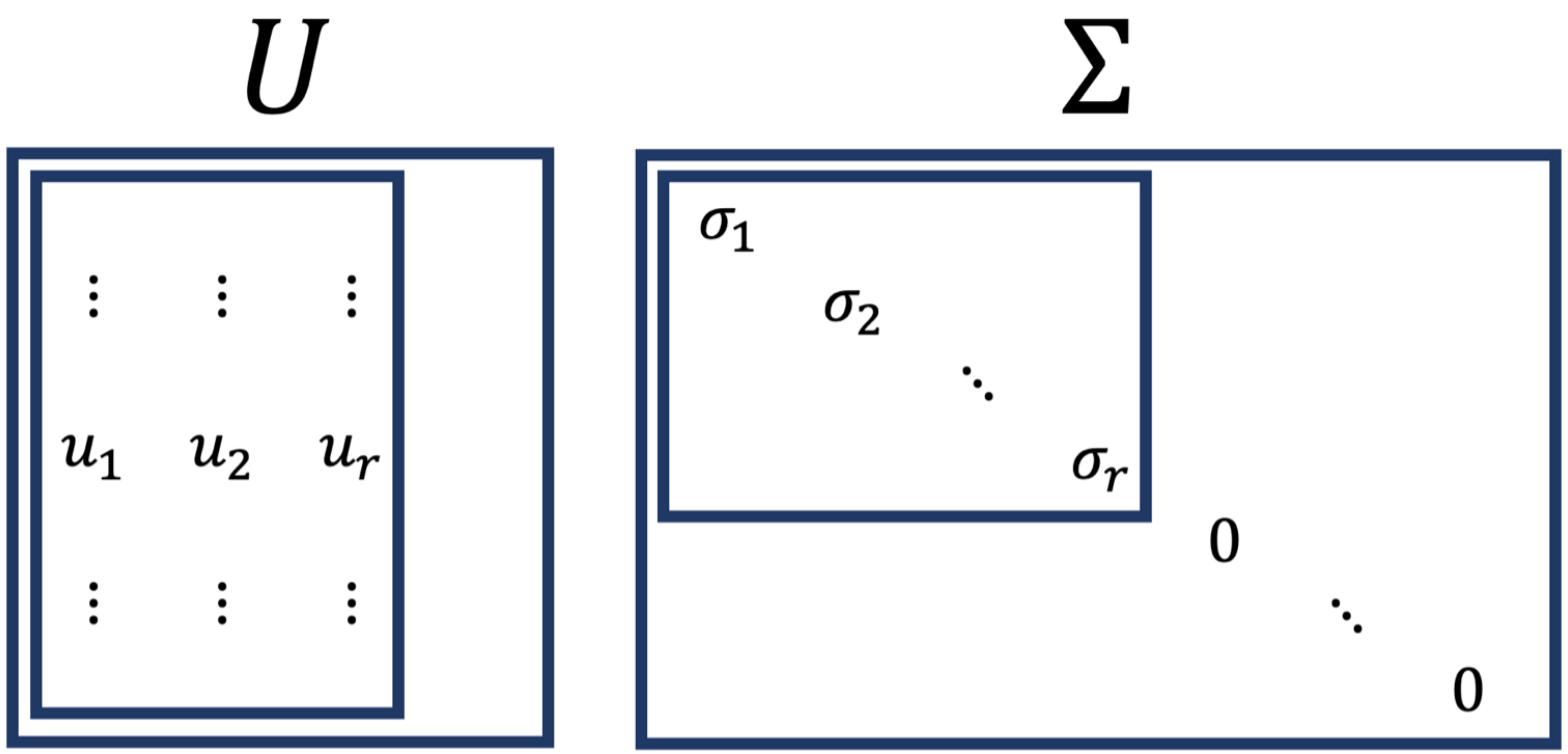
Low-Rank Approximation

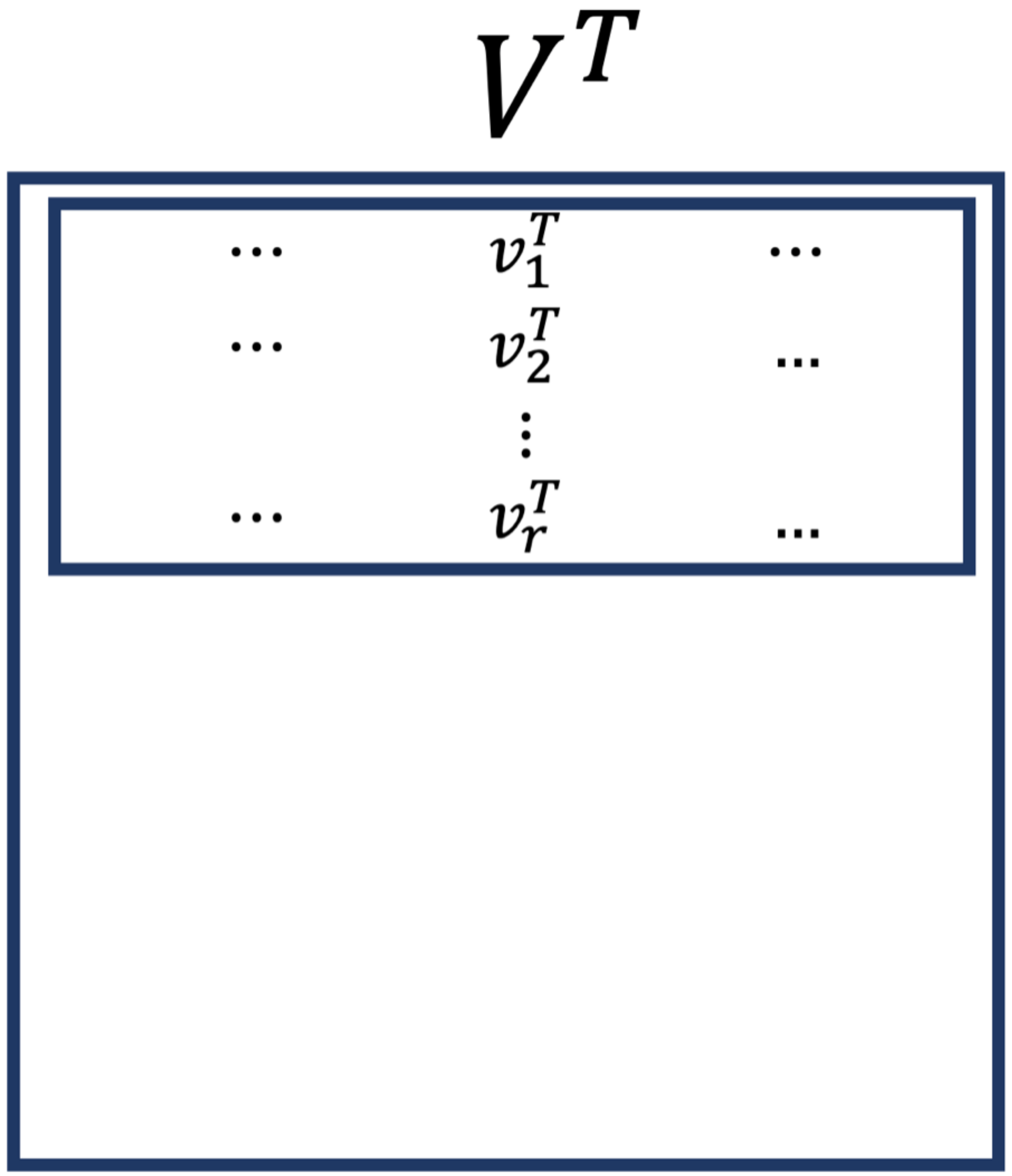
Motivating Question

Given a matrix A, what is the "most similar" matrix to A with a given rank r?

This has applications to ***compression***

Recall: Reduced SVD (The Picture)





If we just want a decomposition of A, we don't need all the 0 singular valudes in Σ

Matrix Norms

$$\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2}$$

Def. The **Frobenius norm** of a matrix is given by the above expression

Low-Rank Approximation

$$A^{(k)} = \text{argmin}_{\{B:\text{rank}(B)=r\}} \|A - B\|_F$$

Given a norm, we can define a constrained optimization problem for the "most similar" matrix of a given rank

An Alternative View of the SVD

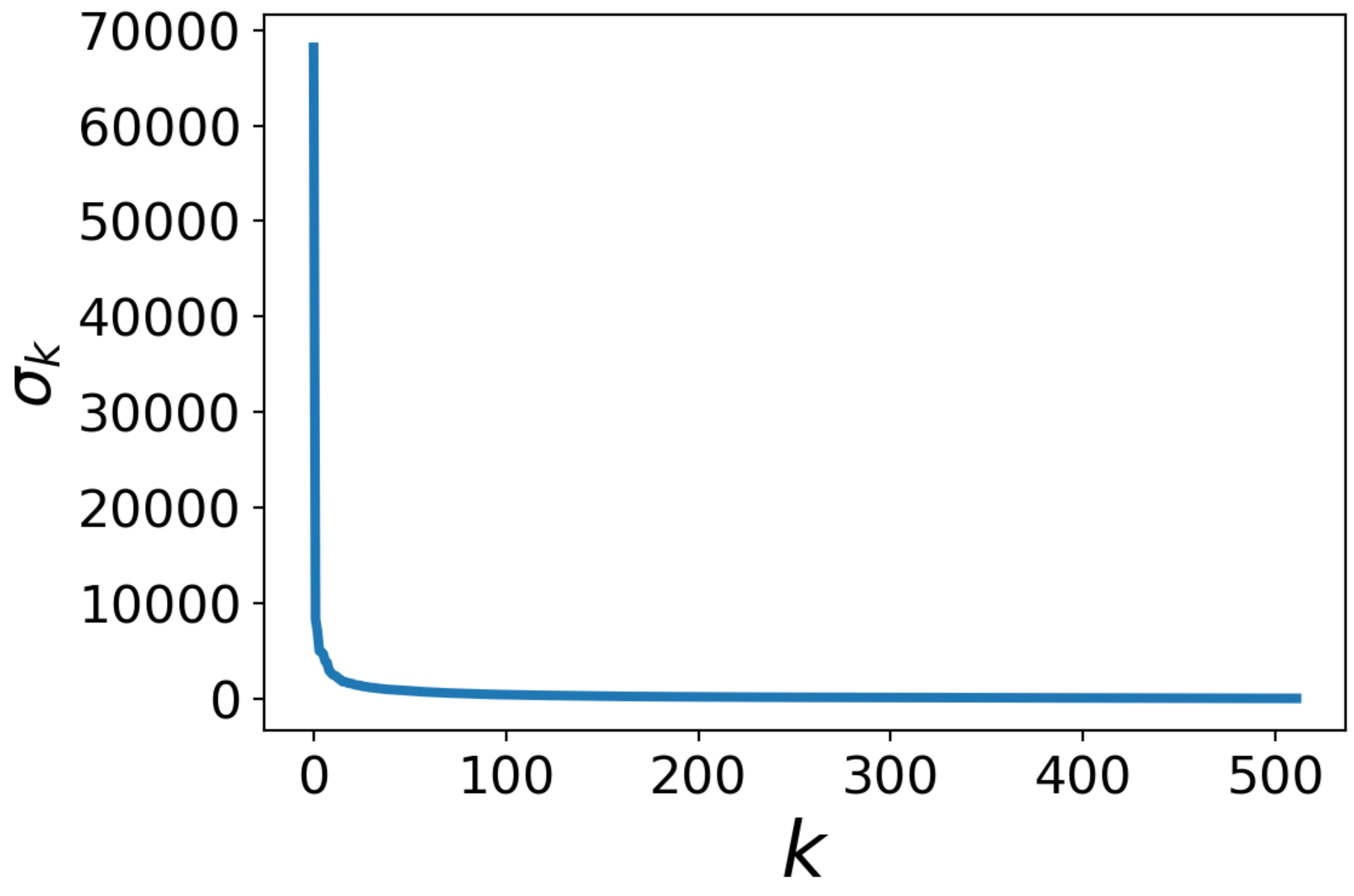
Low-Rank Approximation and SVD

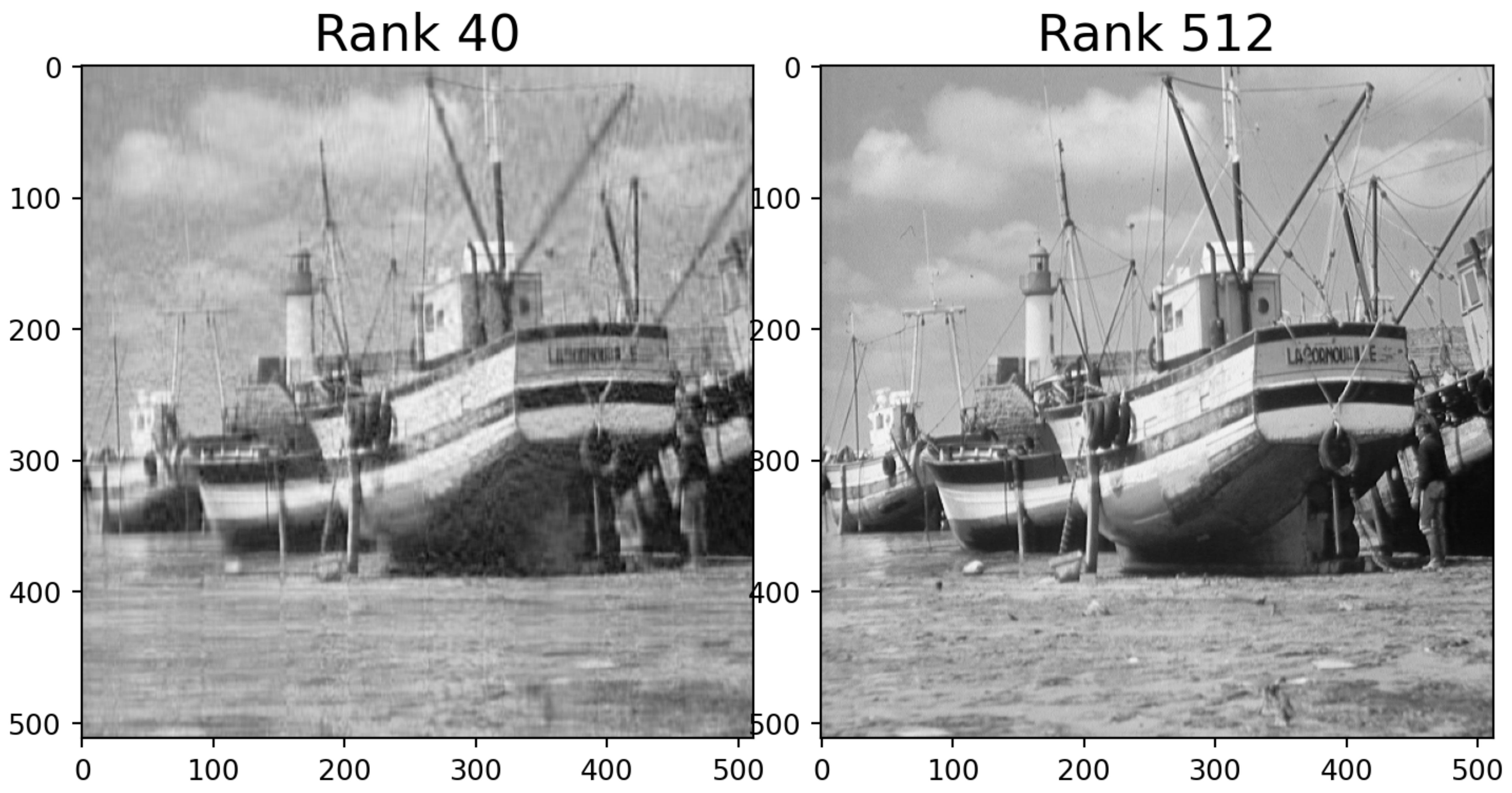
Thm. For $A \in \mathbb{R}^{m \times n}$ with left singular vectors $\mathbf{u}_1, \ldots, \mathbf{u}_m$ and right singular vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$

$$A^{(k)} = \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$$

Example (from the textbook)

singular values drop off rapidly

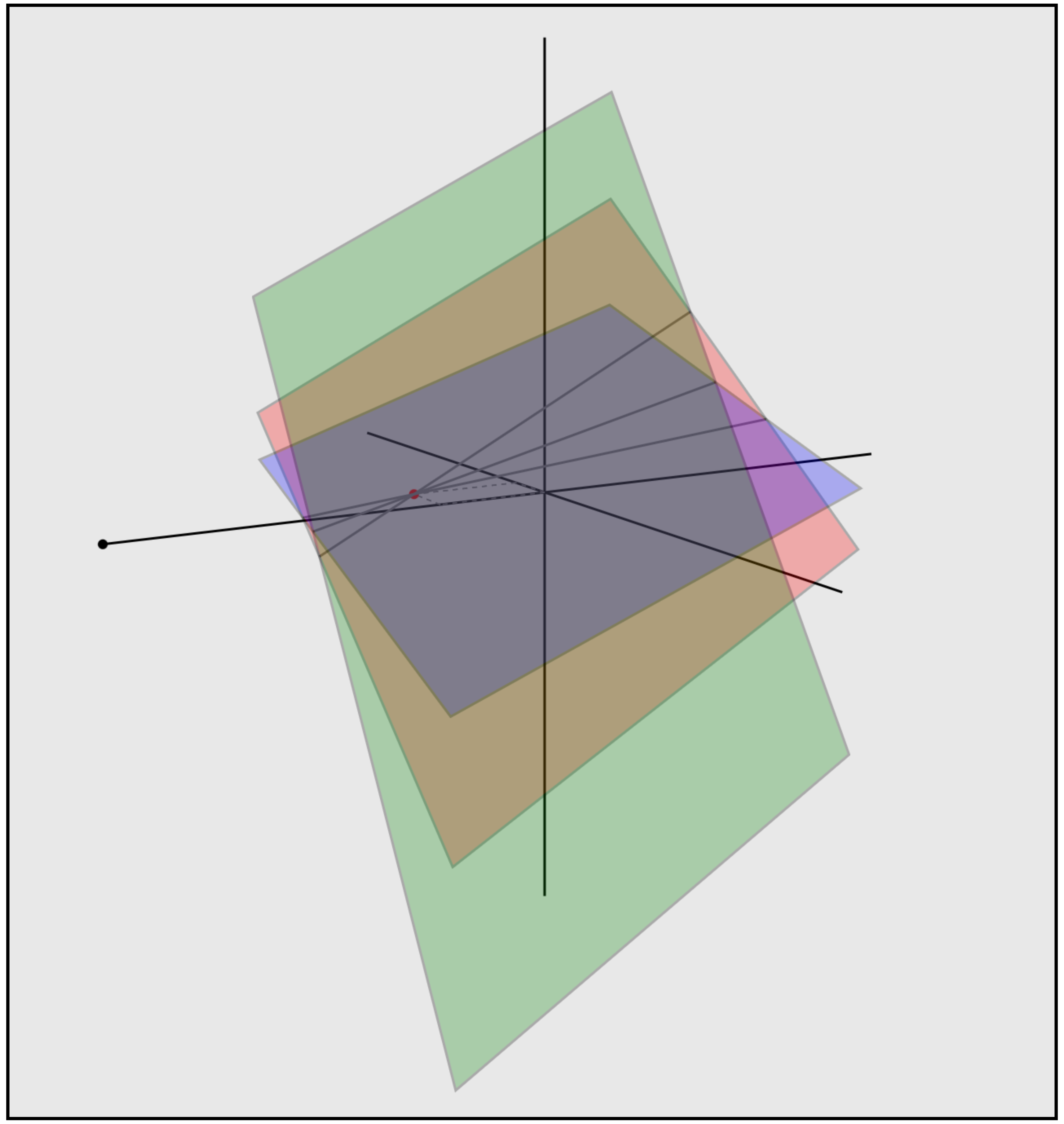




compressed to 8% of orig.

Course Overview

Recall: Linear Systems, Geometrically



Recall: Linear Systems (General-form)

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\
&\vdots \\
a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m
\end{aligned}
$$

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\end{aligned}
$$

Does a system have a solution?

How many solutions are there?

What are its solutions?

Recall: Matrix Representations

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

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augmented matrix

Recall: Elementary Row Operations

scaling	multiply a row by a **NONZERO** number
replacement	add a multiple of one row to another
interchange	switch two rows

Recall: Elementary Row Operations

scaling	multiply a row by a **NONZERO** number
replacement	add a multiple of one row to another
interchange	switch two rows

These operations don't change the solutions

Recall: Row Equivalence

Def. Two matrices are ***row equivalent*** if one can be transformed into the other by a sequence of row operations

$$\begin{bmatrix} 2 & 3 & -6 \\ 4 & -5 & 10 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

Recall: Echelon Form

$$\begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

■ = nonzero, * = anything

Recall: Echelon Form

next leading entry
to the right

$$\begin{bmatrix} 0 & \blacksquare & * & * & * & * & * & * & * & * \\ 0 & 0 & 0 & \blacksquare & * & * & * & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & \blacksquare & * & * & * & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

all-zero rows at
the bottom

$\blacksquare$ = nonzero, $*$ = anything

Recall: Reduced Echelon Form

$$\begin{bmatrix} 0 & 1 & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 1 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Recall: Reduced Echelon Form

leading entries are 1

$$\begin{bmatrix} 0 & \boxed{1} & * & 0 & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & \boxed{1} & 0 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 1 & 0 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 1 & * & * & 0 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & * \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

other column
entries are 0

Recall: Gaussian Elimination

```
FUNCTION GE(A):
  # INPUT: m × n matrix A
  # OUTPUT: row equivalent m × n RREF matrix
  ...
```

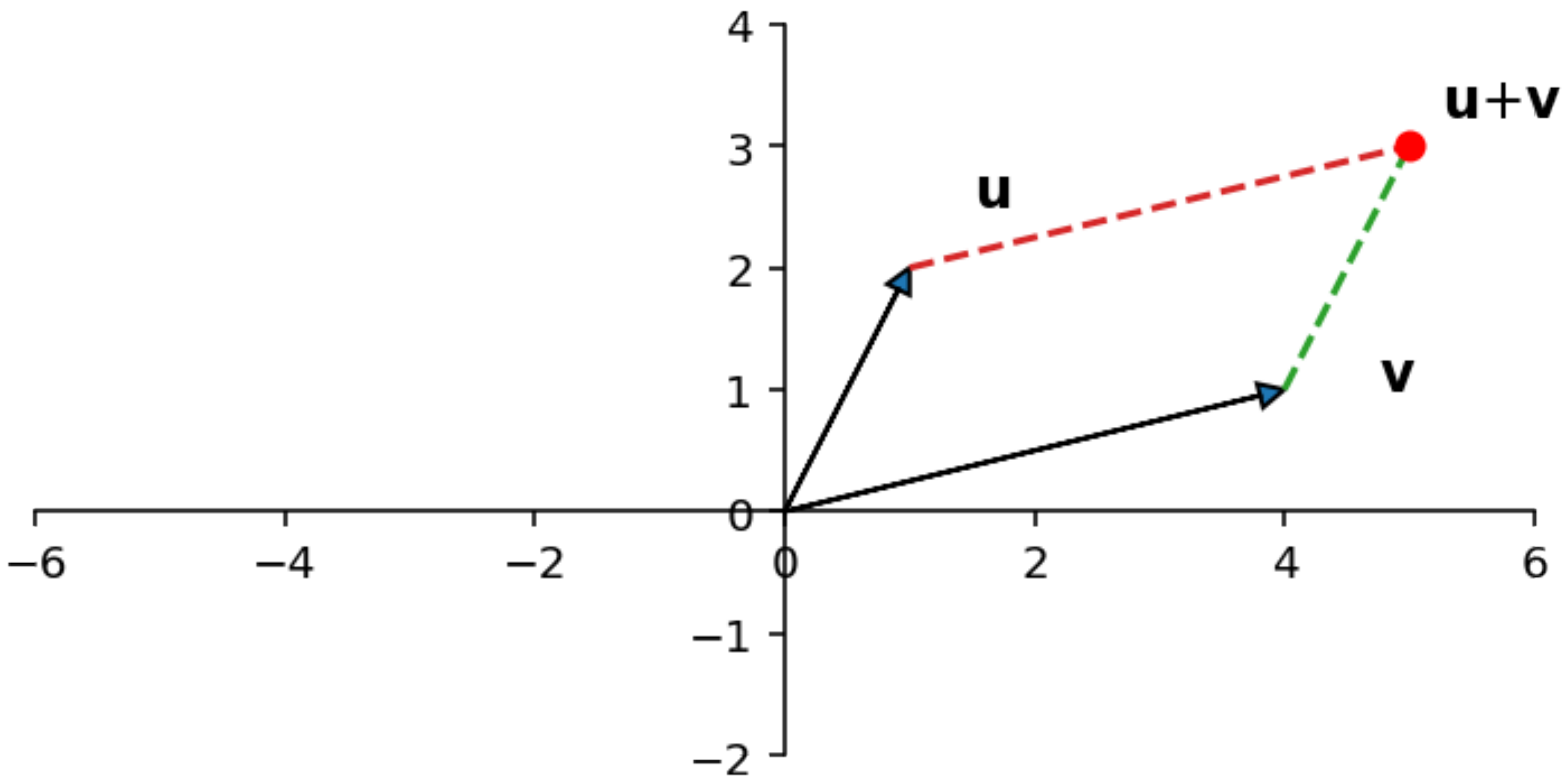
Recall: Gaussian Elimination

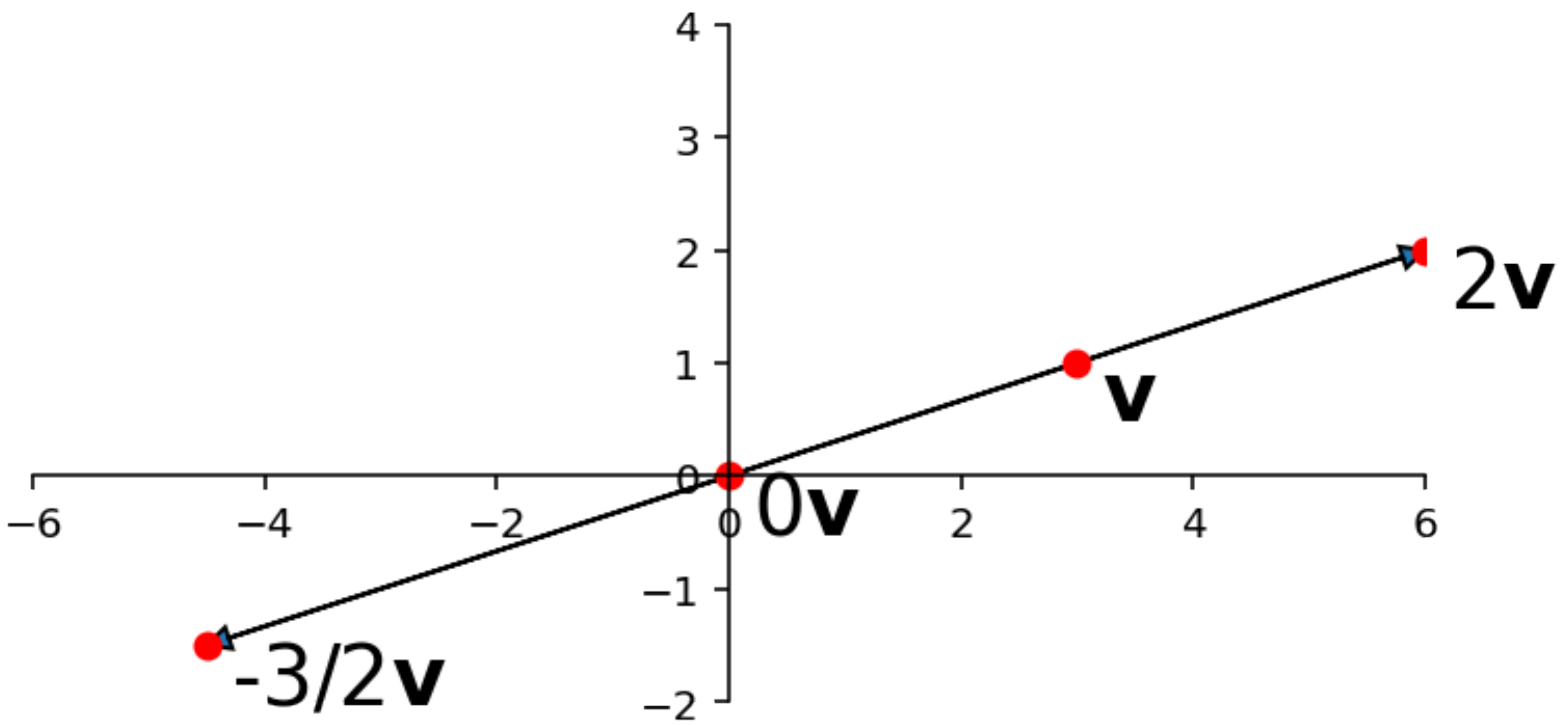
```
FUNCTION fwd_elim(A):
  # INPUT: m × n matrix A
  # OUTPUT: row equivalent m × n echelon form matrix
  ...

FUNCTION back_sub(A):
  # INPUT: m × n echelon form matrix A
  # OUTPUT: row equivalent m × n RREF matrix
  ...

FUNCTION GE(A):
  RETURN back_sub(fwd_elim(A))
```

Recall: Column Vectors

Def. a ***column vector*** is a matrix with a single column

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$





Recall: Vector "Interface"

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`equality`	`what does it mean for two vectors to be equal?`

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addition	what does $\mathbf{u} + \mathbf{v}$ (adding two vectors mean?

Recall: Vector "Interface"

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scaling — what does $a\mathbf{v}$ (multiplying a vector by a real number) mean?

Recall: Vector "Interface"

equality — what does it mean for two vectors to be equal?

addition — what does $\mathbf{u} + \mathbf{v}$ (adding two vectors mean?

scaling — what does $a\mathbf{v}$ (multiplying a vector by a real number) mean?

What properties do they need to satisfy?

Recall: Algebraic Properties

For any vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and any real numbers c, d:

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$$

$$\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$$

$$\mathbf{u} + (-\mathbf{u}) = -\mathbf{u} + \mathbf{u} = \mathbf{0}$$

$$c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$$

$$(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$$

$$c(d\mathbf{u}) = (cd)\mathbf{u}$$

$$1\mathbf{u} = \mathbf{u}$$

Recall: Linear Combinations

Def. a ***linear combination*** of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ is a vector of the form

$$\alpha_1 \mathbf{v}_1 + \alpha_1 \mathbf{v}_2 + \ldots + \alpha_n \mathbf{v}_n$$

where $\alpha_1, \alpha_2, \ldots, \alpha_n$ are in $\mathbb{R}$

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where $\underbrace{\alpha_1, \alpha_2, \ldots, \alpha_n}_{\text{weights}}$ are in $\mathbb{R}$

Recall: Vector Equations

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

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system of linear equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{1m} \end{bmatrix} + x_2 \begin{bmatrix} a_{21} \\ a_{21} \\ \vdots \\ a_{2m} \end{bmatrix} + \ldots + x_n \begin{bmatrix} a_{n1} \\ a_{n2} \\ \vdots \\ a_{nm} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

vector equation

Recall: Vector Equations

$$\left[\begin{array}{ccccc} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{array}\right]$$

augmented matrix

$$\begin{gathered} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{gathered}$$

system of linear equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{1m} \end{bmatrix} + x_2 \begin{bmatrix} a_{21} \\ a_{21} \\ \vdots \\ a_{2m} \end{bmatrix} + \dots + x_n \begin{bmatrix} a_{n1} \\ a_{n2} \\ \vdots \\ a_{nm} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

vector equation

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augmented matrix

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{aligned}$$

system of linear equations

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{1m} \end{bmatrix} + x_2 \begin{bmatrix} a_{21} \\ a_{21} \\ \vdots \\ a_{2m} \end{bmatrix} + \ldots + x_n \begin{bmatrix} a_{n1} \\ a_{n2} \\ \vdots \\ a_{nm} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

vector equation

Recall: Matrix-Vector Multiplication

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix}$$

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a linear combination of the columns where
$\mathbf{s}$ defines the weights

Recall: Algebraic Properties

Matrix–vector multiplication satisfies the following two properties:

1. $A(\mathbf{u}+\mathbf{v}) = A\mathbf{u} + A\mathbf{v}$ (additivity)

2. $A(c\mathbf{v}) = c(A\mathbf{v})$ (homogeneity)

Recall: Four Representations

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix}$$

augmented matrix

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

matrix equation

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

system of linear equations

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vector equation

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matrix equation

they all have the same solution sets

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n &= b_m \end{aligned}$$

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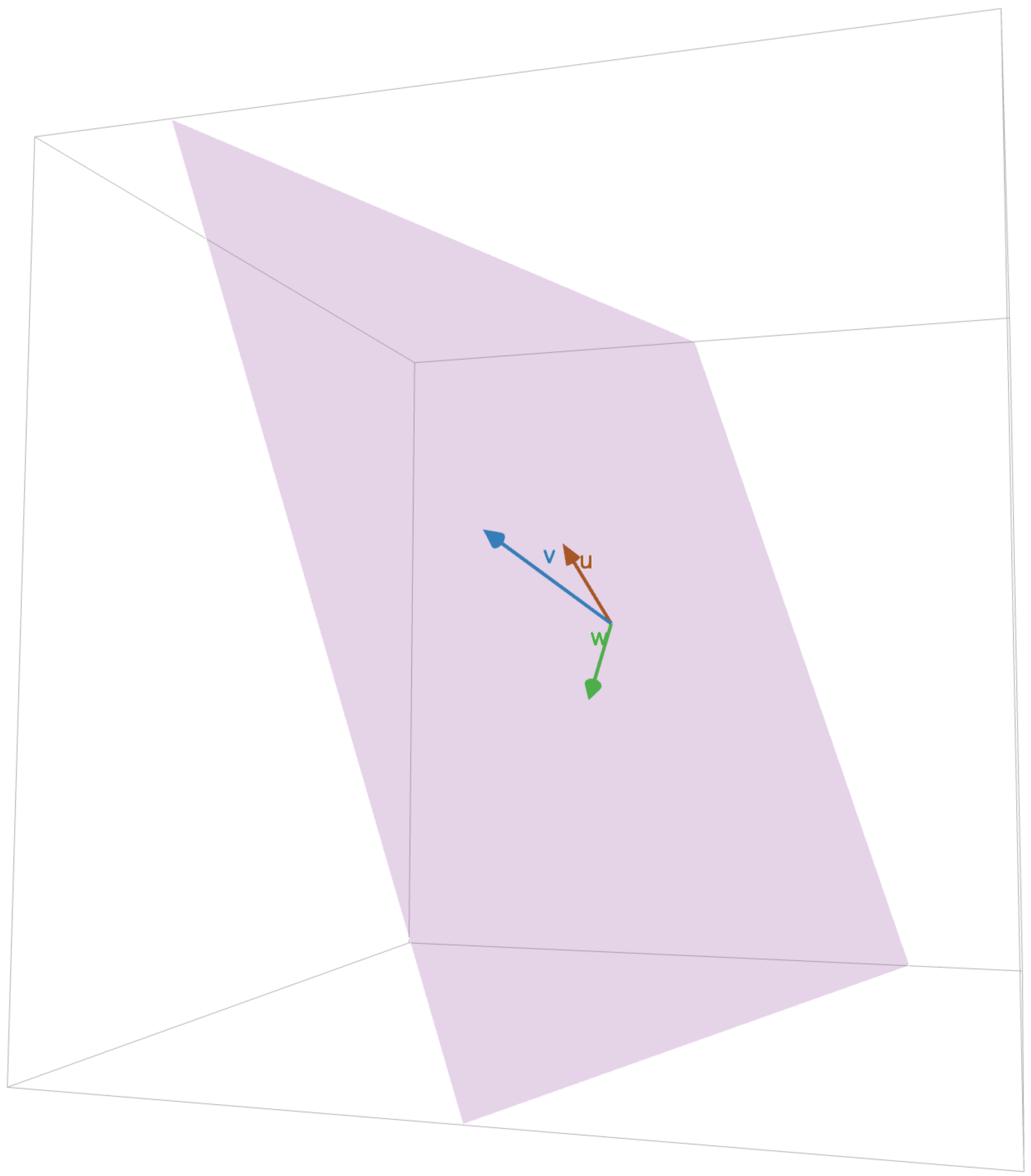
vector equation

Recall: Spans

Def. the ***span*** of a set of vectors is the set of all possible linear combinations of them



$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\} = \{\alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 + \ldots\alpha_n\mathbf{v}_n : \alpha_1, \alpha_2, \ldots, \alpha_n \text{ are in } \mathbb{R}\}$$

Recall: Homogenous Linear Systems

Def. A system of linear equations is called ***homogeneous*** if it can be expressed as

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \ldots + x_n\mathbf{a}_n = \mathbf{0}$$

Recall: Linear Independence

Def. A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is ***linearly independent*** if the vectors equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \ldots + x_n\mathbf{v}_n = \mathbf{0}$$

has exactly one solution (the trivial solution)

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has exactly one solution (the trivial solution)

The columns of A are linearly independent if $A\mathbf{x} = \mathbf{0}$ has exactly one solution

Recall: Linear Dependence

Definition. A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is ***linearly dependent*** if the vectors equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \ldots + x_n\mathbf{v}_n = \mathbf{0}$$

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has a nontrivial solution

A set of vectors is linearly dependent if there is a nontrivial linear combination of the vectors which equals $\mathbf{0}$

Recall: Increasing Span Criterion

Thm. $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ are linearly independent if and only if for all $i \leq n$,

$$\mathbf{v}_i \notin \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{i-1}\}$$

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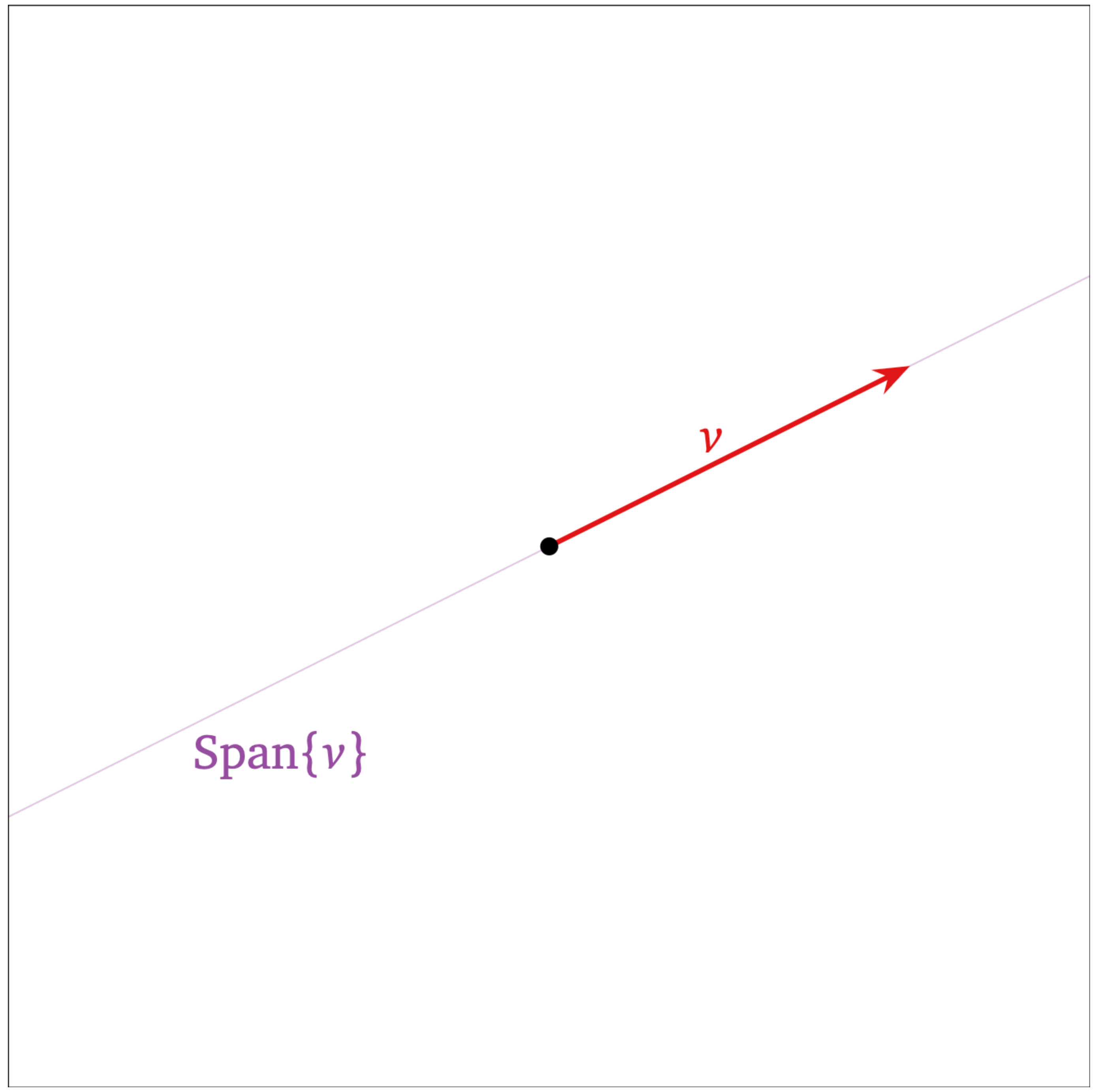
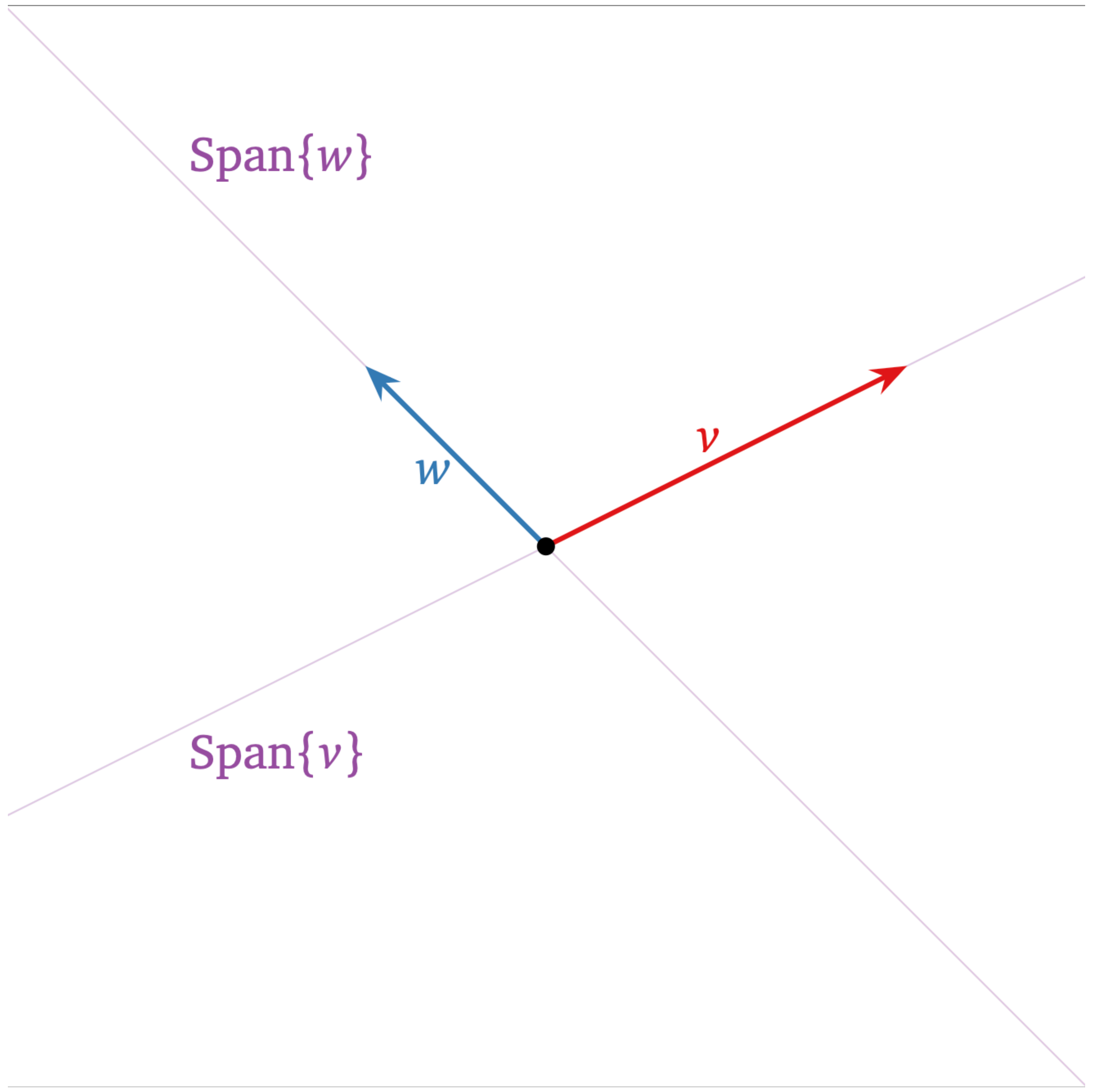
As we add vectors, the span gets larger

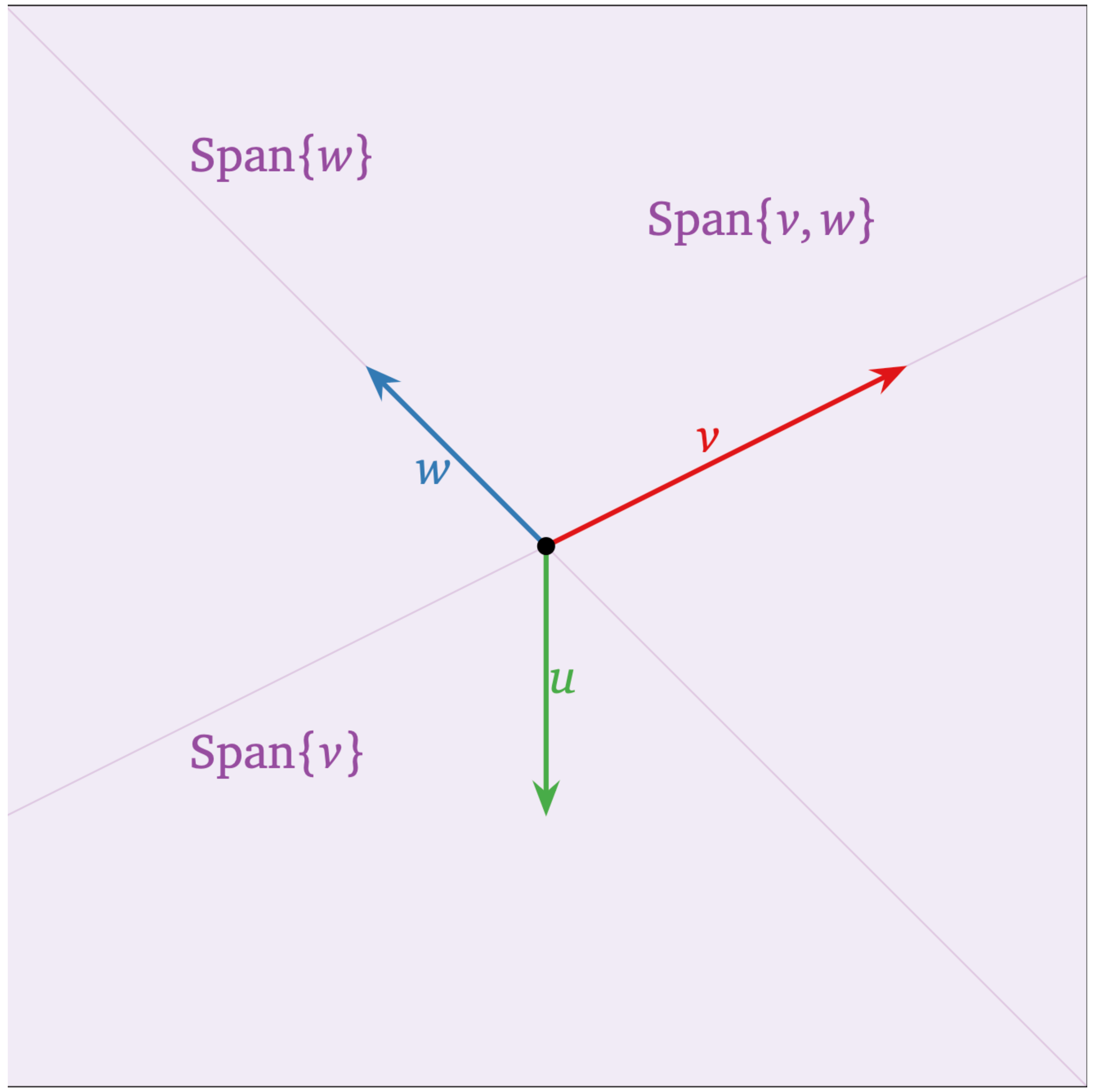
Recall: The Picture



span of 1 vector
a line



span of 2 vector
a plane



span of 3 vector
still a plane

image source

Recall: Linear Independence and Full Span

The columns of a $(m \times n)$ matrix span all of $\mathbb{R}^n$ if there is a pivot in every row

The columns of a matrix are linearly independent if there is a pivot in every column

Recall: Transformations in General

Definition. A ***transformation*** T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a function which maps every vector $\mathbf{v}$ in $\mathbb{R}^n$ to a vector $T(\mathbf{v})$ in $\mathbb{R}^m$

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$$T : \mathbb{R}^n \to \mathbb{R}^m$$

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$$T : \underbrace{\mathbb{R}^n}_{\text{domain}} \to \underbrace{\mathbb{R}^m}_{\text{codomain}}$$

Recall: Image and Range

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Def. For a vector $\mathbf{v}$, the ***image*** of $\mathbf{v}$ under the transformation T is the vector $T(\mathbf{v})$

Def. The ***range*** of a transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is the set of all possible images under T

$$\mathrm{ran}(T) = \{T(\mathbf{v}) : v \in \mathbb{R}^n\}$$

Recall: Matrix Transformations

Matrices *transform* vectors into vectors the span of its columns

$$\mathbf{x} \mapsto A\mathbf{x}$$

map a vector $\mathbf{v}$ to the vector $A\mathbf{v}$

Recall: Linear Transformations

Def. A transformation $T:\mathbb{R}^m \to \mathbb{R}^n$ is ***linear*** if it satisfies the following two properties

1. $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

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Matrix transformations are linear transformations

Recall: Matrix of Linear Transformation

Thm. For any linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$, the matrix

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix}$$

is the <u>unique</u> matrix that implements T

Recall: Onto Transformations

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Def. A transformation $T: \mathbb{R}^n \to \mathbb{R}^m$ is ***onto*** if any vector $\mathbf{b}$ in $\mathbb{R}^m$ is the image of at least one vector $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)

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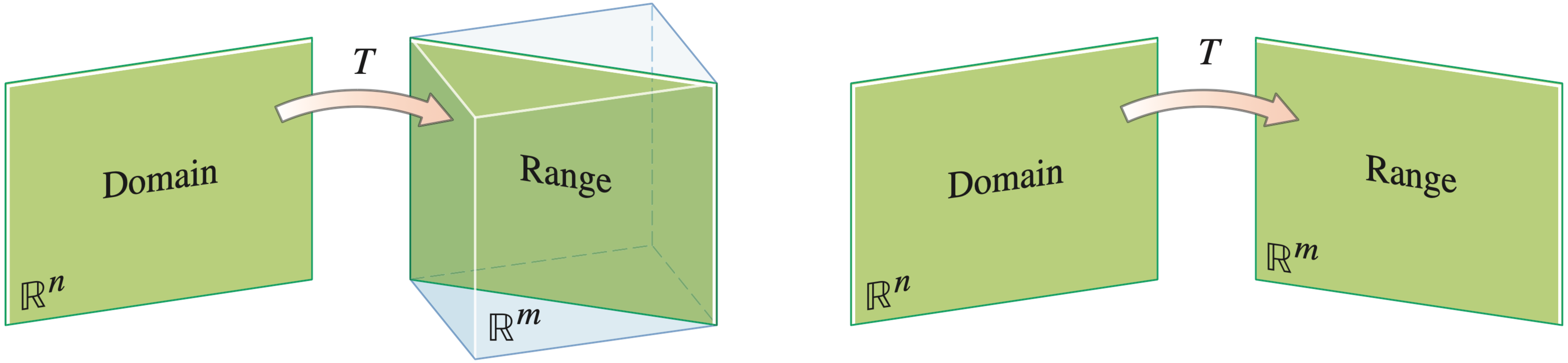

T is *not* onto $\mathbb{R}^m$ $\qquad$ T is onto $\mathbb{R}^m$

image source: Linear Algebra and its Applications. Lay, Lay, and McDonald

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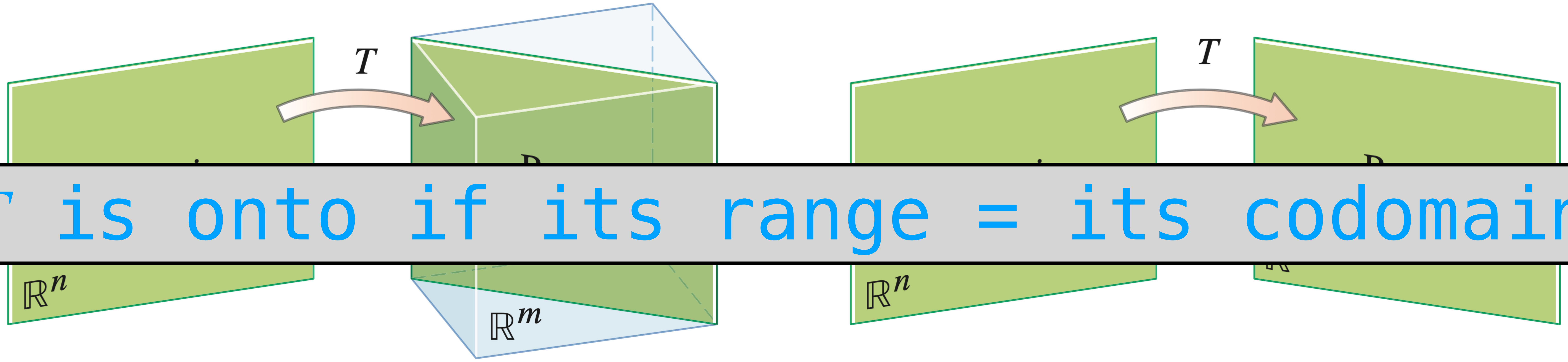

T is onto if its range = its codomain

T is *not* onto $\mathbb{R}^m$ $\qquad$ T is onto $\mathbb{R}^m$

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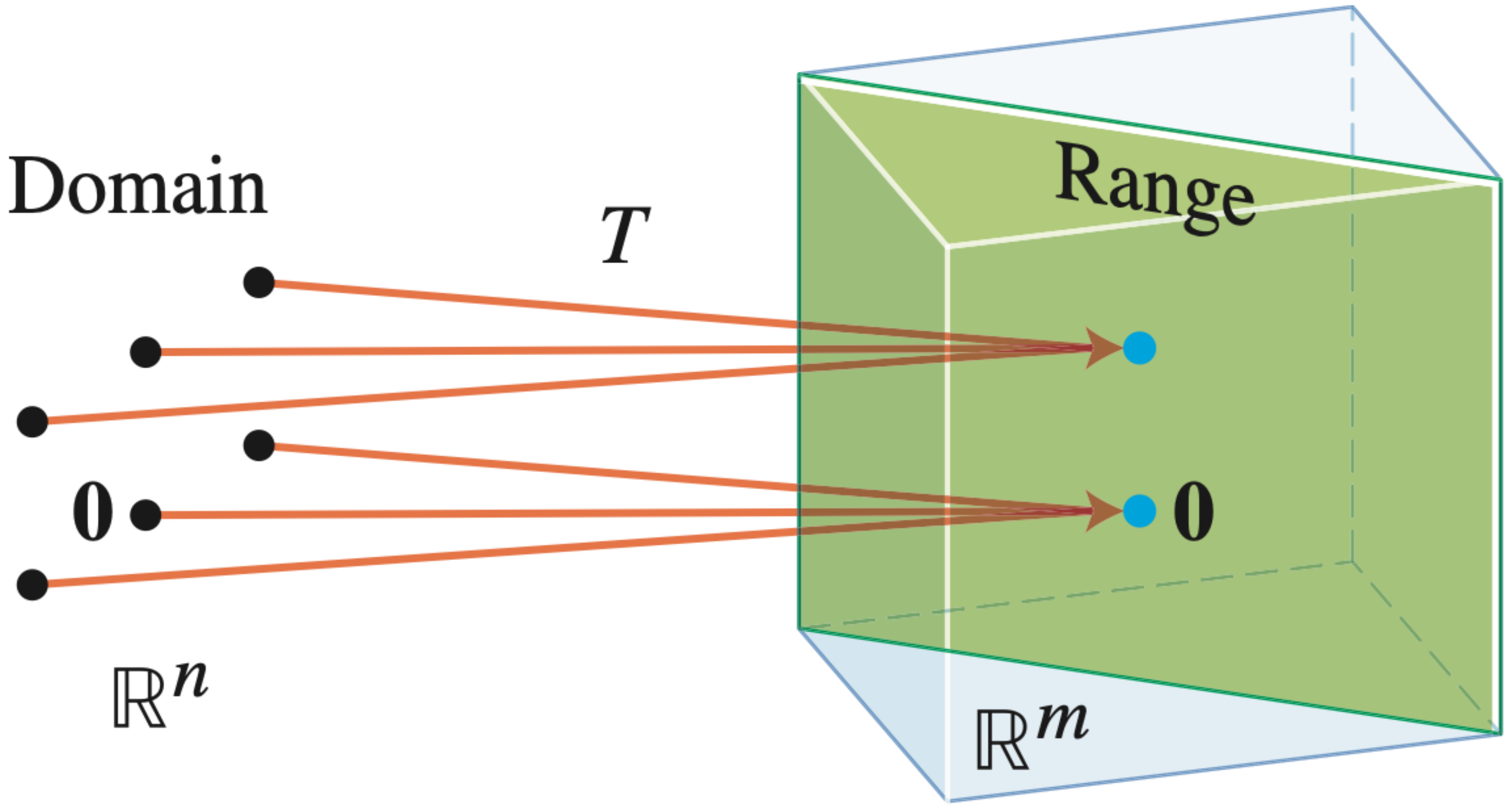
Recall: One-to-one Transformations

Recall: One-to-one Transformations

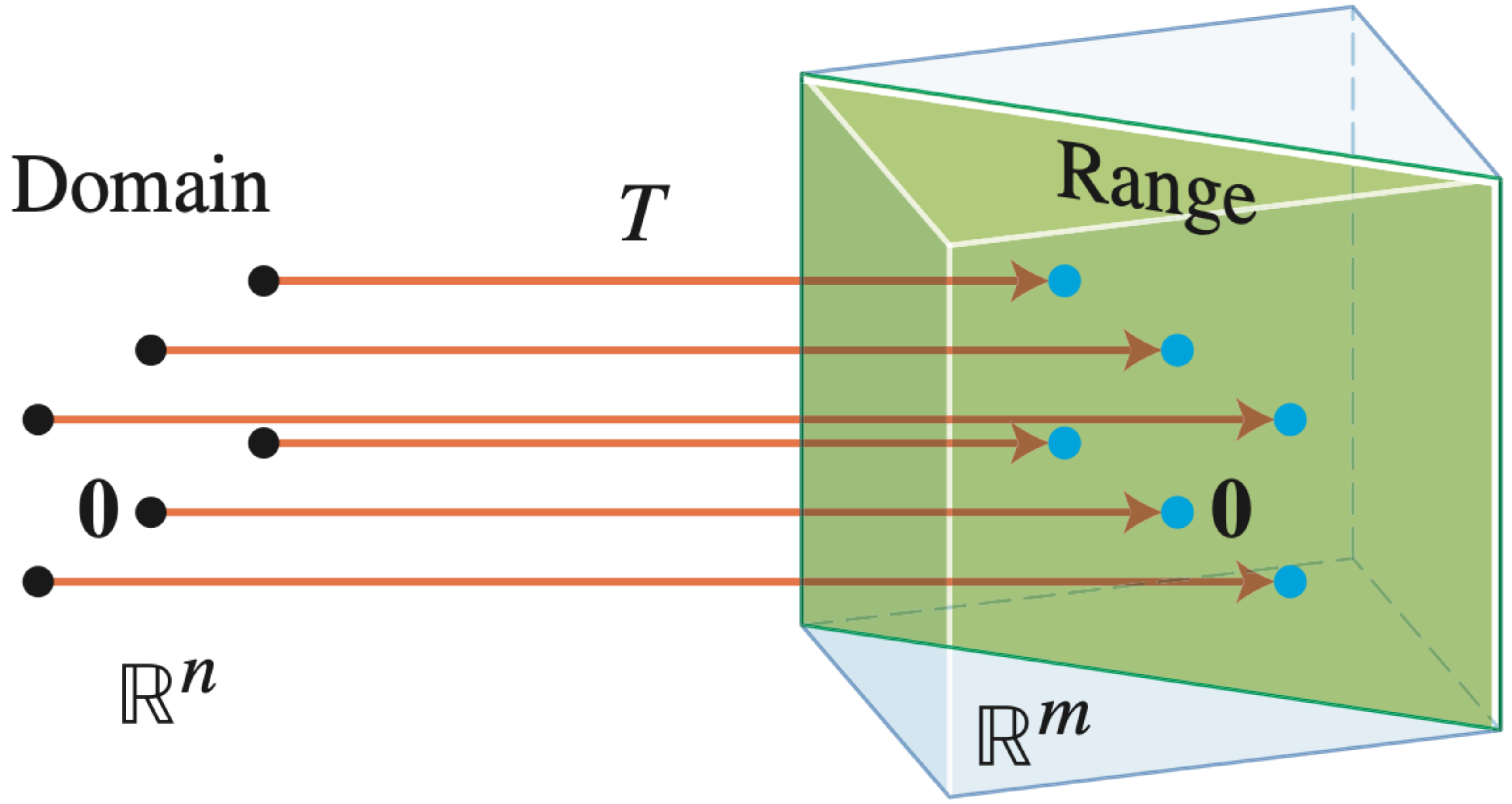
Def. A transformation $T:\mathbb{R}^n \to \mathbb{R}^m$ is ***one–to–one*** if any vector $\mathbf{b}$ in $\mathbb{R}^m$ is the image of at most one vector $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)

Recall: One-to-one Transformations

Def. A transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is ***one-to-one*** if any vector $\mathbf{b}$ in $\mathbb{R}^m$ is the image of at most one vector $\mathbf{v}$ in $\mathbb{R}^n$ (where $T(\mathbf{v}) = \mathbf{b}$)



T is *not* one-to-one



T is one-to-one

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Recall: Onto

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Thm. The following are equivalent for linear $T : \mathbb{R}^n \to \mathbb{R}^m$ implemented by A

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Thm. The following are equivalent for linear $T : \mathbb{R}^n \to \mathbb{R}^m$ implemented by A

- T is onto
- $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$

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- $\text{range}(T) = \text{codomain}(T)$

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» A has a pivot position in every <u>row</u>

Recall: One-to-One

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» $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
» $A\mathbf{x} = \mathbf{0}$ has only the trivial solution

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» The columns of A are linearly independent
» A has a pivot position in every <u>column</u>

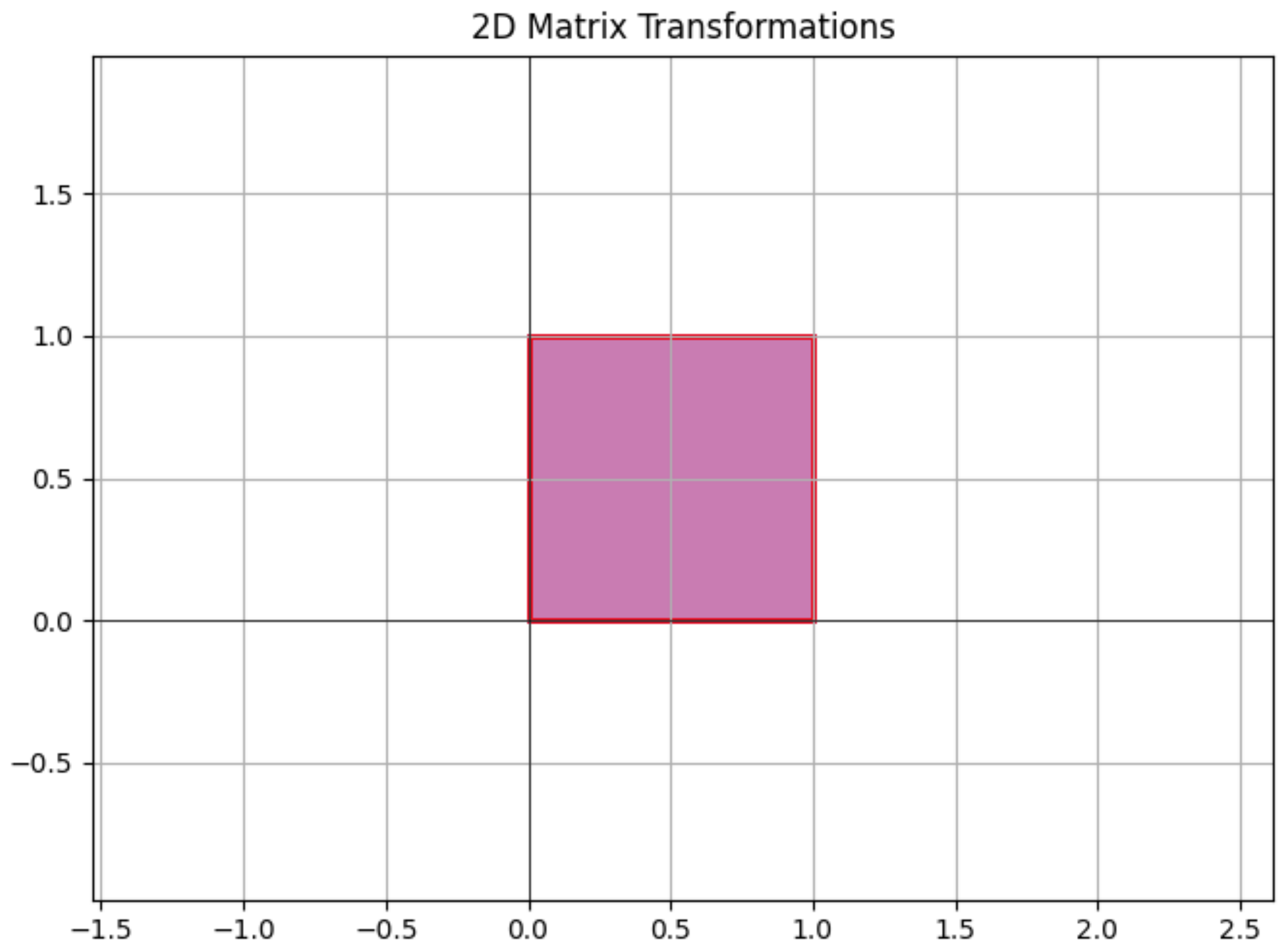
Recall: Matrix Multiplication

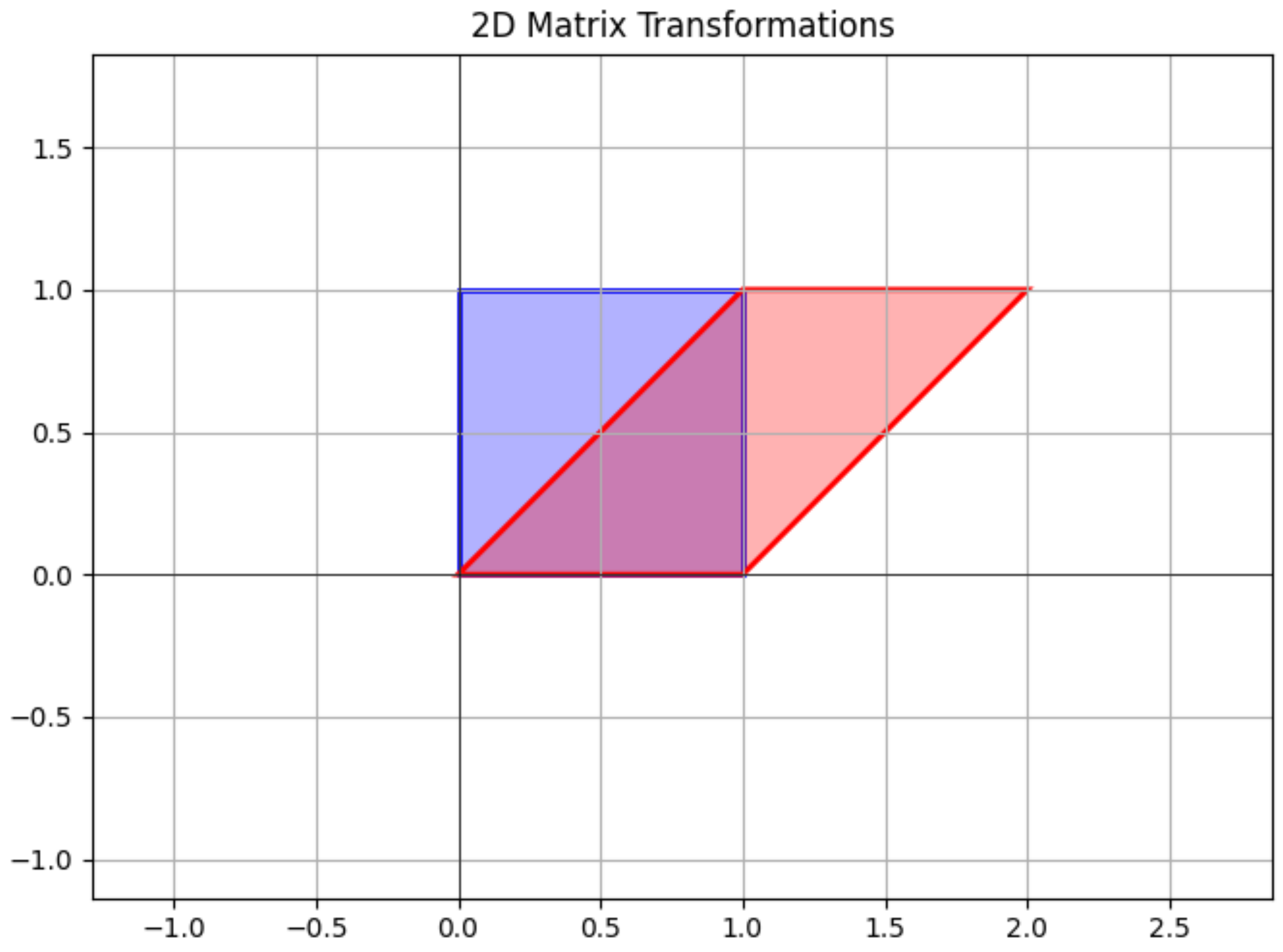
Def. For $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$ with columns $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_p$ the product $AB \in \mathbb{R}^{m\times p}$ is given by

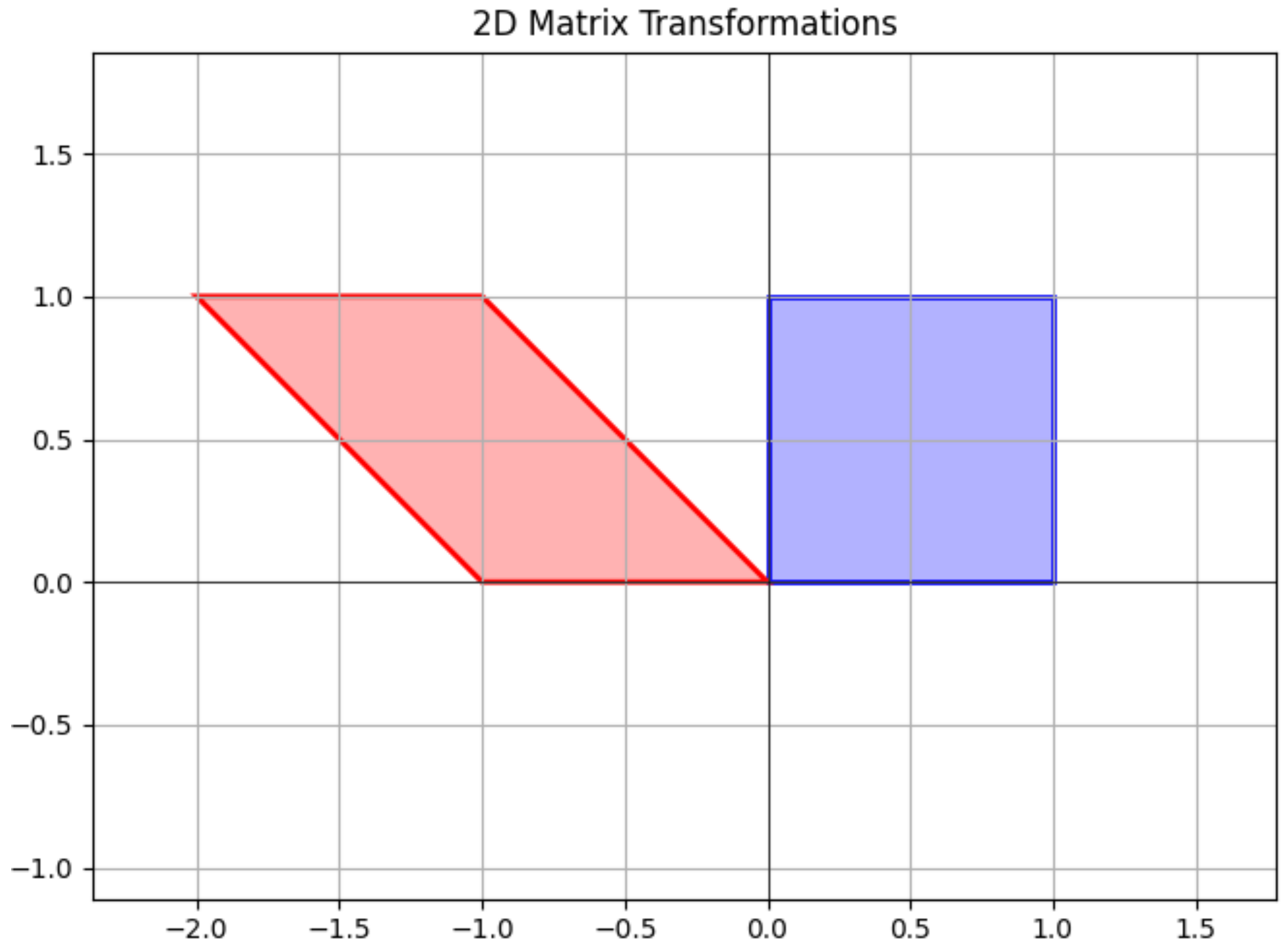
$$AB = A\begin{bmatrix}\mathbf{b}_1 & \mathbf{b}_2 & \ldots & \mathbf{b}_p\end{bmatrix} = \begin{bmatrix}A\mathbf{b}_1 & A\mathbf{b}_2 & \ldots & A\mathbf{b}_p\end{bmatrix}$$

Replace each column of B with A multiplied by that column

Recall: Matrix Multiplication is Composition







shear

reflect

Recall: Matrix "Interface"

multiplication	what does AB mean when A and B are matrices?
addition	what does $A+B$ mean when A and B are matrices?
scaling	what does cA mean when A is matrix and c is a real number?

Recall: Transpose

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 5 & 5 & 5 & 5 \end{bmatrix}$$

Recall: Transpose

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Recall: Algebraic Properties

$(A^T)^T = A$

$(A + B)^T = A^T + B^T$

$(cA)^T = cA^T$ (where c is a scalar)

$(AB)^T = B^T A^T$

Recall: Algebraic Properties

$(A^T)^T = A$

$(A + B)^T = A^T + B^T$

$(cA)^T = cA^T$ (where c is a scalar)

$(AB)^T = B^T A^T$ Important: the order reverses!

Recall: Matrix Inverses

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Def. For a matrix $A \in \mathbb{R}^{n\times n}$, an **inverse** of A is a matrix $B \in \mathbb{R}^{n\times n}$ such that

$$AB = I_n \text{ and } BA = I_n$$

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A is **invertible** if it has an inverse. Otherwise it is **singular**

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Example. $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$

Recall: Invertible Matrix Theorem

The following are logically equivalent:

1. A is invertible
2. A^T is invertible
3. $A\mathbf{x} = \mathbf{b}$ has at least one solution for any $\mathbf{b}$
4. $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
5. $A\mathbf{x} = \mathbf{b}$ has a unique solution for any $\mathbf{b}$
6. A has n pivots (per row and per column)
7. A is row equivalent to I
8. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution
9. The columns of A are linearly independent
10. The columns of A span $\mathbb{R}^n$
11. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is onto
12. $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one
13. $\mathbf{x} \mapsto A\mathbf{x}$ is a one-to-one correspondence
14. $\mathbf{x} \mapsto A\mathbf{x}$ is invertible

These all express the **same thing**

!! only for square matrices !!

Recall: LU Factorization

Given a $m \times n$ matrix A, we are going to factorize A as

echelon form of A

$$A = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ * & 1 & 0 & 0 \\ * & * & 1 & 0 \\ * & * & * & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}}_{U}$$

Note. This applies to non-square matrices

Recall: Gaussian Elimination and Elementary Matrices

$$A \sim E_1 A \sim E_2 E_1 A \sim \ldots \sim E_k E_{k-1} \ldots E_2 E_1 A$$

Recall: Gaussian Elimination and Elementary Matrices

$$A \sim E_1A \sim E_2E_1A \sim \ldots \sim E_kE_{k-1}\ldots E_2E_1A$$

This exactly tells us that if B is the final echelon form we get then

$B = (E_kE_{k-1}\ldots E_2E_1)A = EA$

where E implements a <u>sequence</u> of row operations. So:

Recall: Gaussian Elimination and Elementary Matrices

$$A \sim E_1 A \sim E_2 E_1 A \sim \ldots \sim E_k E_{k-1} \ldots E_2 E_1 A$$

This exactly tells us that if B is the final echelon form we get then

$$B = \overbrace{(E_k E_{k-1} \ldots E_2 E_1)}^{\text{Invertible}} A = EA$$

where E implements a <u>sequence</u> of row operations. So:

Recall: Gaussian Elimination and Elementary Matrices

$$A \sim E_1A \sim E_2E_1A \sim \ldots \sim E_kE_{k-1}\ldots E_2E_1A$$

This exactly tells us that if B is the final echelon form we get then

$$B = \underbrace{(E_kE_{k-1}\ldots E_2E_1)}_{\text{Invertible}}A = EA$$

where E implements a <u>sequence</u> of row operations. So:

$$A = E^{-1}B = (E_1^{-1}E_2^{-1}\ldots E_{k-1}^{-1}E_k^{-1})B$$

Recall: Linear Dynamical Systems

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Def. A **(discrete time) linear dynamical system** is a described a $n \times n$ matrix A. It's **evolution function** is the matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$

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The **possible states** of the system are vectors in $\mathbb{R}^n$

Given an **initial state vector** $\mathbf{v}_0$, we can determine the **state vector** of the system after i time steps:

$$\mathbf{v}_i = A\mathbf{v}_{i-1}$$

(linear recurrence relation)

Recall: State Vectors

$$
\begin{aligned}
&\mathbf{v}_0 \\
&\mathbf{v}_1 = A\mathbf{v}_0 \\
&\mathbf{v}_2 = A\mathbf{v}_1 = A(A\mathbf{v}_0) \\
&\mathbf{v}_3 = A\mathbf{v}_2 = A(AA\mathbf{v}_0) \\
&\mathbf{v}_4 = A\mathbf{v}_3 = A(AAA\mathbf{v}_0) \\
&\mathbf{v}_5 = A\mathbf{v}_4 = A(AAAA\mathbf{v}_0) \\
&\quad\vdots
\end{aligned}
$$

The **state vector** $\mathbf{v}_k$ is the state of the system after k time steps

Recall: Markov Chains

Def. A **Markov chain** is a linear dynamical system whose evolution function is given by a *stochastic* matrix

Recall: Regular Stochastic Matrices

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Def. A stochastic matrix A is **regular** if A^k has all positive entries for *some nonnegative* k

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Thm. A regular stochastic matrix P has a unique steady state, and

every probability vector converges to it

Recall: Steady-State Vectors

Def. A **steady-state vector** for a stochastic matrix A is a probability vector $\mathbf{q}$ such that

$$A\mathbf{q} = \mathbf{q}$$

A steady-state vector is *not changed* by the stochastic matrix. They describe *equilibriums*

Recall: Subspace

Def. A **subspace** of $\mathbb{R}^n$ is a set H of vectors in $\mathbb{R}^n$ such that

1. for every $\mathbf{u}$ and $\mathbf{v}$ in H, the vector $\mathbf{u} + \mathbf{v}$ is in H
2. for every $\mathbf{u}$ in H and scalar c, the vector $c\mathbf{u}$ is in H

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!! Subspaces must "live" somewhere !!

Recall: Subspace in $\mathbb{R}^3$

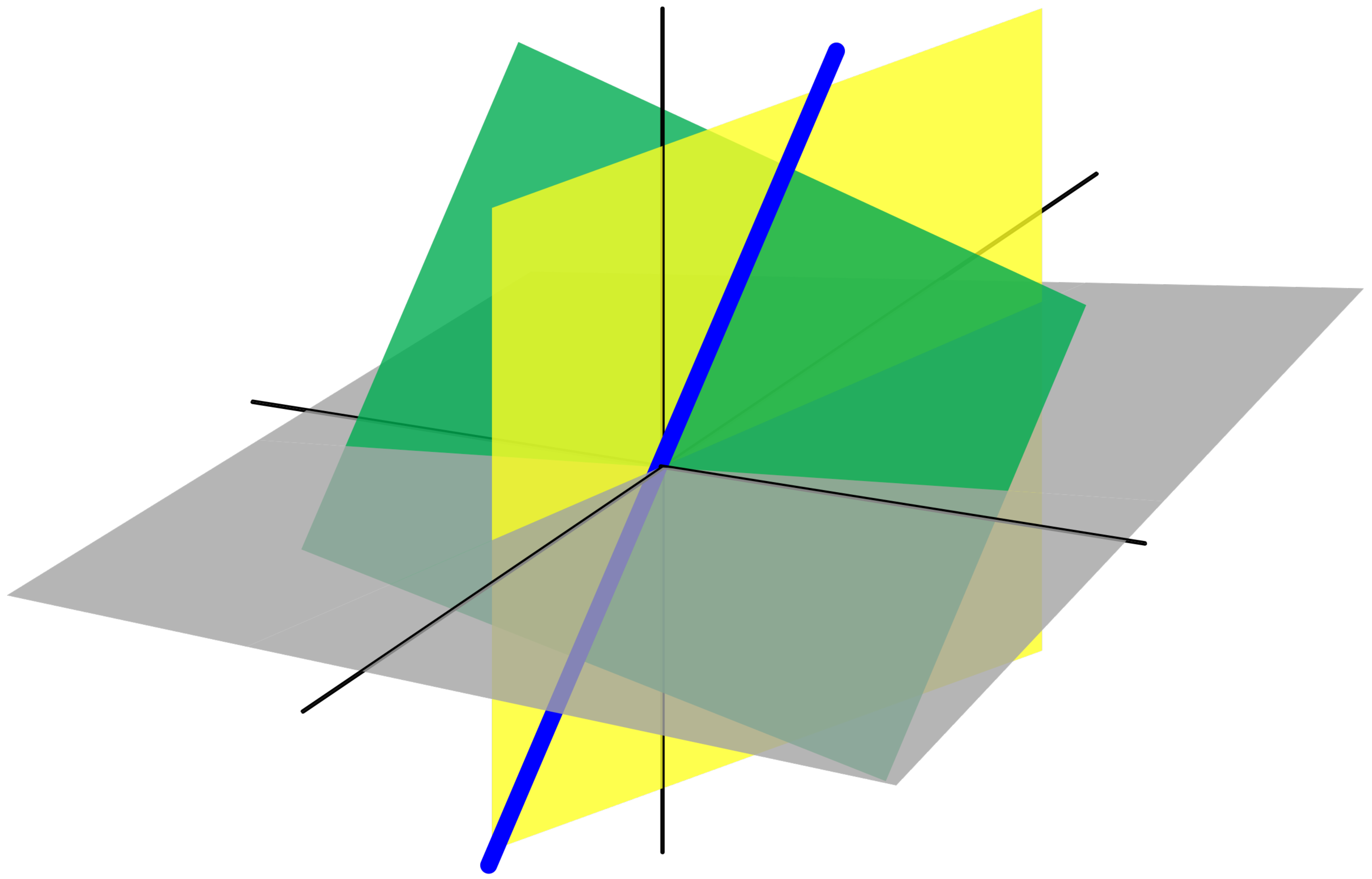
There are only 4 kinds of subspaces of $\mathbb{R}^3$:

1. $\{\mathbf{0}\}$ just the origin

2. lines (through the origin)

3. planes (through the origin)

4. All of $\mathbb{R}^3$



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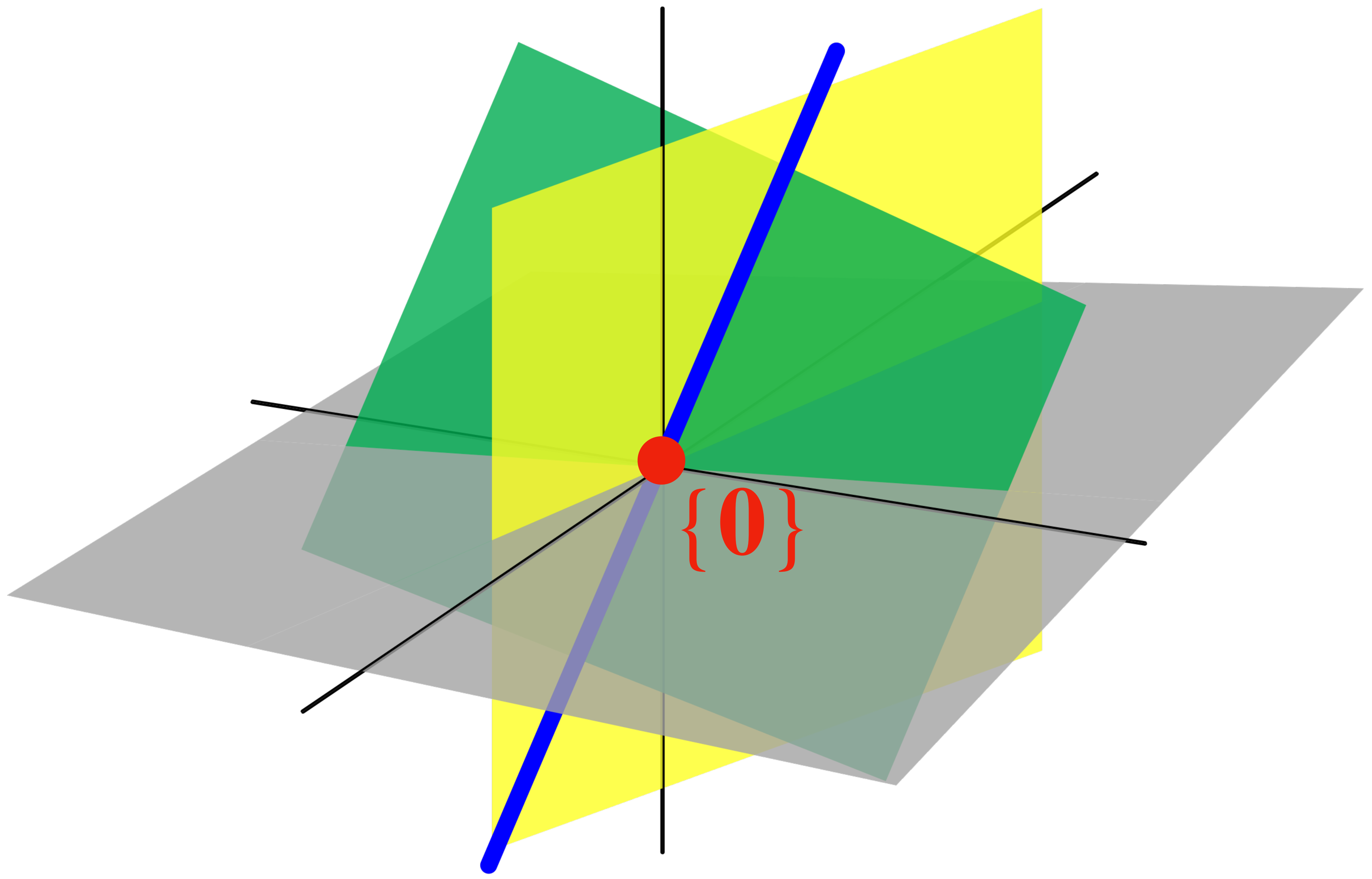

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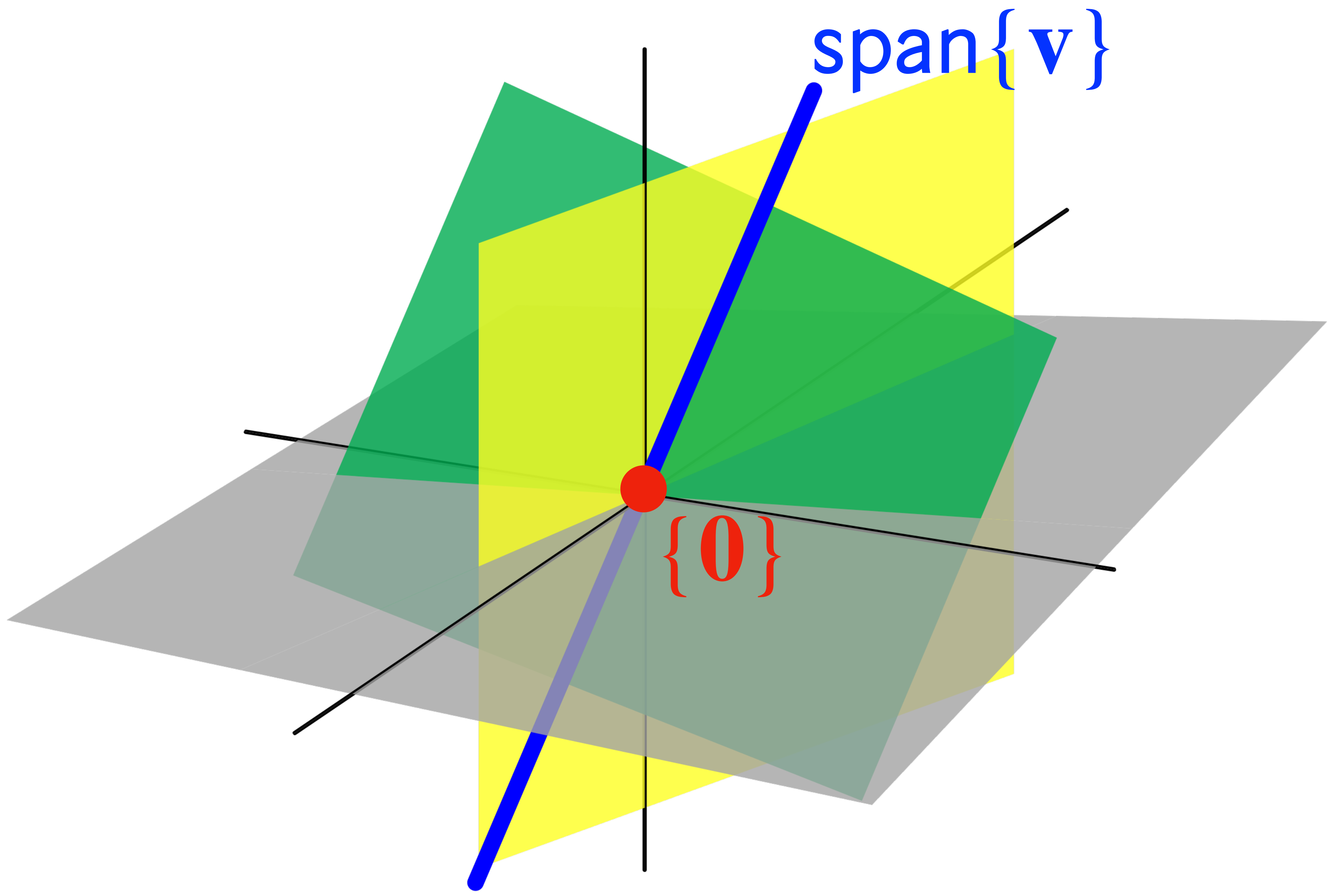

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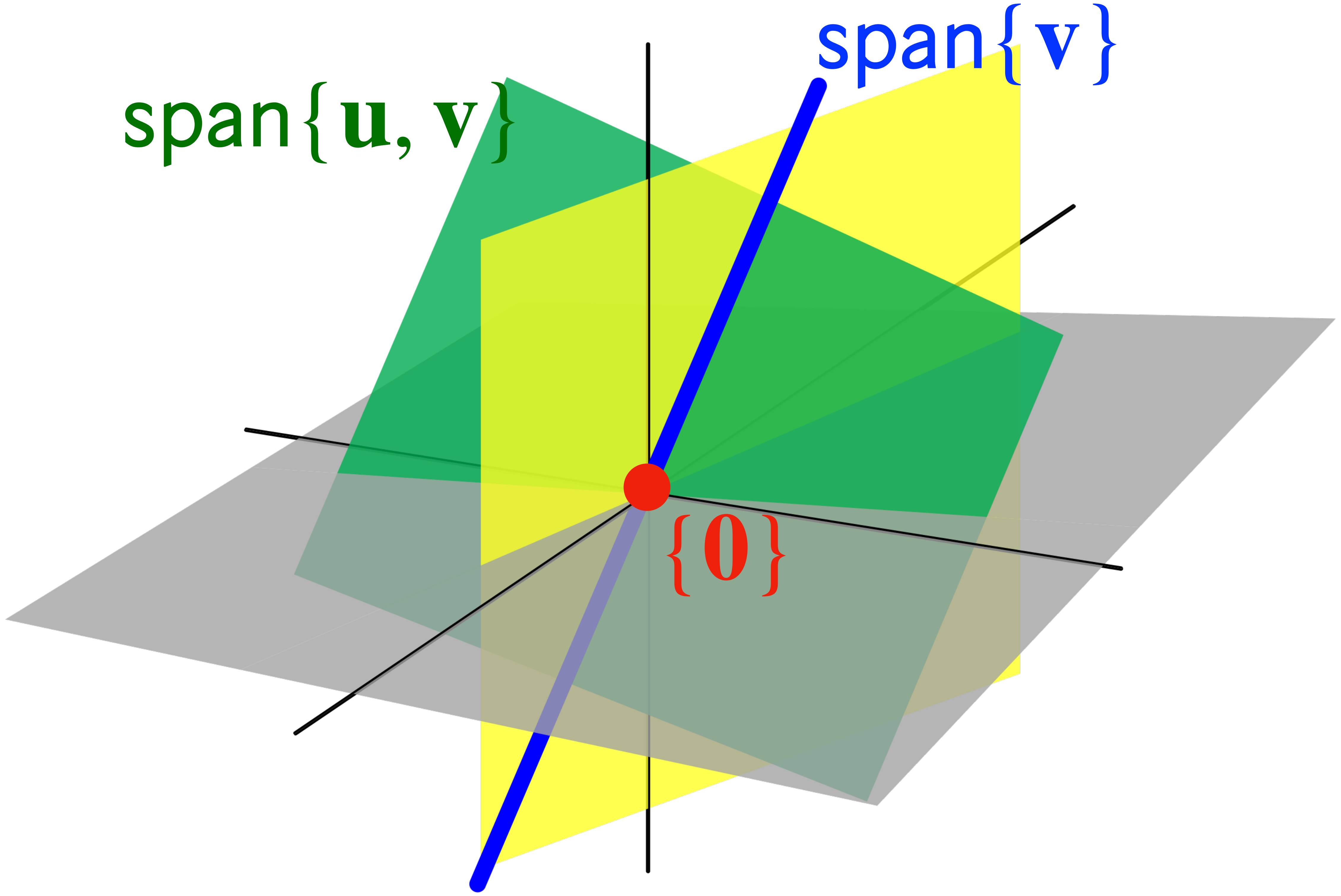

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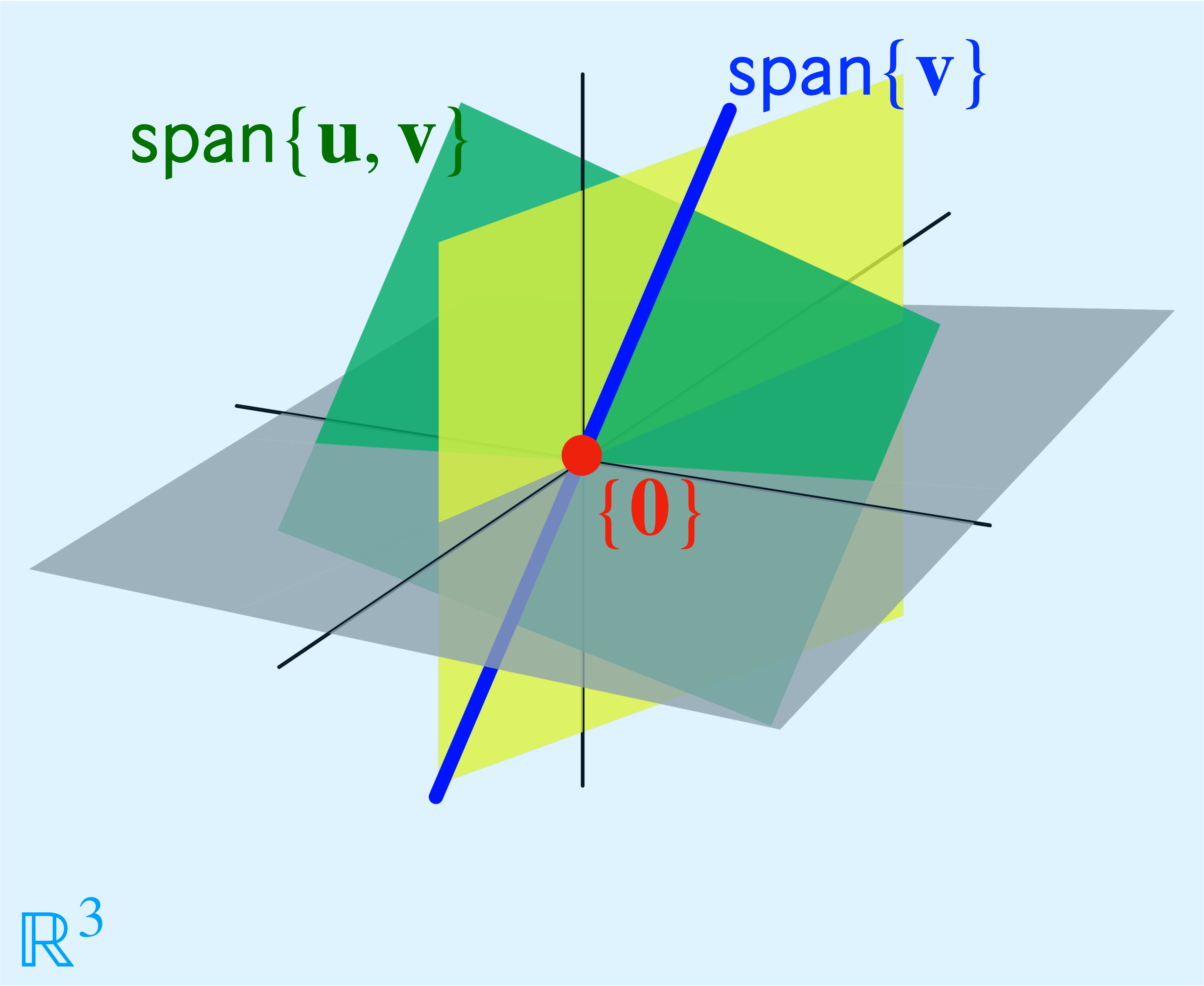

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A basis is a *minimal* set of vectors which spans all of H

Recall: Column Space

$$\begin{array}{cc} & n \\ m & \begin{bmatrix} | & | & \cdots & | & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_{n-1} & \mathbf{a}_n \\ | & | & \cdots & | & | \end{bmatrix} \end{array}$$

$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \ldots c_n\mathbf{a}_n$ **is a vector in** $\mathbb{R}^m$

$\text{Col}(A)$

is a subspace of

$\mathbb{R}^m$

Recall: Null Space

n columns

m rows

$$\underset{m \times n}{A}\underset{n \times 1}{\mathbf{v}} = \underset{m \times 1}{\mathbf{0}}$$

$\mathbf{v}$ **is a vector in** $\mathbb{R}^n$

$\text{Nul}(A)$

is a subspace of

$\mathbb{R}^n$

Recall: Dimension

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Def. The **dimension** of a subspace H of $\mathbb{R}^n$, written $\dim H$, is the *number* of vectors in *any* basis of H

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» a plane (through the origin) is a 2D subspace
» a line (through the origin) is a 1D subspace

Recall: Rank

Def. The **rank** of a matrix A, written $\operatorname{rank} A$, is the dimension of $\operatorname{Col} A$

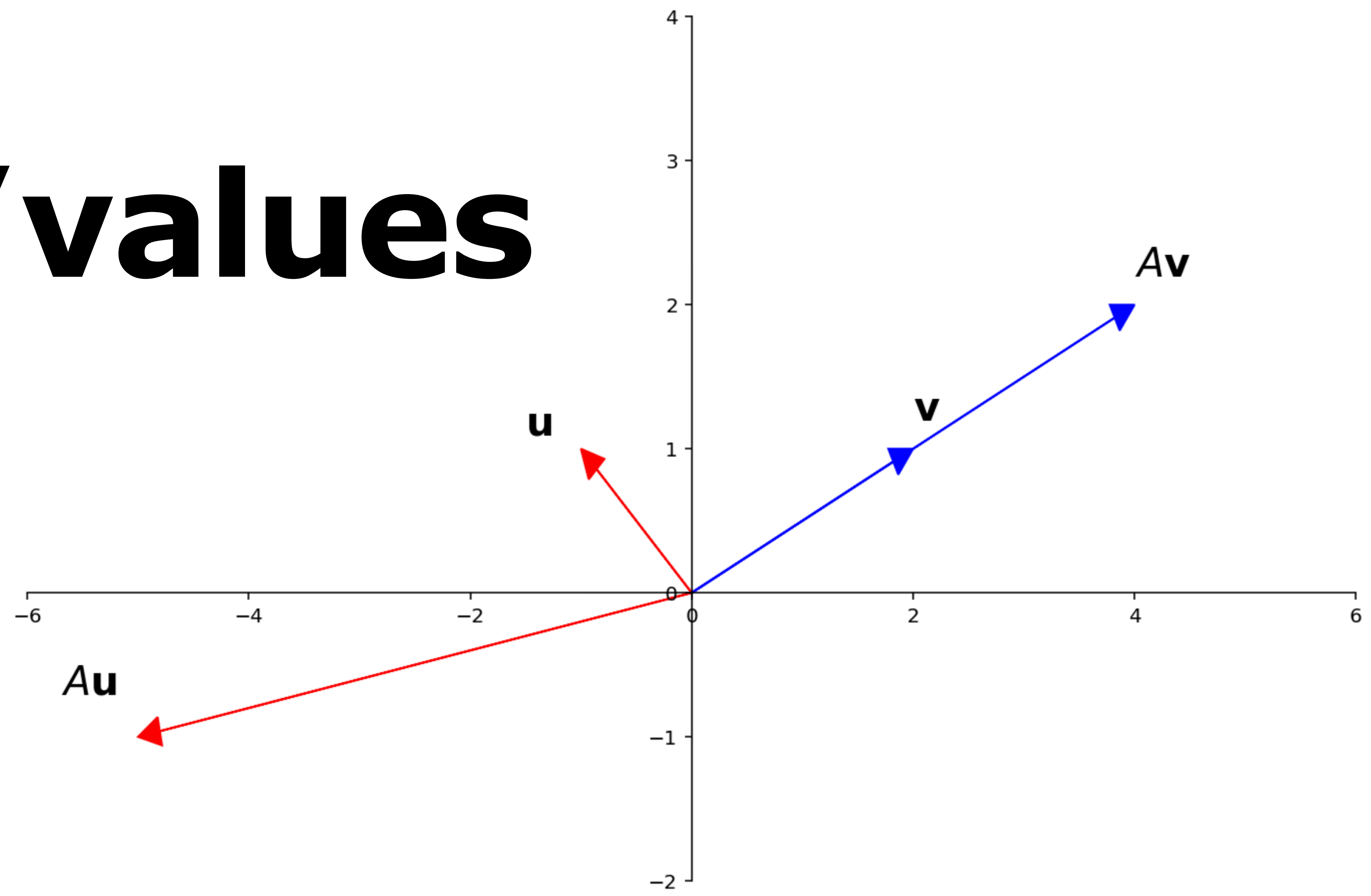
This is just terminology

Recall: Rank-Nullity Theorem

Thm. For $A \in \mathbb{R}^{m \times n}$, $\text{rank}(A) + \underbrace{\dim(\text{Nul}(A))}_{\text{nullity}} = n$

This is incredibly important

Recall: Eigenvectors/values



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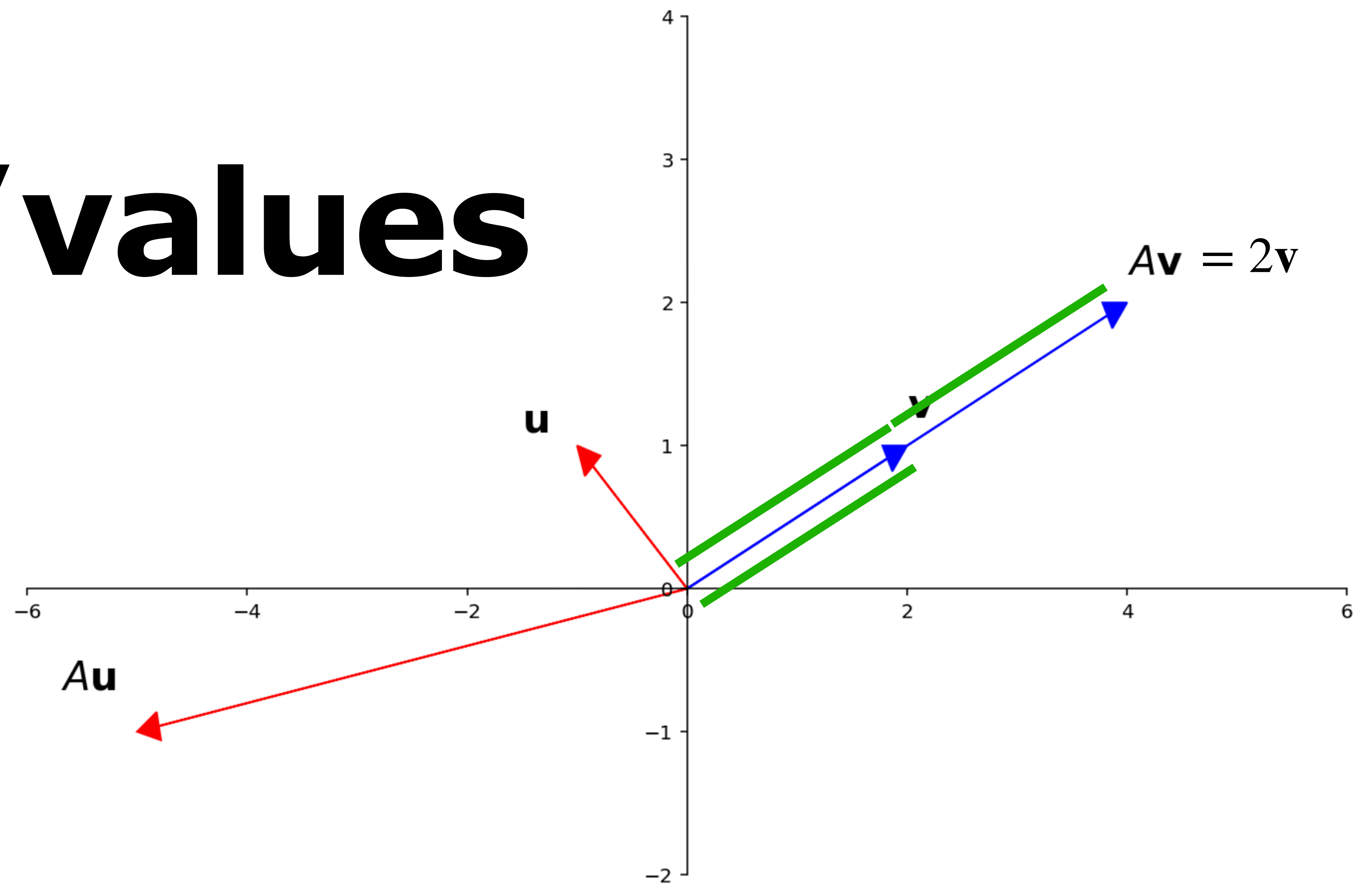

Def. A *nonzero* vector $\mathbf{v} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ are an **eigenvector and eigenvalue** for a $A \in \mathbb{R}^{n \times n}$ if

$$A\mathbf{v} = \lambda\mathbf{v}$$

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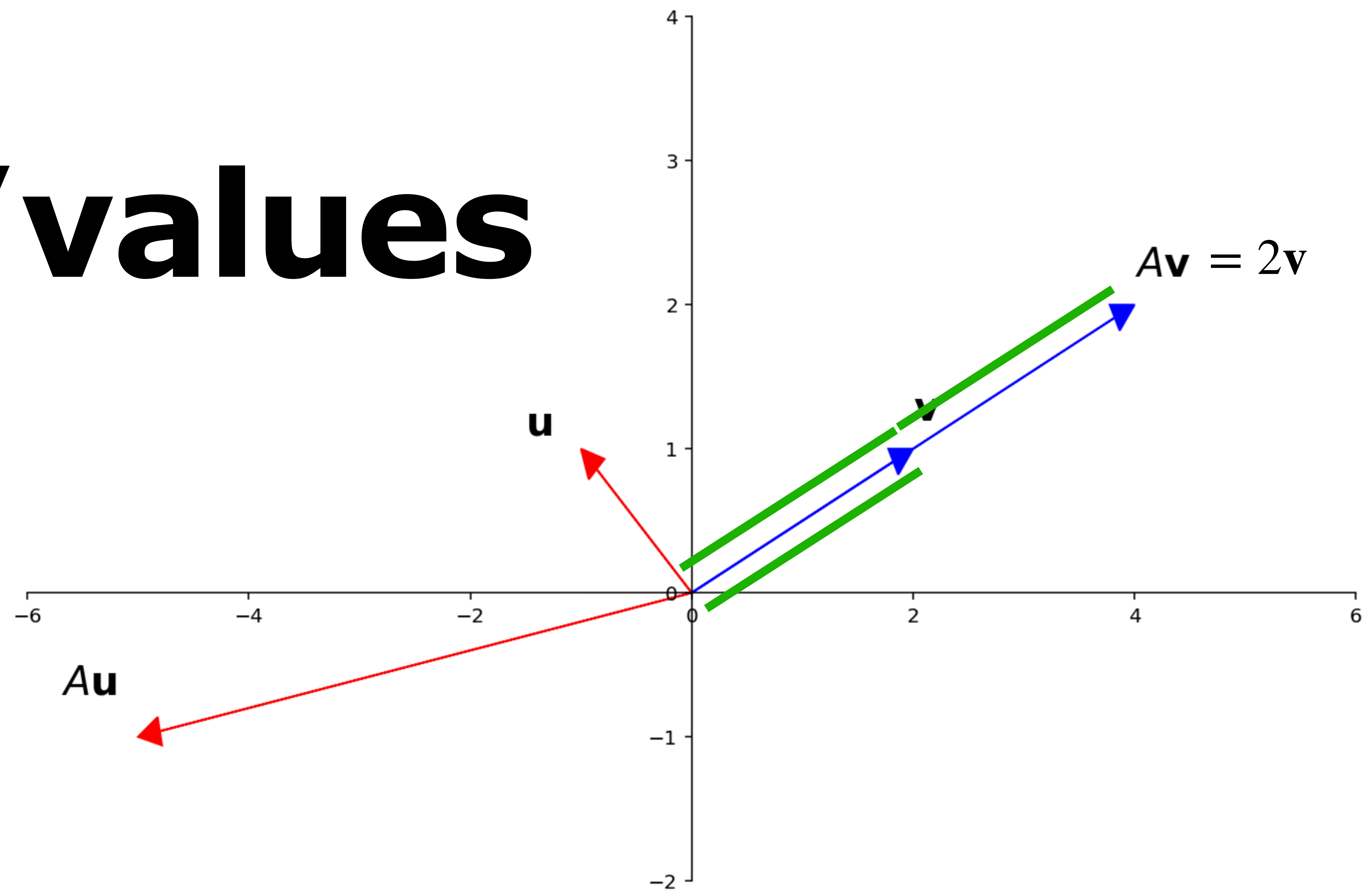

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Eigenvectors are just scaled by A

Recall: Verifying Eigenvectors

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Q. Determine if $\mathbf{v}$ is an eigenvector of A and determine the corresponding eigenvalues

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A. Easy. Work out the matrix–vector multiplication

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Ex.

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ -5 \end{bmatrix} = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = -4 \begin{bmatrix} 6 \\ -5 \end{bmatrix} \quad ✓$$

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 13 \\ 9 \end{bmatrix} \quad ✗$$

Recall: Verifying Eigenvalues

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Q. Find an eigenvector of A whose eigenvalue is λ

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*If we don't need the vector we can just show that $A - \lambda I$ is **not** invertible (by IMT)*

Recall: Finding Eigenspaces

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A. Find a basis for $\mathrm{Nul}(A - \lambda I)$

Recall: Determinants

$$\det(A) = \frac{(-1)^s}{c} U_{11} U_{22} \ldots U_{nn}$$

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$$\det(A) = \frac{(-1)^s}{c} \underbrace{\overbrace{U_{11}U_{22}\dots U_{nn}}^{\text{product of diagonal entries}}}_{0 \text{ if } A \text{ is not invertible}}$$

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This is a polynomial with the eigenvalues of A as roots

So we can "solve" for the eigenvalues in the equation

$$\det(A - \lambda I) = 0$$

Recall: Eigenbases

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*We can represent vectors as **unique** linear combinations of eigenvectors*

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Not all matrices have eigenbases

Recall: The Diagonalization Theorem

$$A = PDP^{-1}$$

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Thm. A is diagonalizable $\Leftrightarrow$ A has an eigenbasis

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The idea:

Recall: The Diagonalization Theorem

$$A = \overset{\text{eigenbasis}}{P} D P^{-1}$$

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The idea:

» The columns of P form an **eigenbasis** for A

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The idea:

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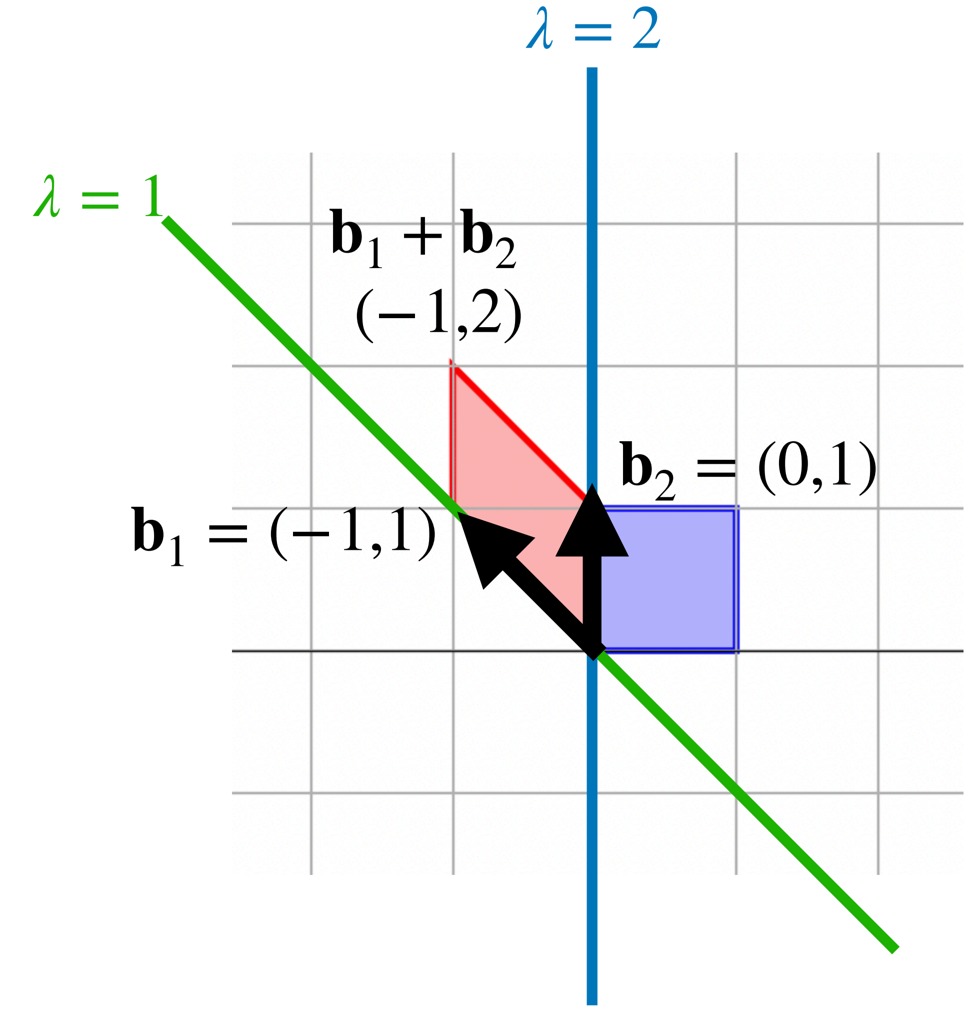
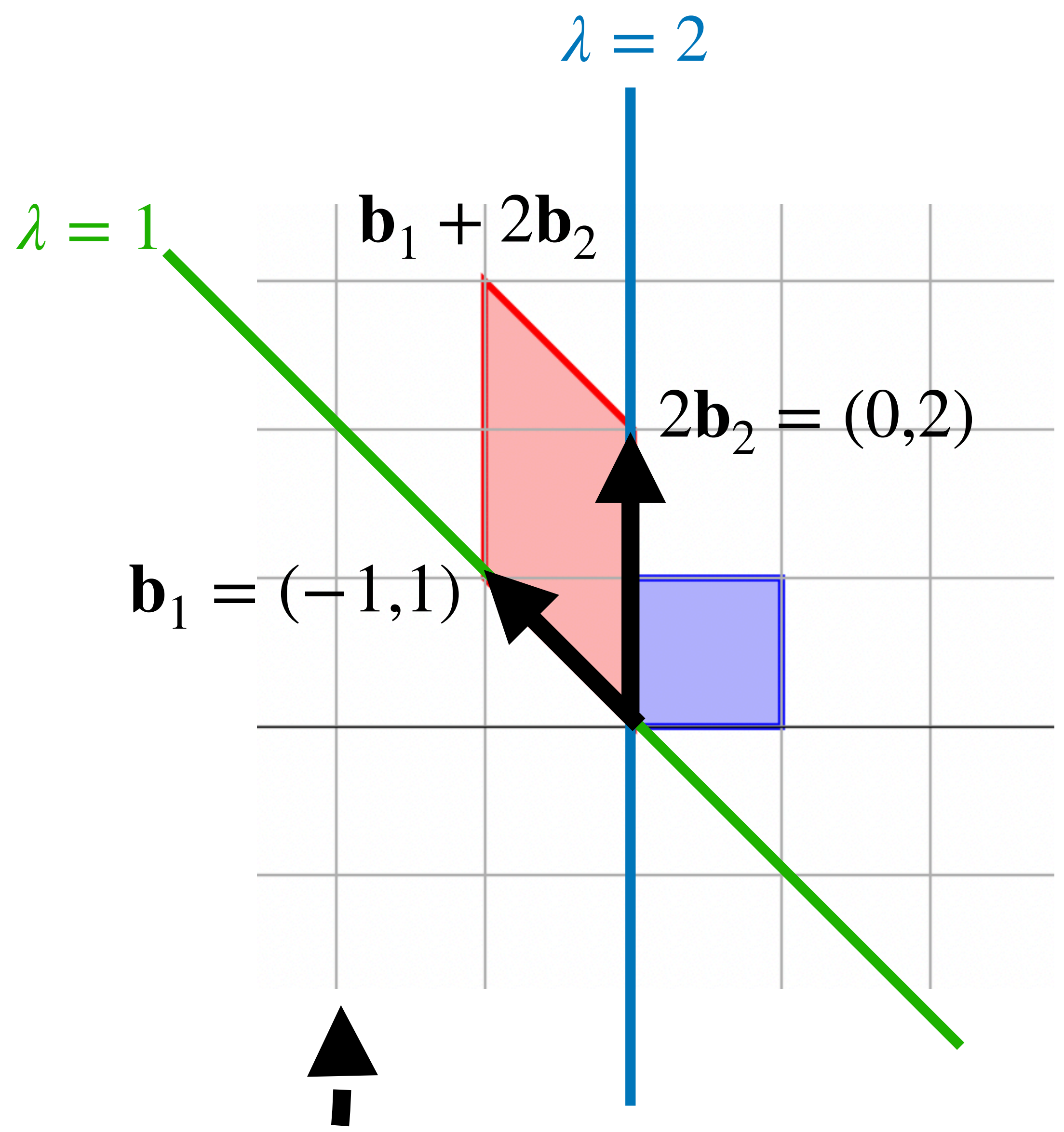
» **The matrix P^{-1} is a change of basis to this eigenbasis of A**

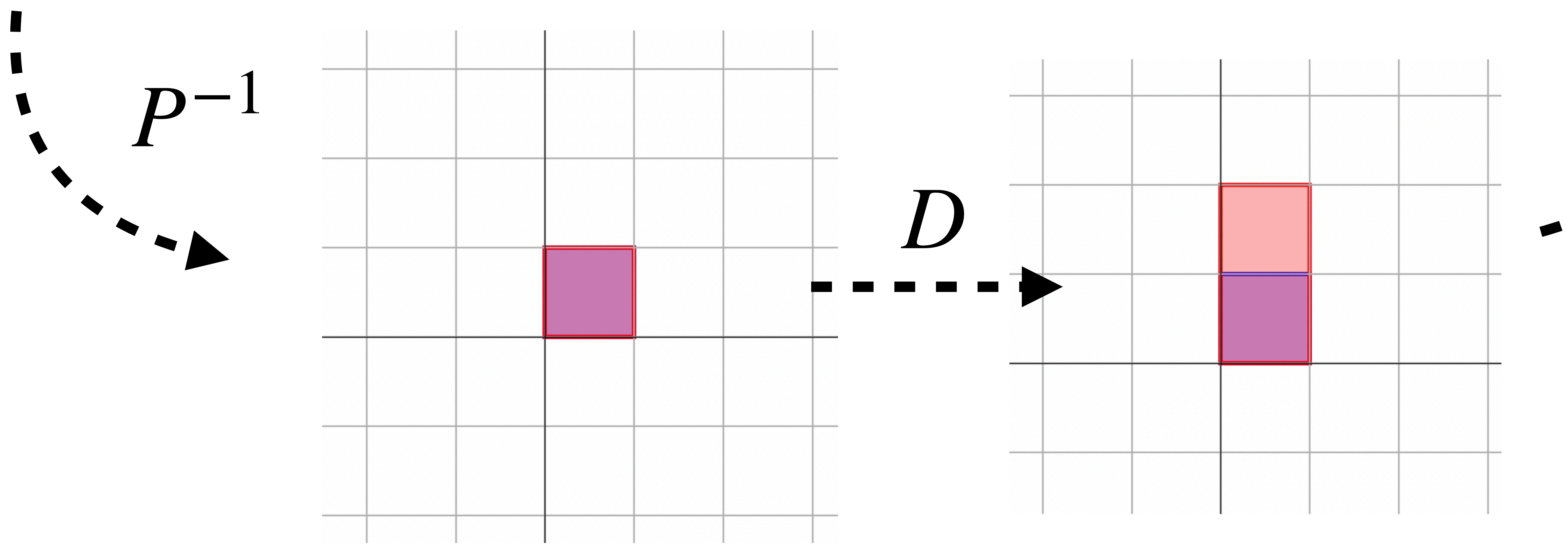
Recall: The Picture



$$A = PDP^{-1}$$

$$\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}^{-1}$$





Recall: Inner Products

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$$\begin{bmatrix} u_1 & u_2 & u_3 & u_4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$$

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Def. The **inner product** of $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$$

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$$\langle \mathbf{u}, \mathbf{v} \rangle = \underbrace{\mathbf{u} \cdot \mathbf{v}}_{\text{dot product}} = \mathbf{u}^T\mathbf{v}$$

Recall: Algebraic Properties

- $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (symmetry)
- $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = (\mathbf{u} \cdot \mathbf{w}) + (\mathbf{v} \cdot \mathbf{w})$
- $(\alpha \mathbf{u}) \cdot v = \alpha(\mathbf{u} \cdot \mathbf{v})$

(linearity)

- $\mathbf{u} \cdot \mathbf{u} \geq 0$ (nonnegativity)
- $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = 0$

Recall: Norms and Inner Products

Def. The ℓ^2 **norm** of a vector $\mathbf{v}$ in $\mathbb{R}^n$ is

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$$

The norm of a vector is the square root of the inner product with itself

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The norm of a vector is the square root of the inner product with itself

It's important that $\mathbf{v}^T\mathbf{v}$ is nonnegative

Recall: Cosines and Unit Vectors

Thm. For $\mathbf{u}$, $\mathbf{v} \in \mathbb{R}^n$ with an angle θ between them

$$\cos\theta = \left\langle \frac{\mathbf{u}}{\|\mathbf{u}\|}, \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\rangle$$

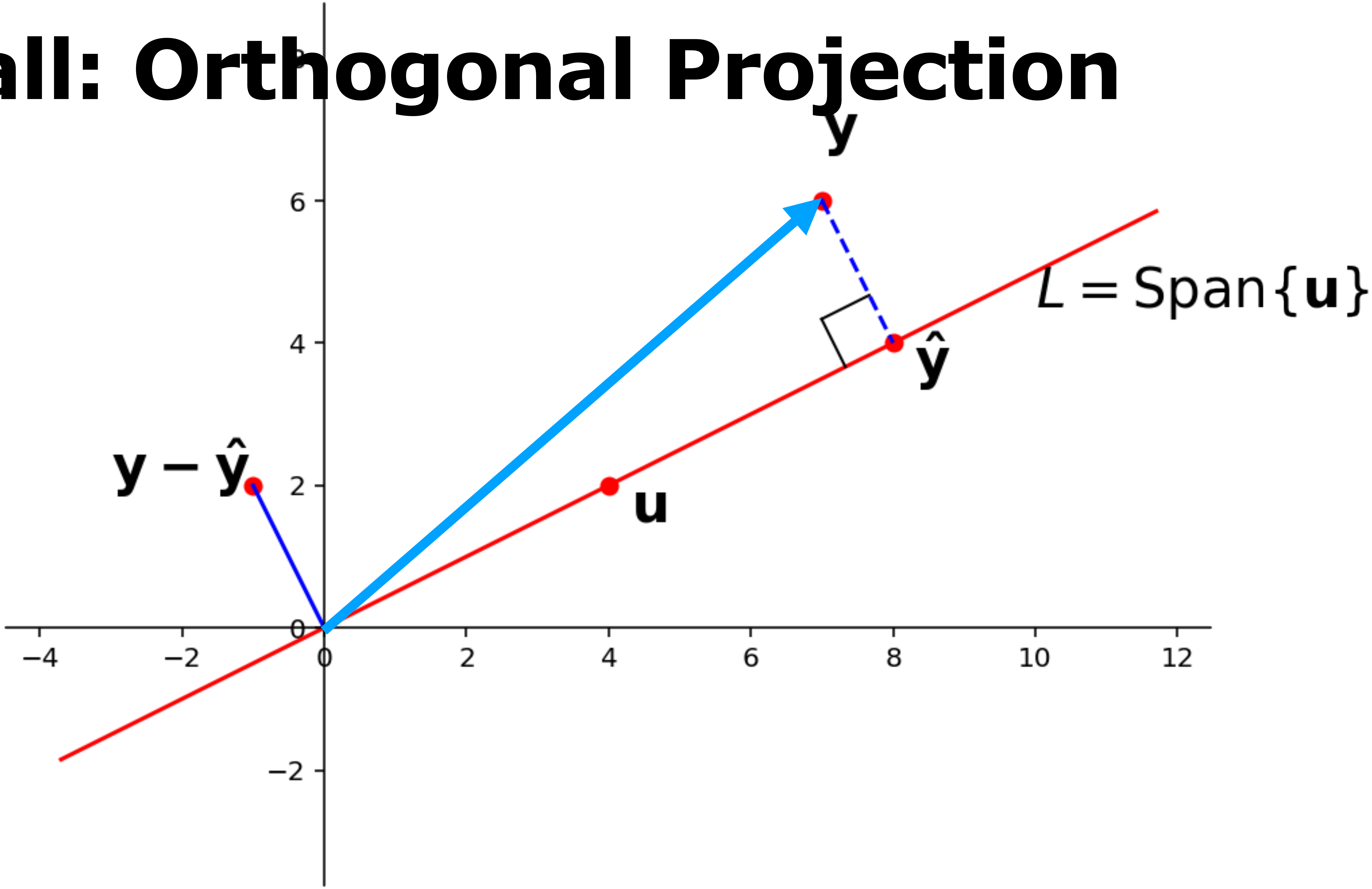
The cosine of the angle between two vectors is the inner product of their ℓ^2 normalizations

Recall: Orthogonality

Def. Vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$

This definition gives an easy computational way to determine orthogonality

Recall: Orthogonal Projection



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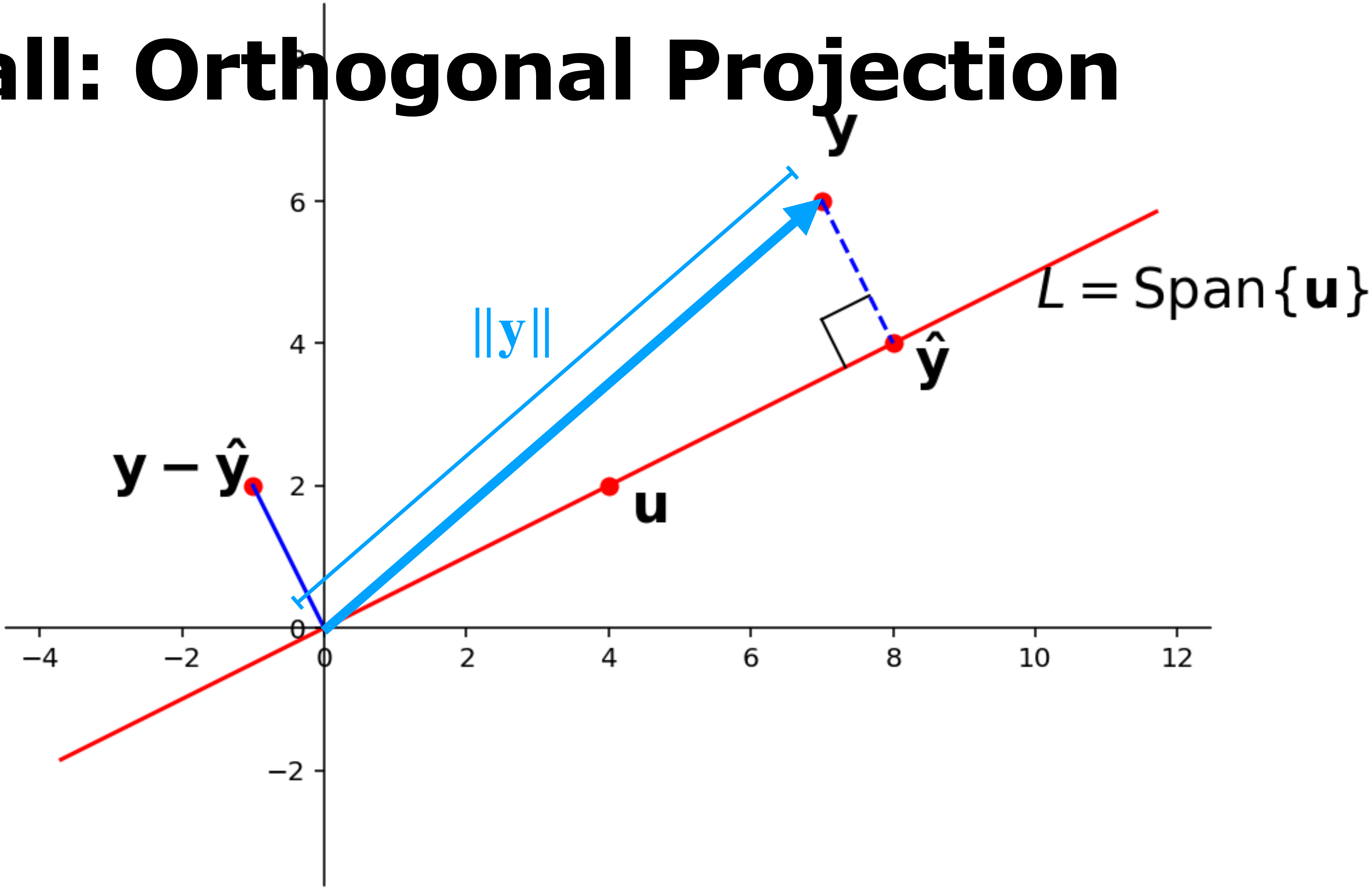

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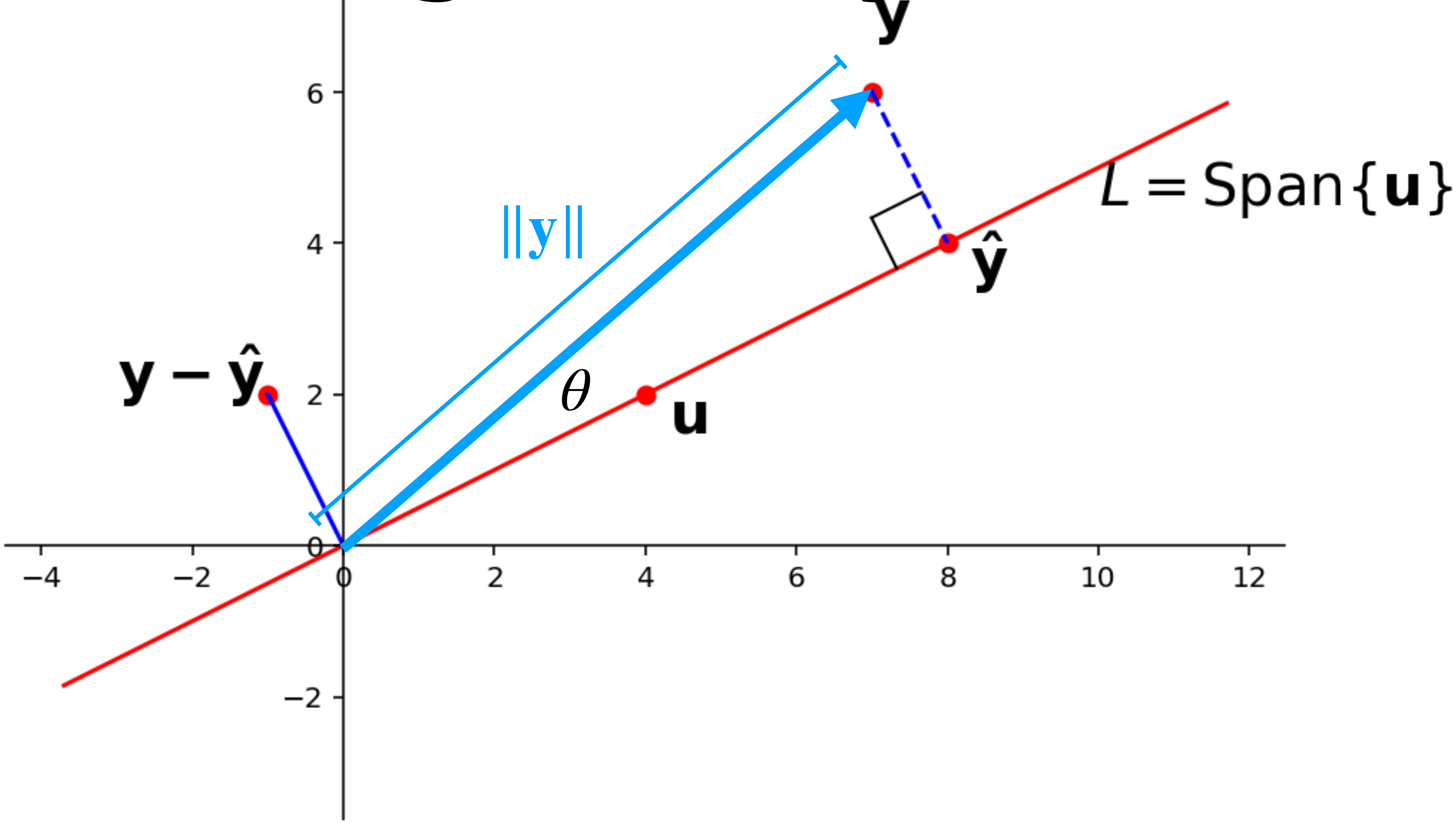
$$\cos\theta = \frac{\|\hat{\mathbf{y}}\|}{\|\mathbf{y}\|}$$



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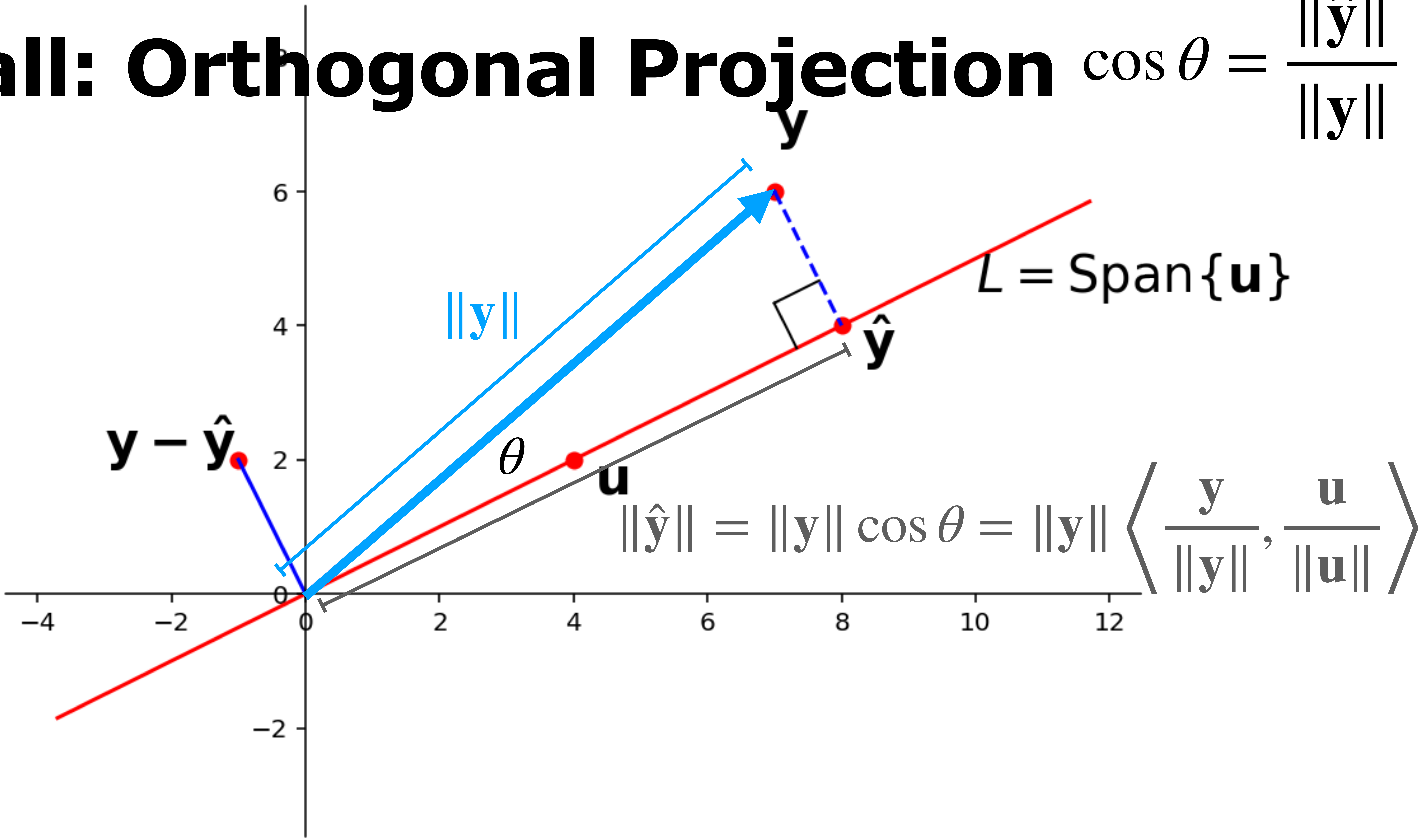

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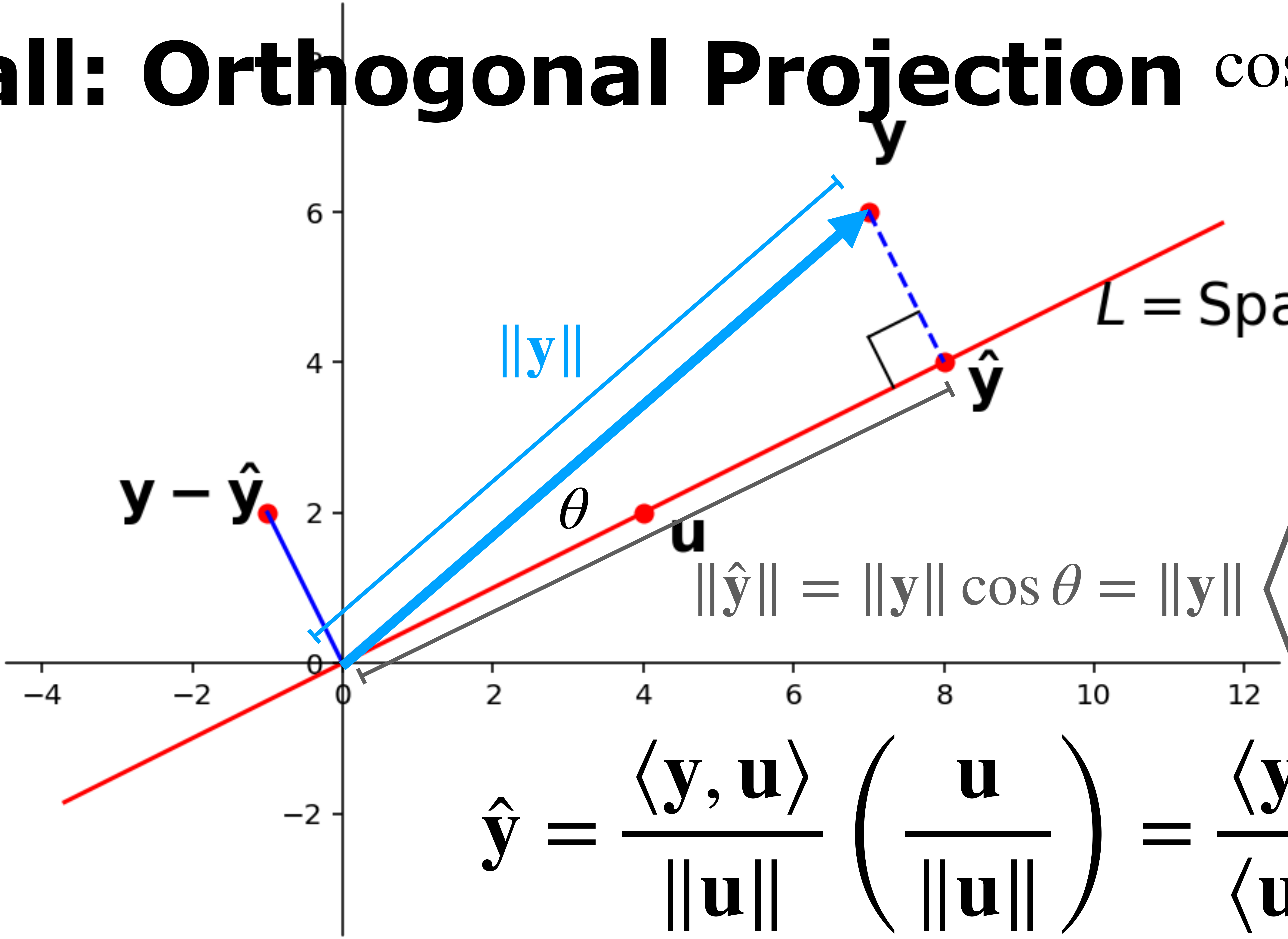


$$\|\hat{\mathbf{y}}\| = \|\mathbf{y}\|\cos\theta = \|\mathbf{y}\|\left\langle \frac{\mathbf{y}}{\|\mathbf{y}\|}, \frac{\mathbf{u}}{\|\mathbf{u}\|} \right\rangle$$

$$\hat{\mathbf{y}} = \frac{\langle \mathbf{y}, \mathbf{u} \rangle}{\|\mathbf{u}\|}\left(\frac{\mathbf{u}}{\|\mathbf{u}\|}\right) = \frac{\langle \mathbf{y}, \mathbf{u} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle}\mathbf{u}$$

Recall: Orthogonal Sets

Def. A set $\{u_1, u_2, \ldots, u_p\}$ of vectors from $\mathbb{R}^n$ is an **orthogonal set** if every pair of distinct vectors is orthogonal, i.e., if $i \neq j$ then

$$\langle u_i, u_j \rangle = 0$$

Each vector is pairwise/mutually perpendicular

Recall: Least Squares

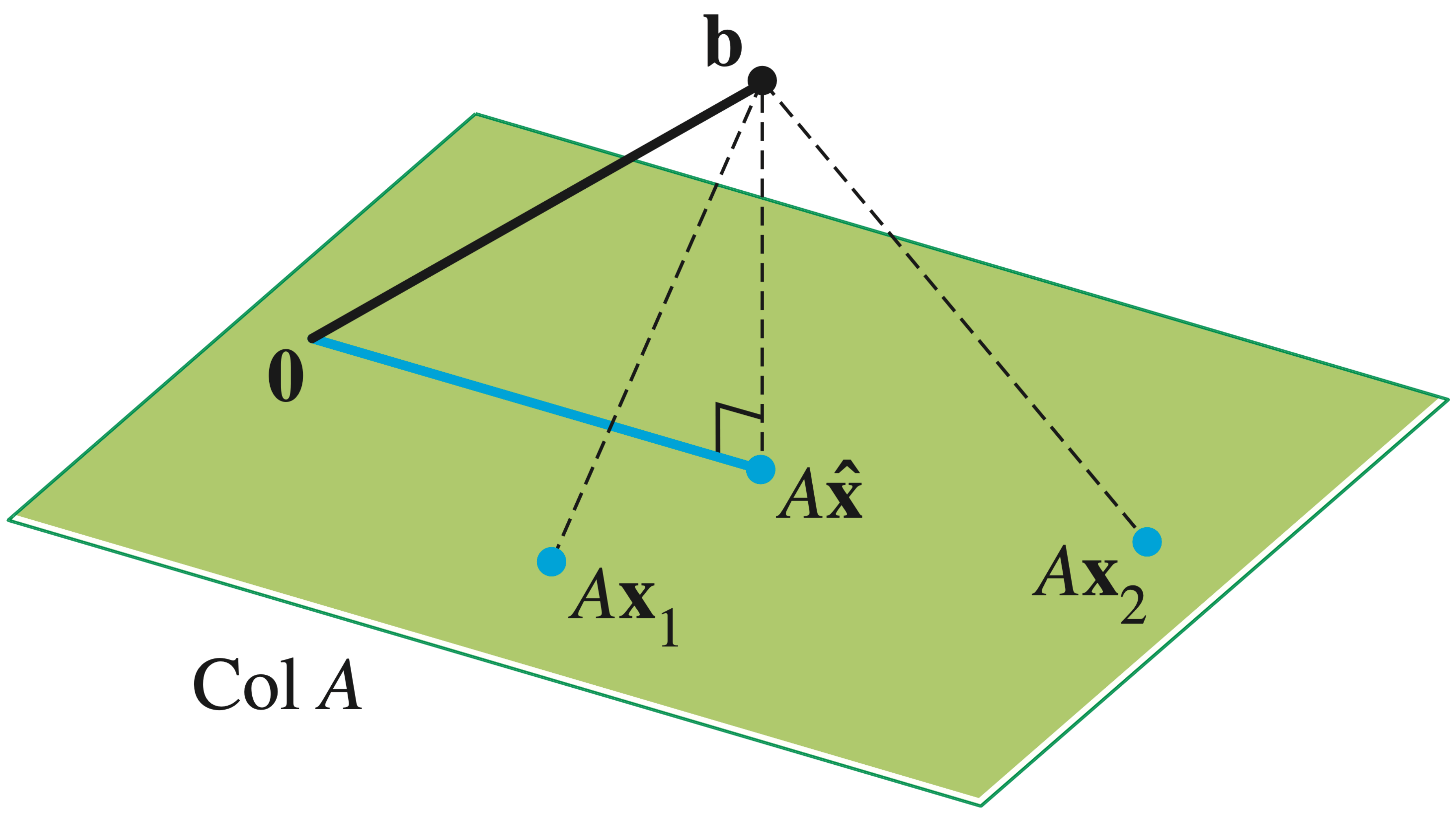
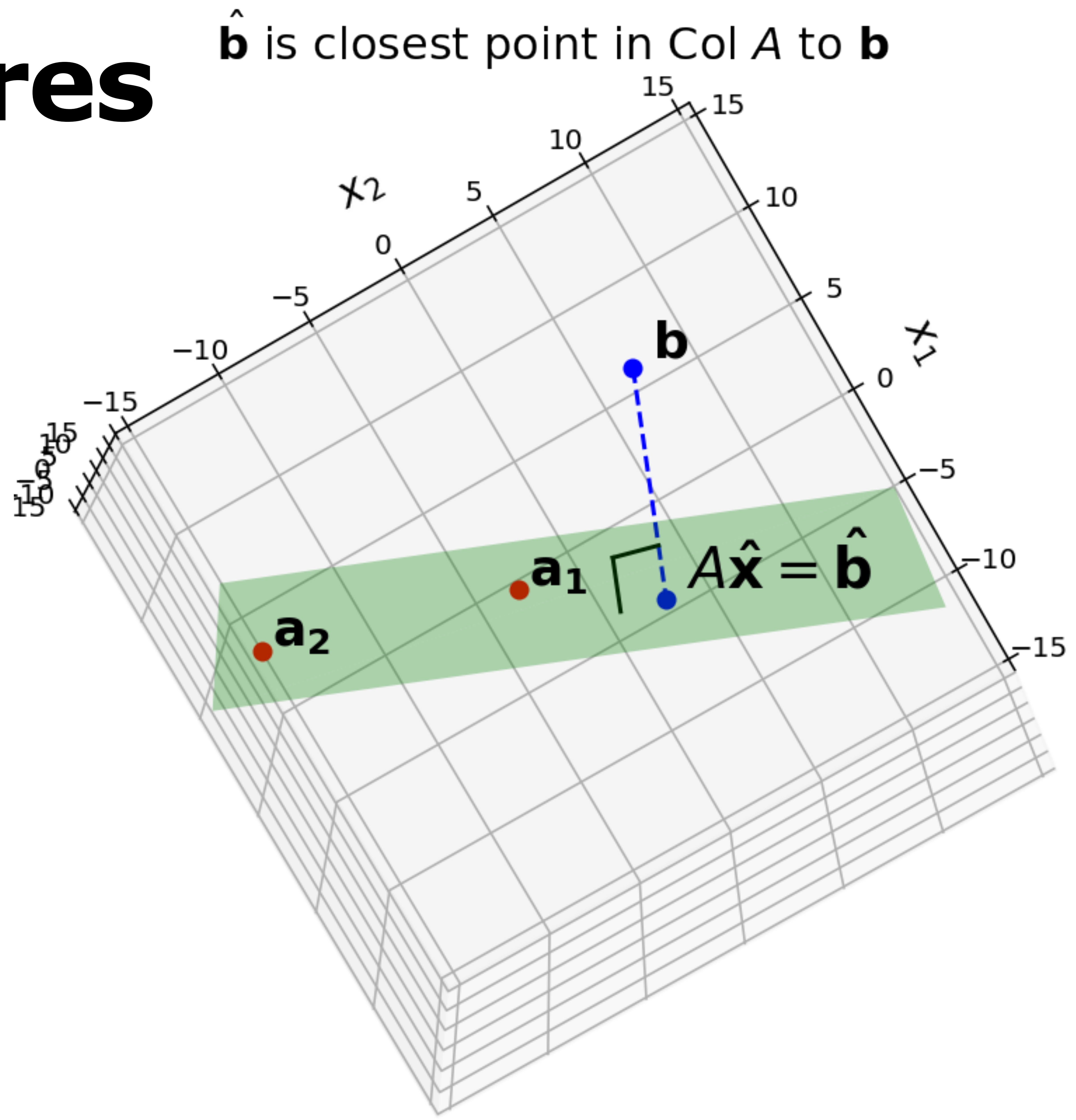


Figure 22.8



Recall: The Normal Equations

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Thm. The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ is the same as the set of solutions to

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In particular, this set of solutions is nonempty

Recall: The Normal Equations

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$$A^T A\mathbf{x} = A^T\mathbf{b}$$

In particular, this set of solutions is nonempty

(We just showed that if $\hat{\mathbf{x}}$ is a least squares solution then $A^T A\hat{\mathbf{x}} = A^T\mathbf{b}$)

Recall: Unique Least Squares Solutions

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

If A has linearly independent columns, then its unique least squares solution is the above

Recall: Projecting onto a subspace

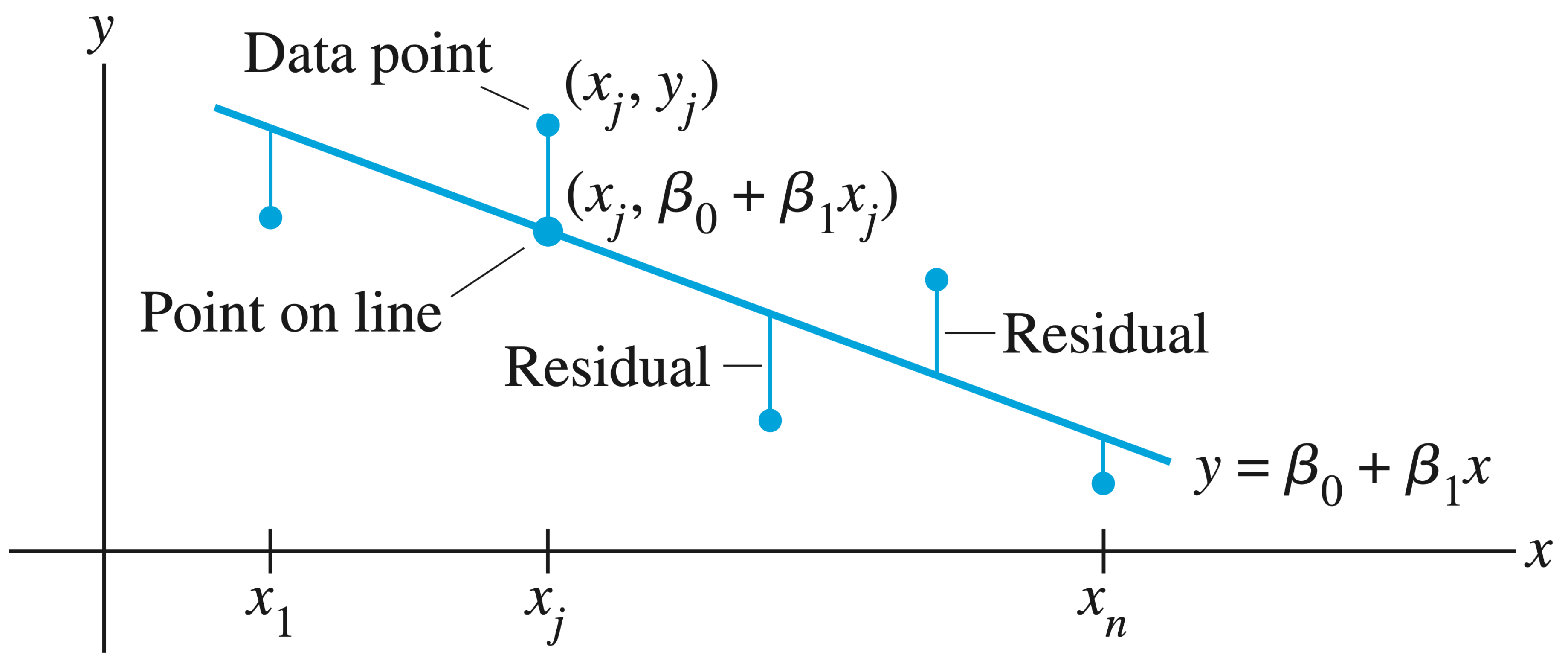
$$\hat{\mathbf{b}} = A\hat{\mathbf{x}} = A(A^TA)^{-1}A^T\mathbf{b}$$

If the columns of A are linearly independent, then **they form a basis**

Said another way: if $\mathscr{B}$ is a basis, then we can construct a matrix A whose columns are the vectors in $\mathscr{B}$

This means we can find arbitrary projections

Recall: Linear Models



data point

$$\{(x_1, y_1), \ldots, (x_i, y_i), \ldots, (x_n, y_n)\}$$

dataset

independent variable

$$(x_i, y_i)$$

dependent variable (label)

model parameters/ regression coefficients

$$f(x) = \beta_0 + \beta_1 x$$

model

observation

residual

$$\sum_{i=1}^{n} (y_i - f(x_i))^2$$

prediction

Recall: General Linear Regression

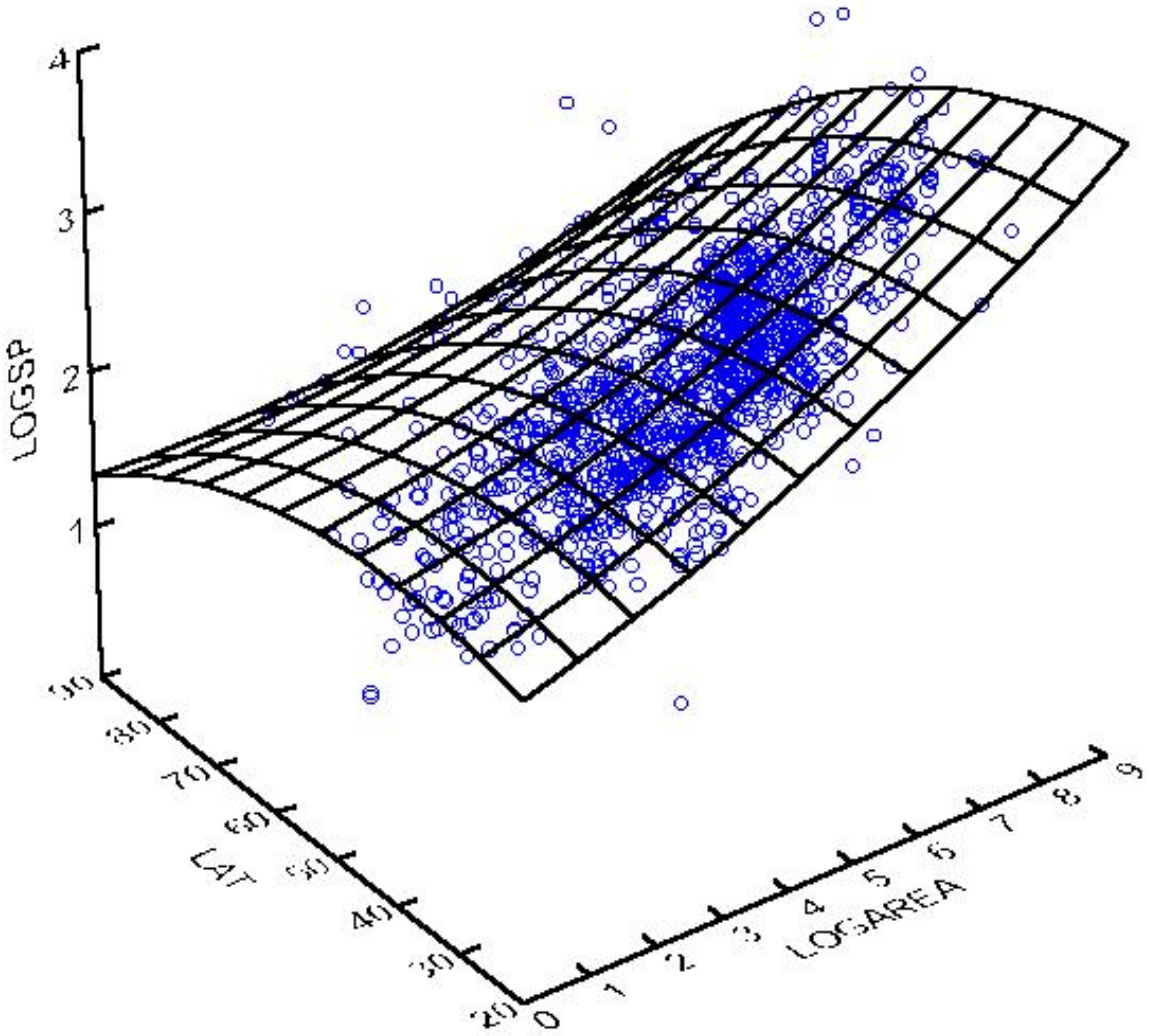
Dataset: $\{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_m, y_m)\}$ where $\mathbf{x}_i \in \mathbb{R}^n$ and $y_i \in \mathbb{R}$

Given a function

$$f_{\beta_1,\ldots,\beta_k} : \mathbb{R}^n \to \mathbb{R}$$

which is *linear in the parameters* $\beta_1, \ldots \beta_k$, find values for $\beta_1, \ldots, \beta_k$ which minimize

$$\sum_{i=1}^{k} (f_{\beta_1,\ldots,\beta_k}(\mathbf{x}_i) - y_i)^2$$



https://ordination.okstate.edu/MULTIPLE.htm

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$$\beta_1\phi_1(\mathbf{x}_1) + \ldots + \beta_k\phi_k(\mathbf{x}_1) = y_1$$

$$\beta_1\phi_1(\mathbf{x}_2) + \ldots + \beta_k\phi_k(\mathbf{x}_2) = y_2$$

$$\vdots$$

$$\beta_1\phi_1(\mathbf{x}_2) + \ldots + \beta_k\phi_k(\mathbf{x}_2) = y_2$$

Step 1: Set up an (almost assuredly inconsistent) system of linear equations in terms of the variables $\beta_1, \ldots, \beta_k$

Recall: General Linear Regression

This is still linear in the β's

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design matrix

$$\begin{array}{cccc} X & \vec{\beta} & & \mathbf{y} \\ \begin{bmatrix} \phi_1(\mathbf{x}_1) & \phi_2(\mathbf{x}_1) & \ldots & \phi_k(\mathbf{x}_1) \\ \phi_1(\mathbf{x}_2) & \phi_2(\mathbf{x}_2) & \ldots & \phi_k(\mathbf{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_1(\mathbf{x}_m) & \phi_2(\mathbf{x}_m) & \ldots & \phi_k(\mathbf{x}_m) \end{bmatrix} & \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix} & = & \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_k \end{bmatrix} \end{array}$$

Step 2: Rewrite the system as a matrix equation

Recall: General Linear Regression

Dataset: $\{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_m, y_m)\}$ where $\mathbf{x}_i \in \mathbb{R}^n$ and $y_i \in \mathbb{R}$

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$$\sum_{i=1}^{k} (f_{\beta_1,\ldots,\beta_k}(\mathbf{x}_i) - y_i)^2$$

$$\hat{\vec{\beta}} = (X^T X)^{-1} X^T \mathbf{y}$$

Step 3: Find the least squares solution of this system and use as the parameters of your model

Recall: Orthonormality

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Def. A set $\{u_1, u_2, \ldots, u_p\}$ is an **orthonormal set** if of it an orthogonal set of *unit* vectors

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ortho·normal

orthogonal/perpendicular

normalized/made unit vectors

Recall: Inverses of Orthogonal Matrices

Thm. If a matrix $U \in \mathbb{R}^{n \times n}$ is orthogonal then it is invertible and

$$U^{-1} = U^T$$

Recall: Orthogonal Diagonalizability

Def. A matrix A is **orthogonally diagonalizable** if there is a diagonal matrix D and matrix P such that

$$A = PDP^{-1} = PDP^{T}$$

P must be an *orthonormal matrix*

Symmetric matrices are orthogonally diagonalizable

Recall: Quadratic Forms

Def. Quadratic forms can be represented as

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

where A is *symmetric*

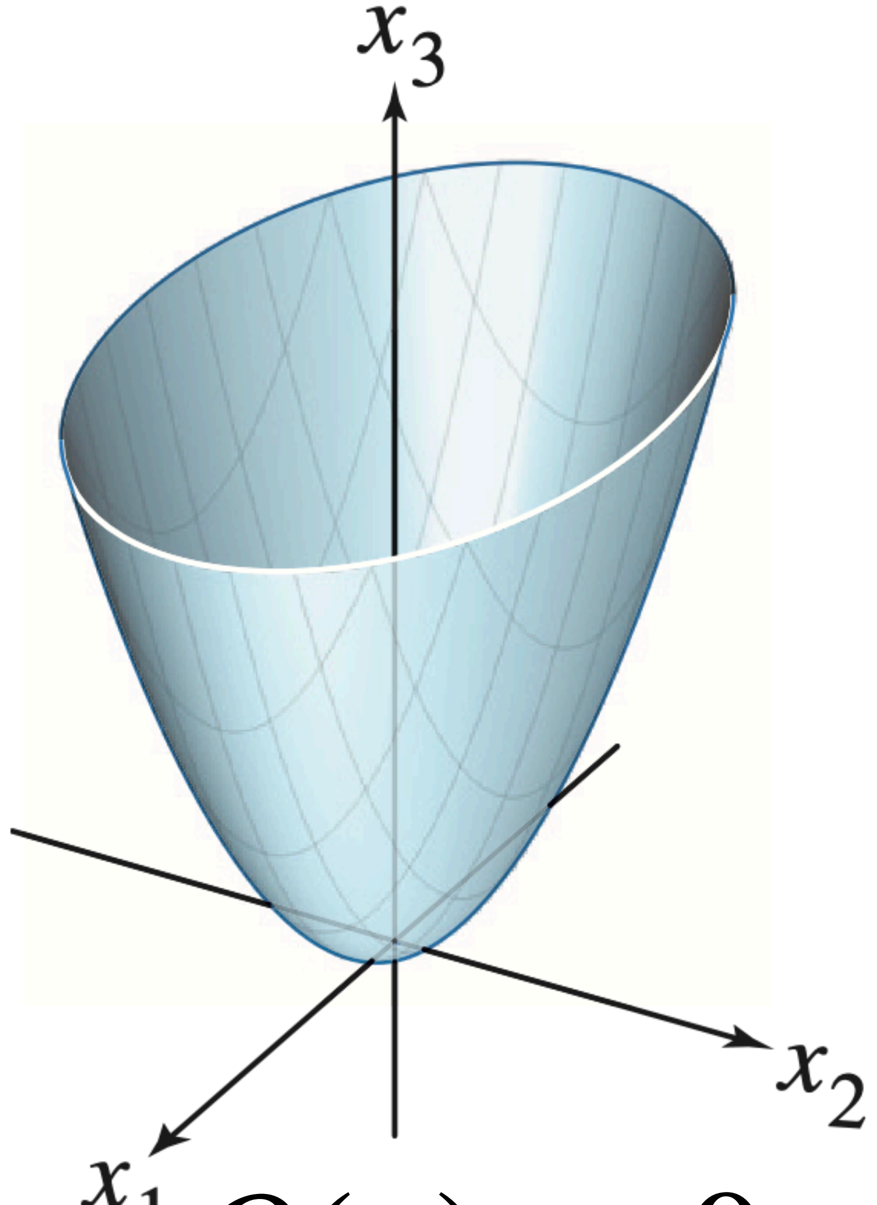
Definiteness

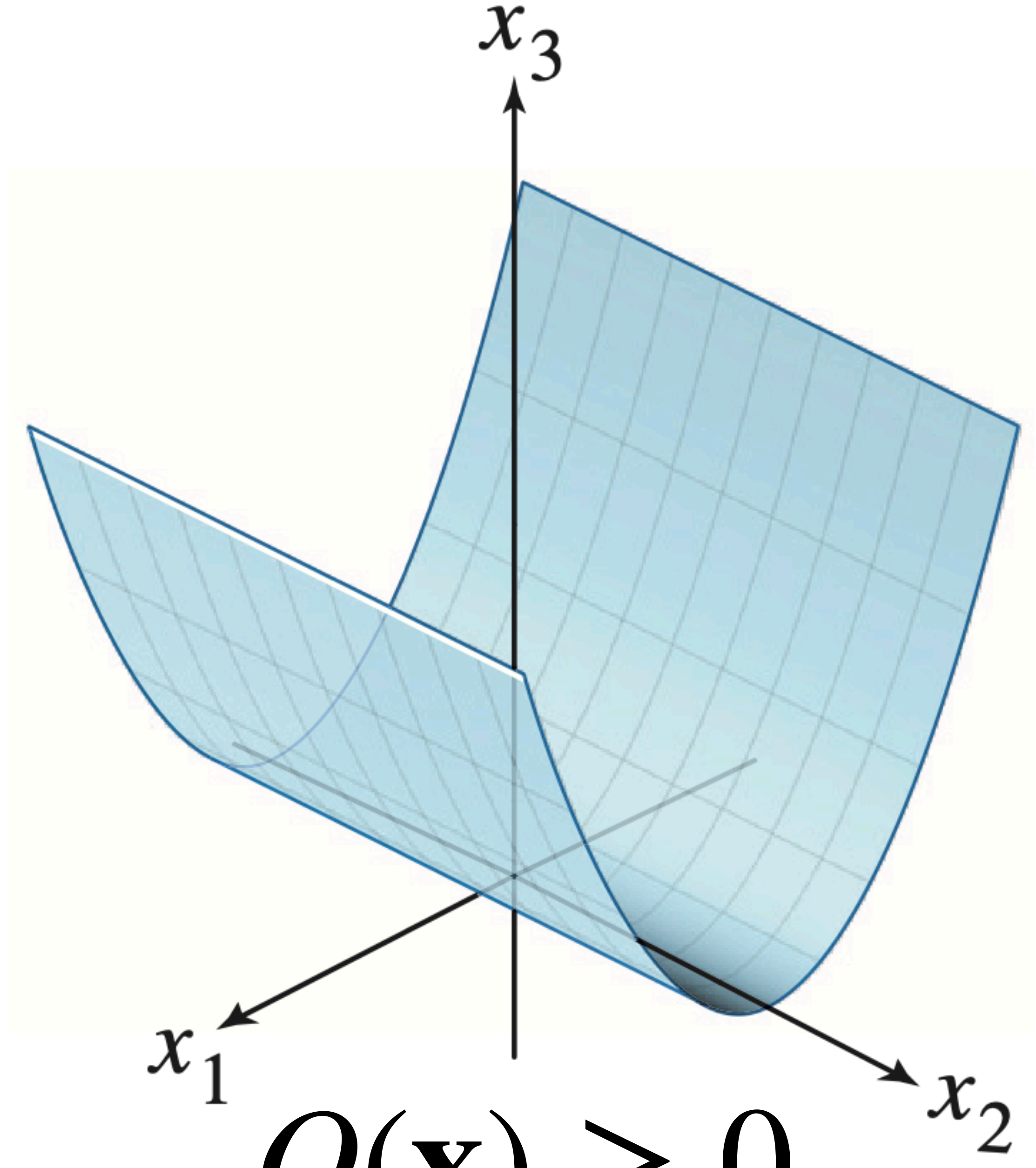
all nonneg. eigenvals

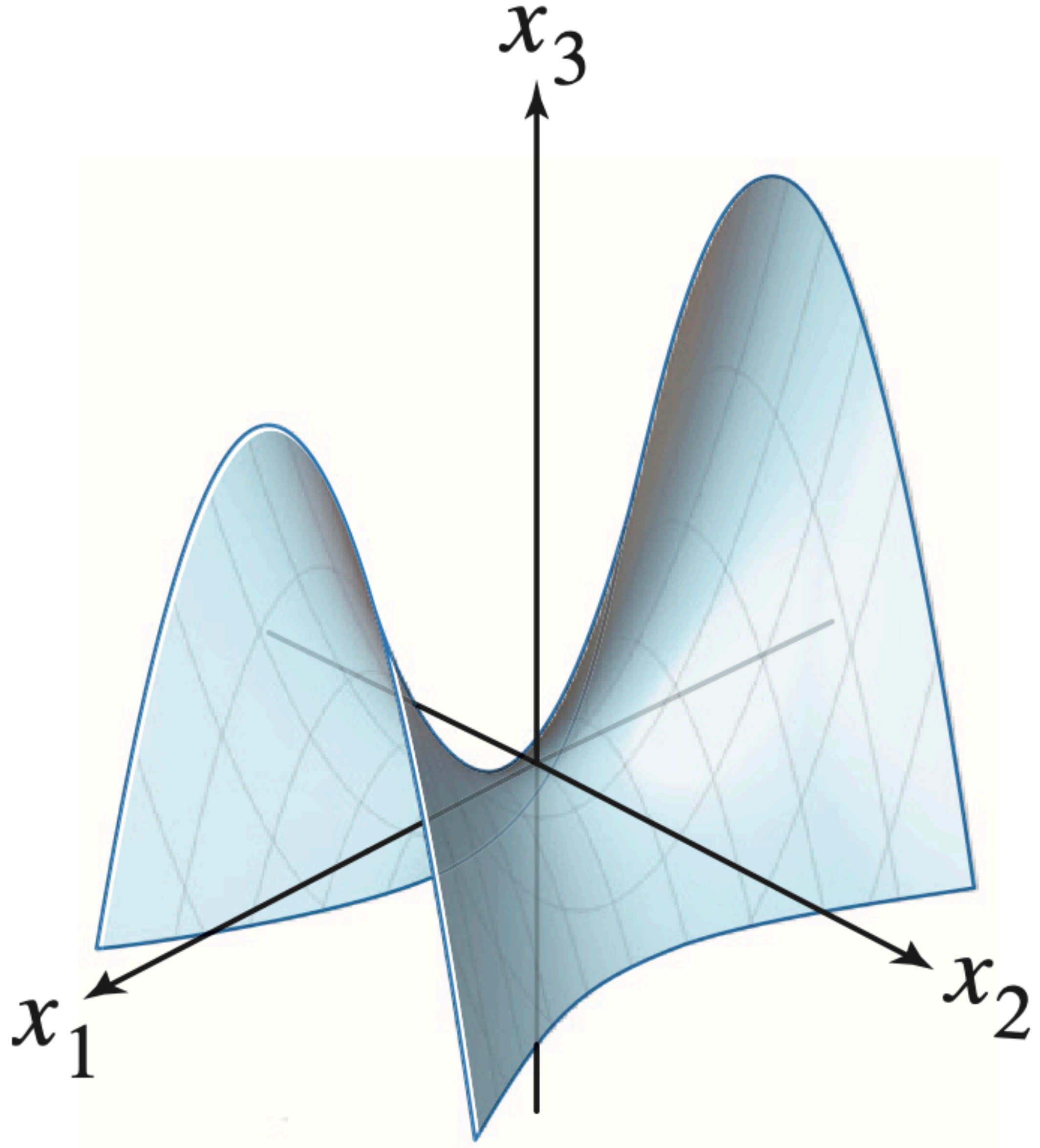
positive semidefinite

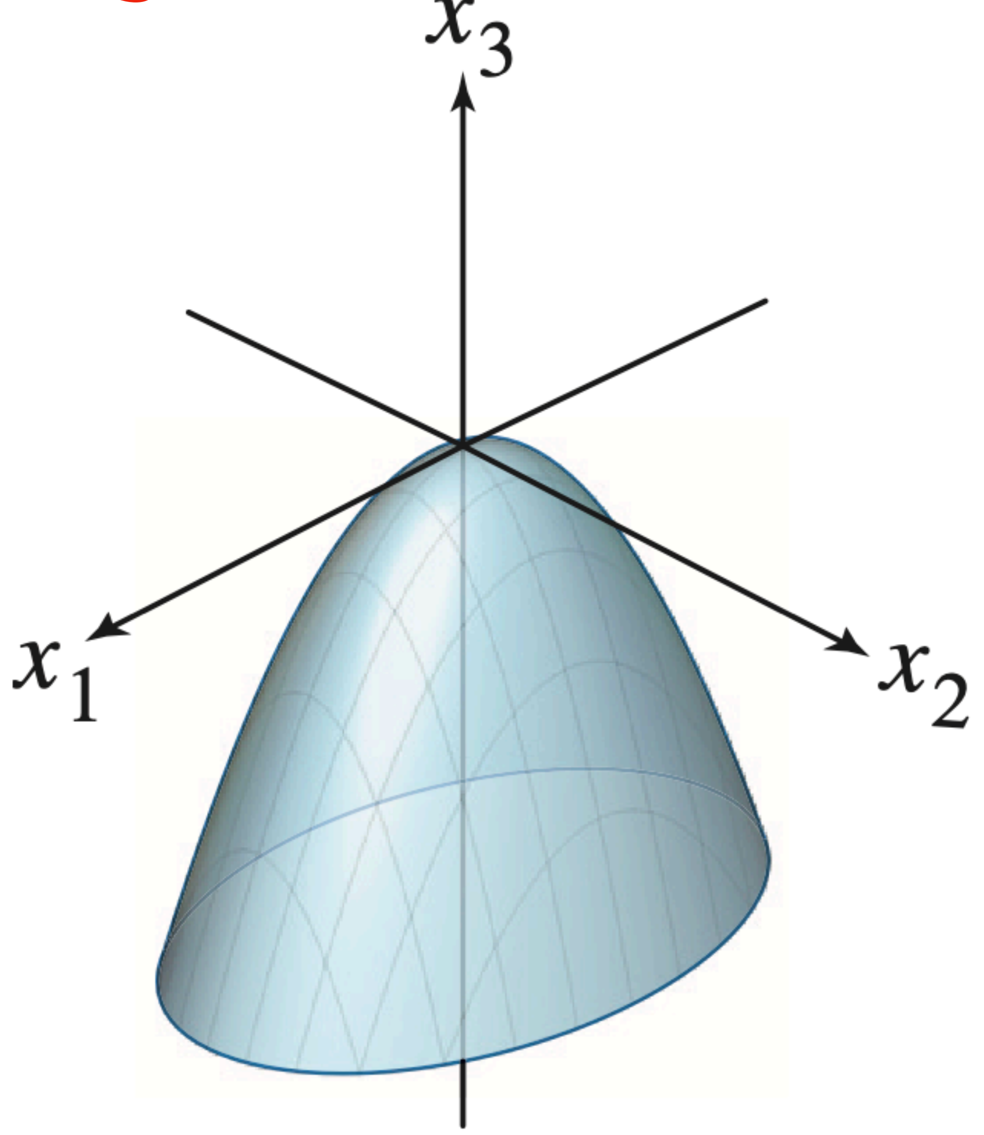
all neg. eigenvals

negative definite









$Q(\mathbf{x}) > 0$ $\quad$ $Q(\mathbf{x}) \geq 0$ $\quad$ $Q(\mathbf{x})$ can be + & − $\quad$ $Q(\mathbf{x}) < 0$

positive definite

all pos. eigenvals

indefinite

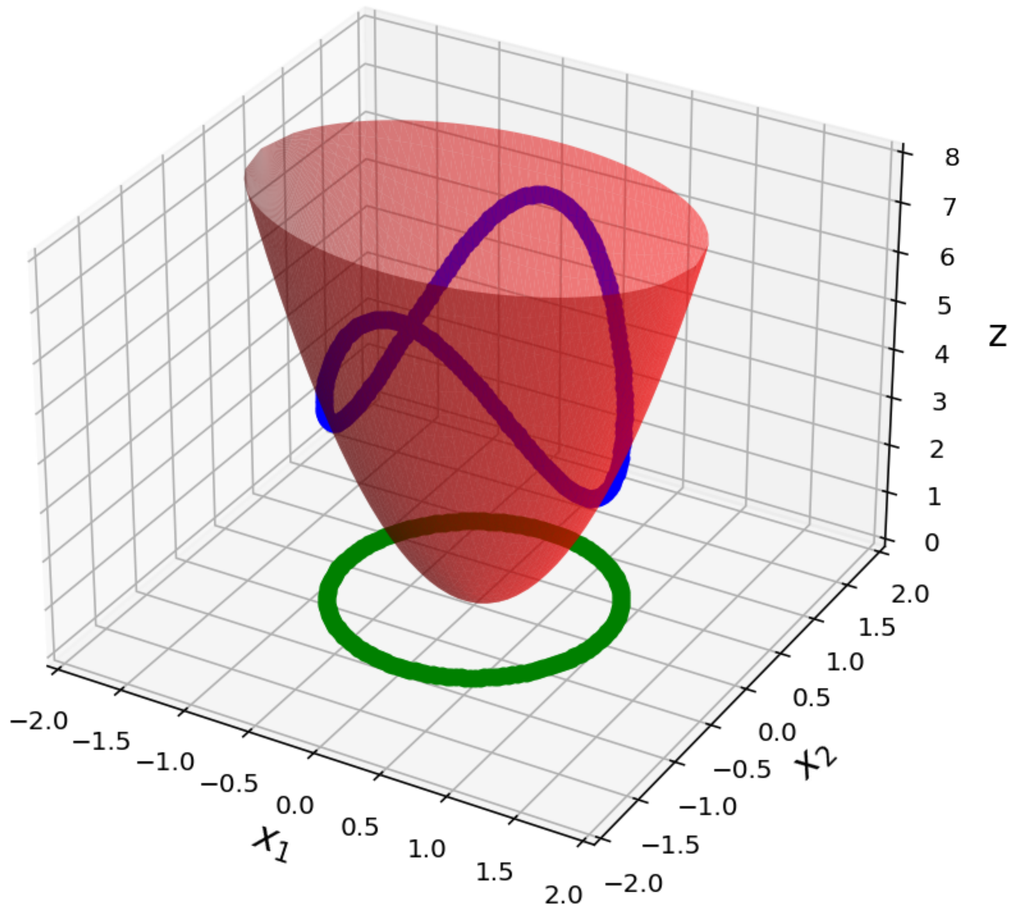
pos. and neg. eigenvals

Constrained Optimization and Eigenvalues

Thm. For a symmetric matrix A with *largest* eigenvalue λ_1 and *smallest* eigenvalue λ_n

$$\max_{\|\mathbf{x}\|=1} \mathbf{x}^T A \mathbf{x} = \lambda_1 \qquad \min_{\|\mathbf{x}\|=1} \mathbf{x}^T A \mathbf{x} = \lambda_n$$

No matter what A looks like, this holds



Singular Value Decomposition

Thm. For $A \in \mathbb{R}^{m\times n}$, there are *orthogonal* matrices $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$ such that

$$A = \overbrace{U}^{m\times m} \underbrace{\Sigma}_{m\times n} \overbrace{V^T}^{n\times n}$$

where diagonal entries* of Σ are $\sigma_1, \ldots, \sigma_n$ the singular values of A

* these are diagonal entries in a *non-square* matrix

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left singular right singular

$$A = \underbrace{\overset{m \times m}{U}}\ \underset{m \times n}{\Sigma}\ \overset{n \times n}{V^T}$$

where diagonal entries* of Σ are $\sigma_1, \ldots, \sigma_n$ the singular values of A

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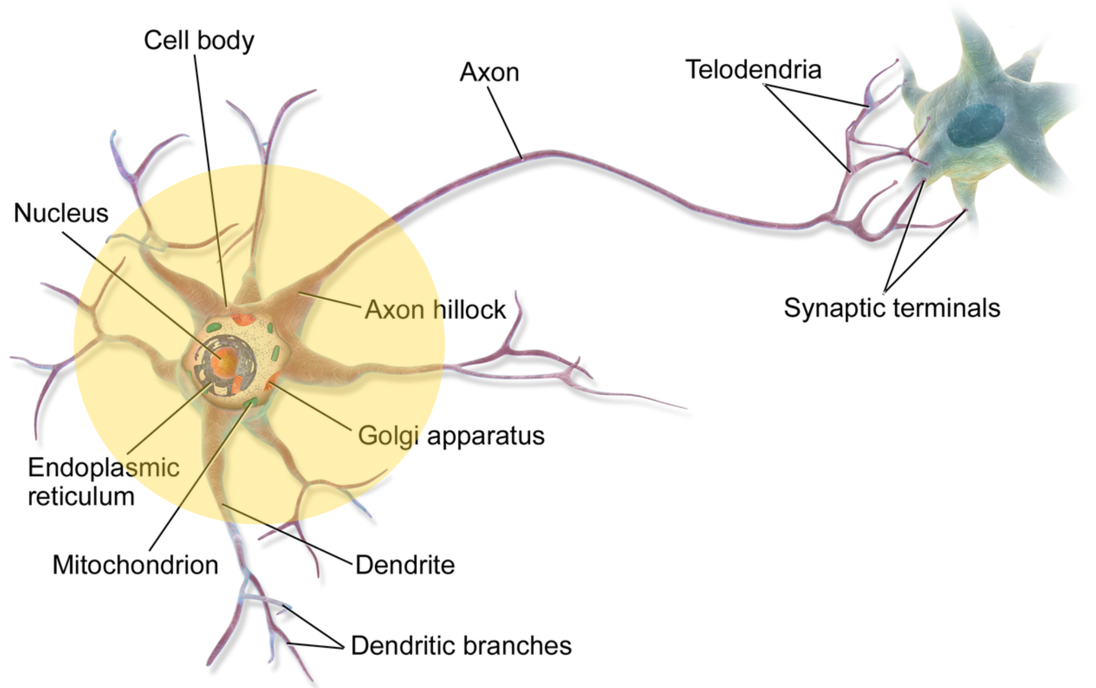
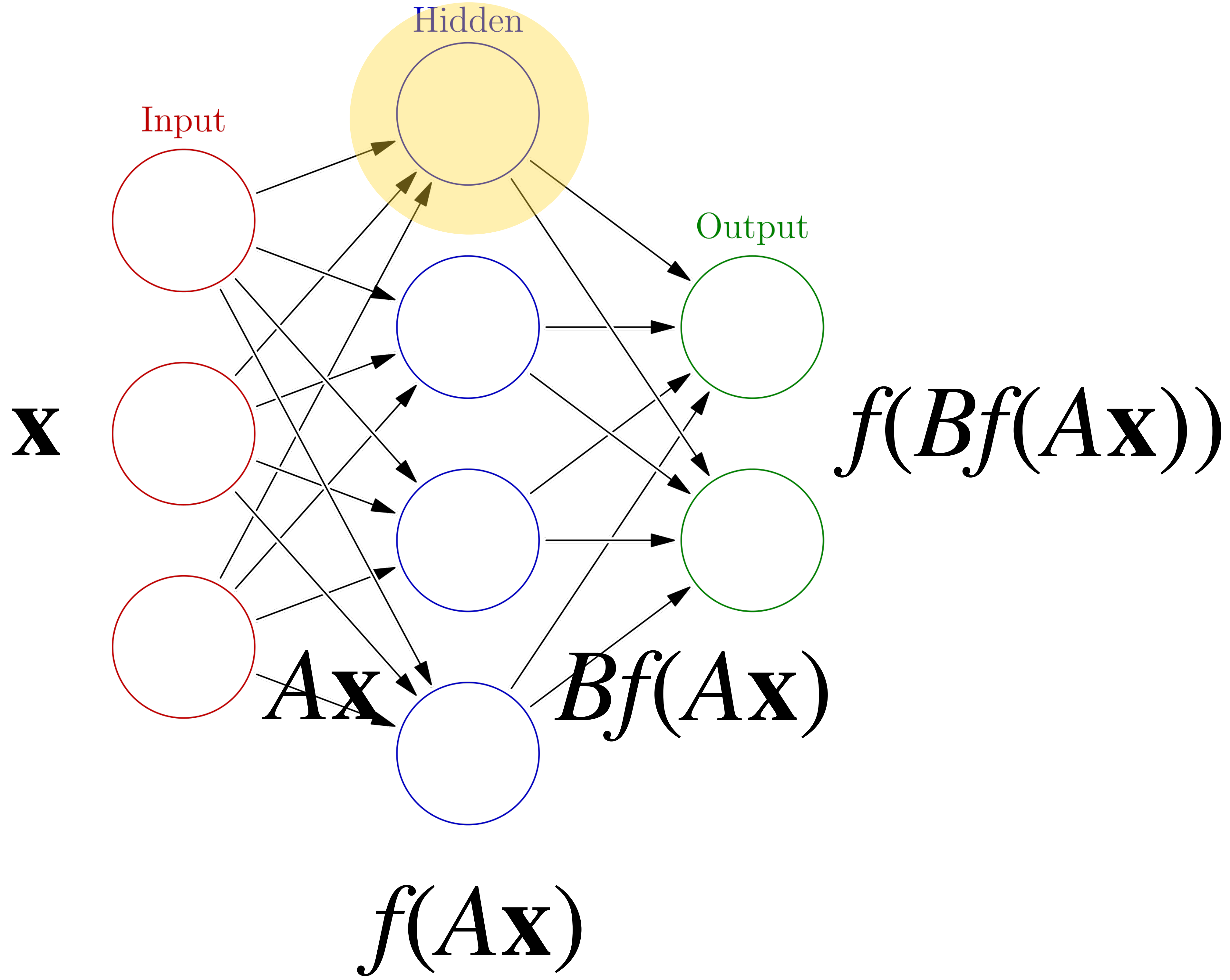
let's do some problems

What's next?
(Closing Remarks)

Non-Linearity

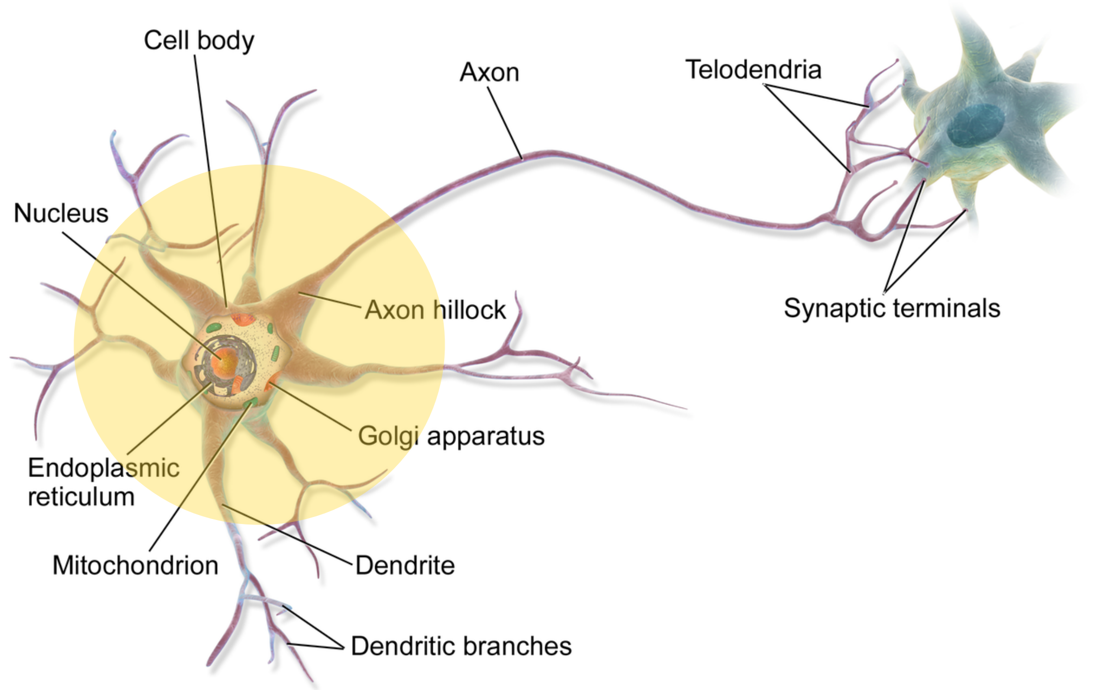


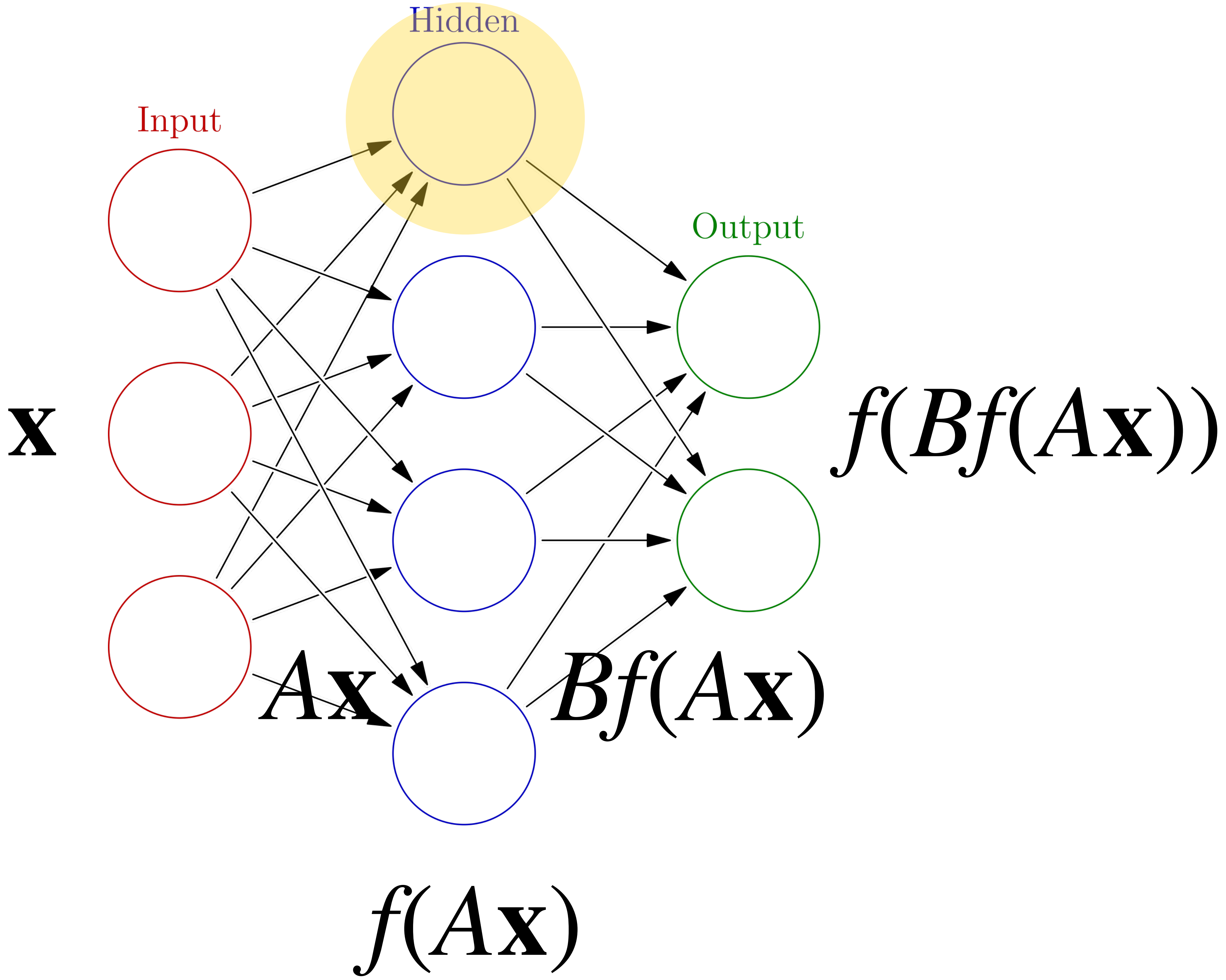


https://commons.wikimedia.org/wiki/File:Blausen_0657_MultipolarNeuron.png
https://commons.wikimedia.org/wiki/File:Colored_neural_network.svg

Non-Linearity

Neural networks are models of artificial neurons bundles





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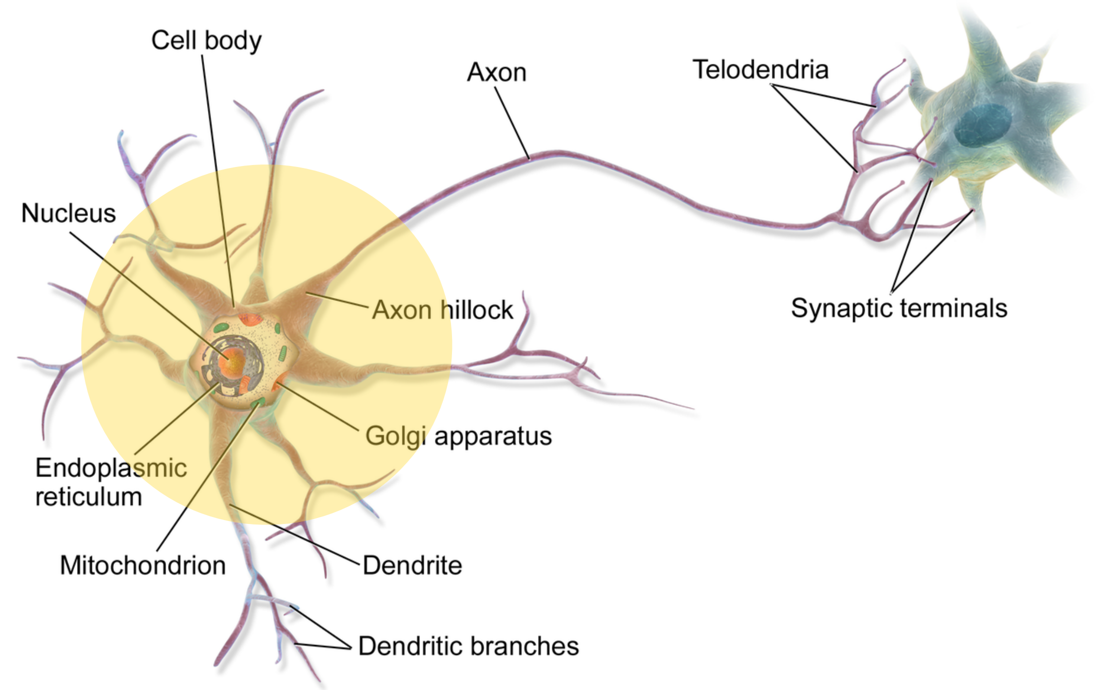
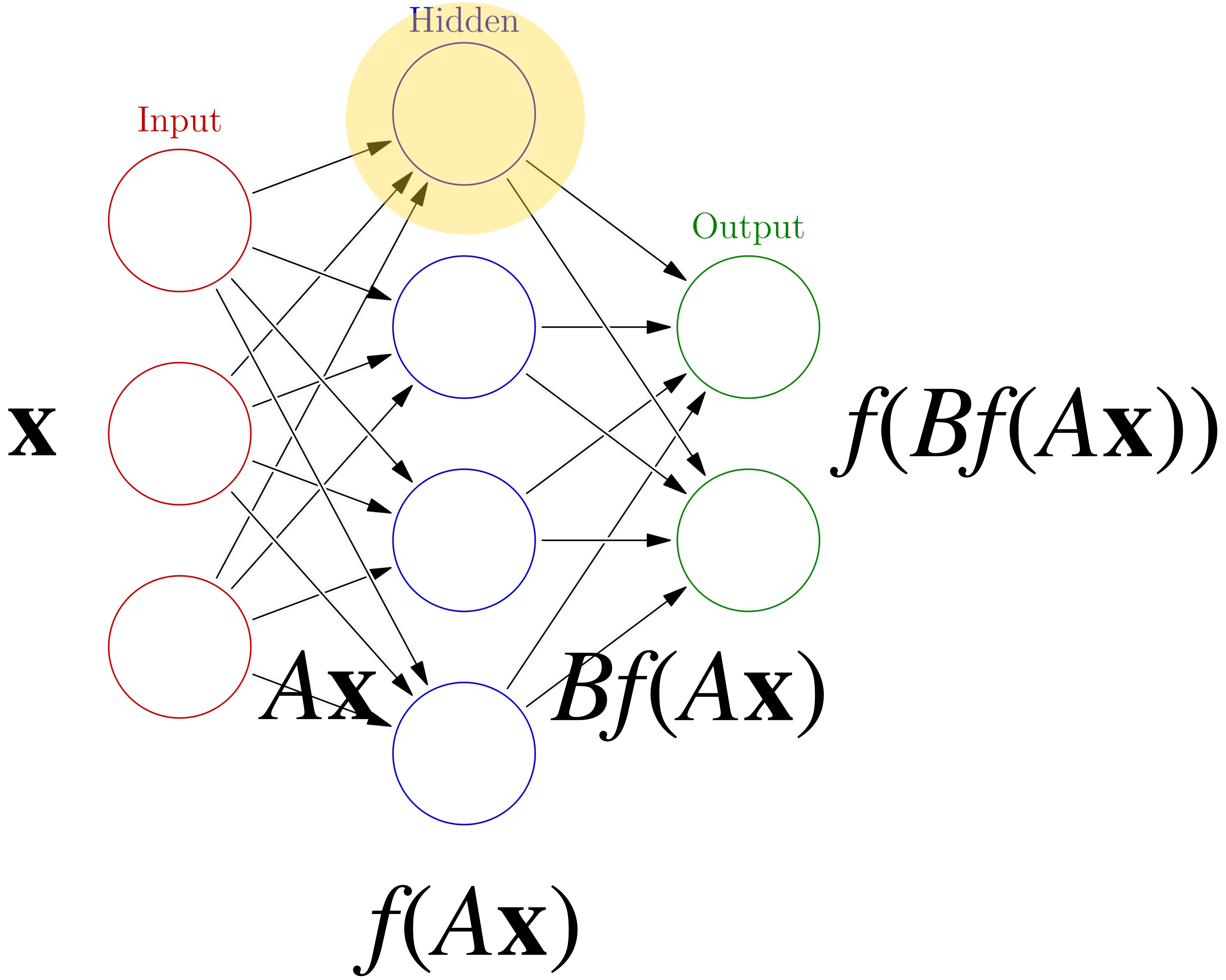


Neural networks are models of artificial neurons bundles

Given an input vector $\mathbf{x}$, it is transformed into a *hidden* vector $A\mathbf{x}$ by a linear transformation, and then an *activation function* f is applied to the result



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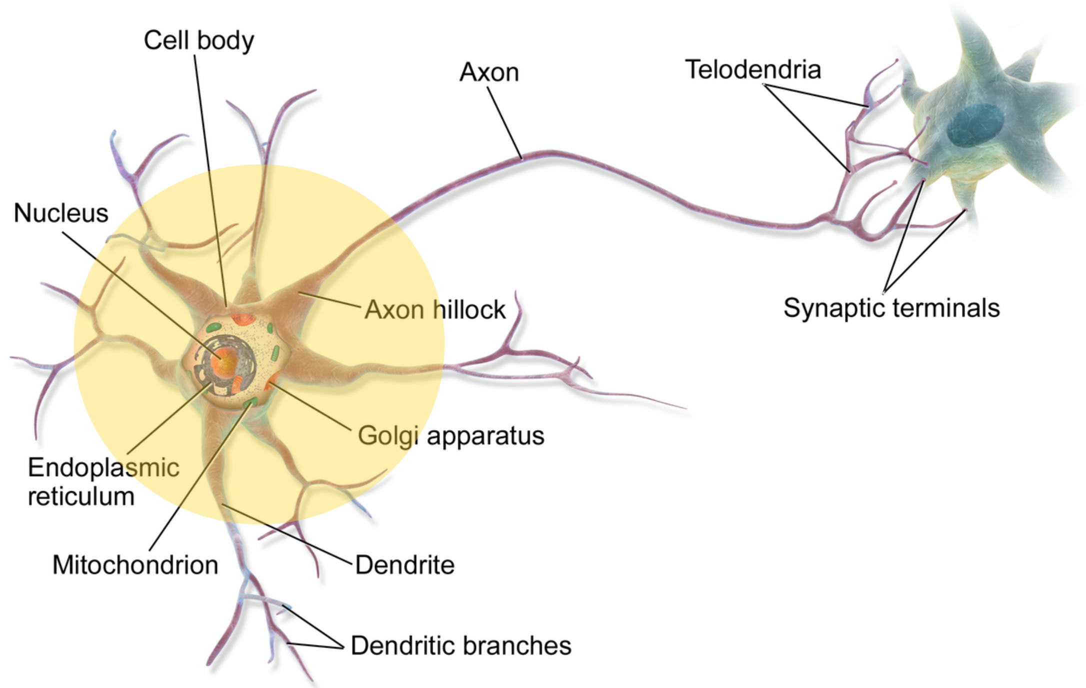

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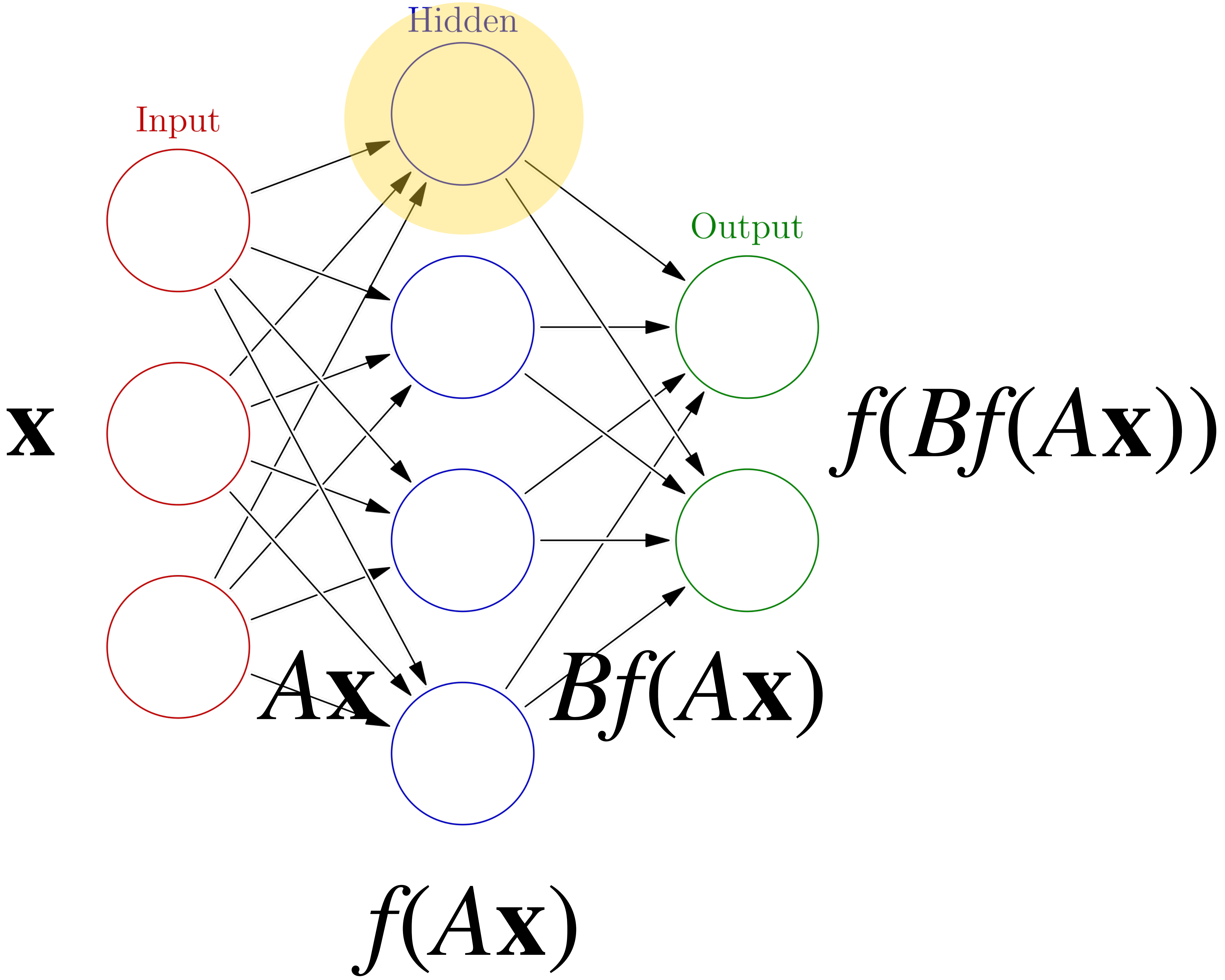

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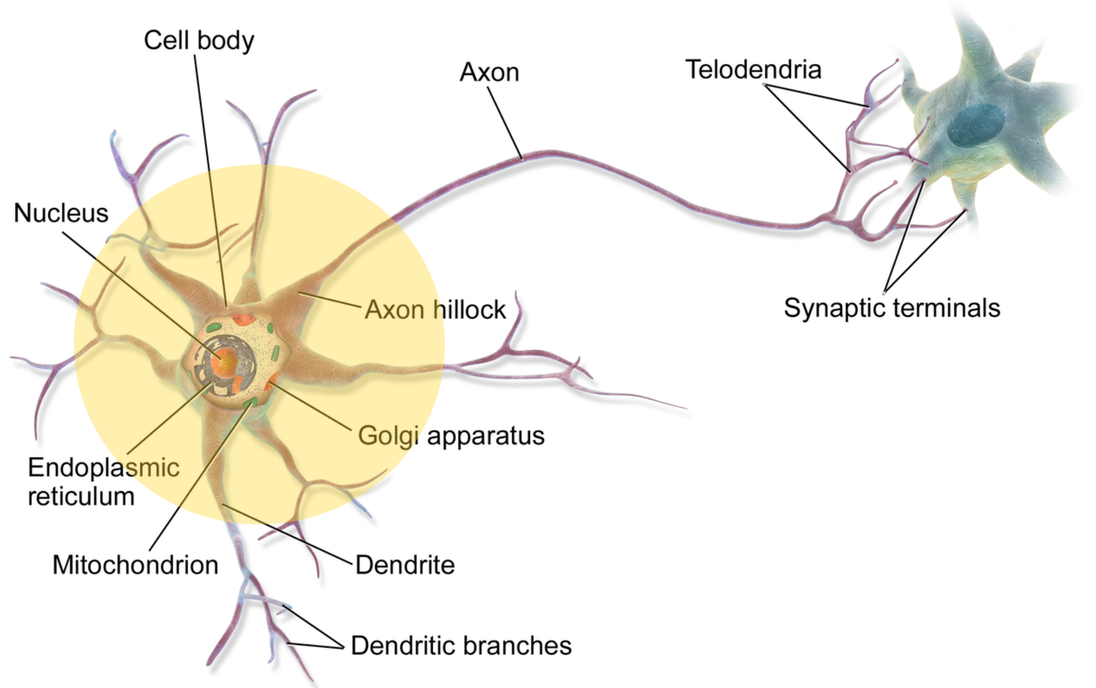

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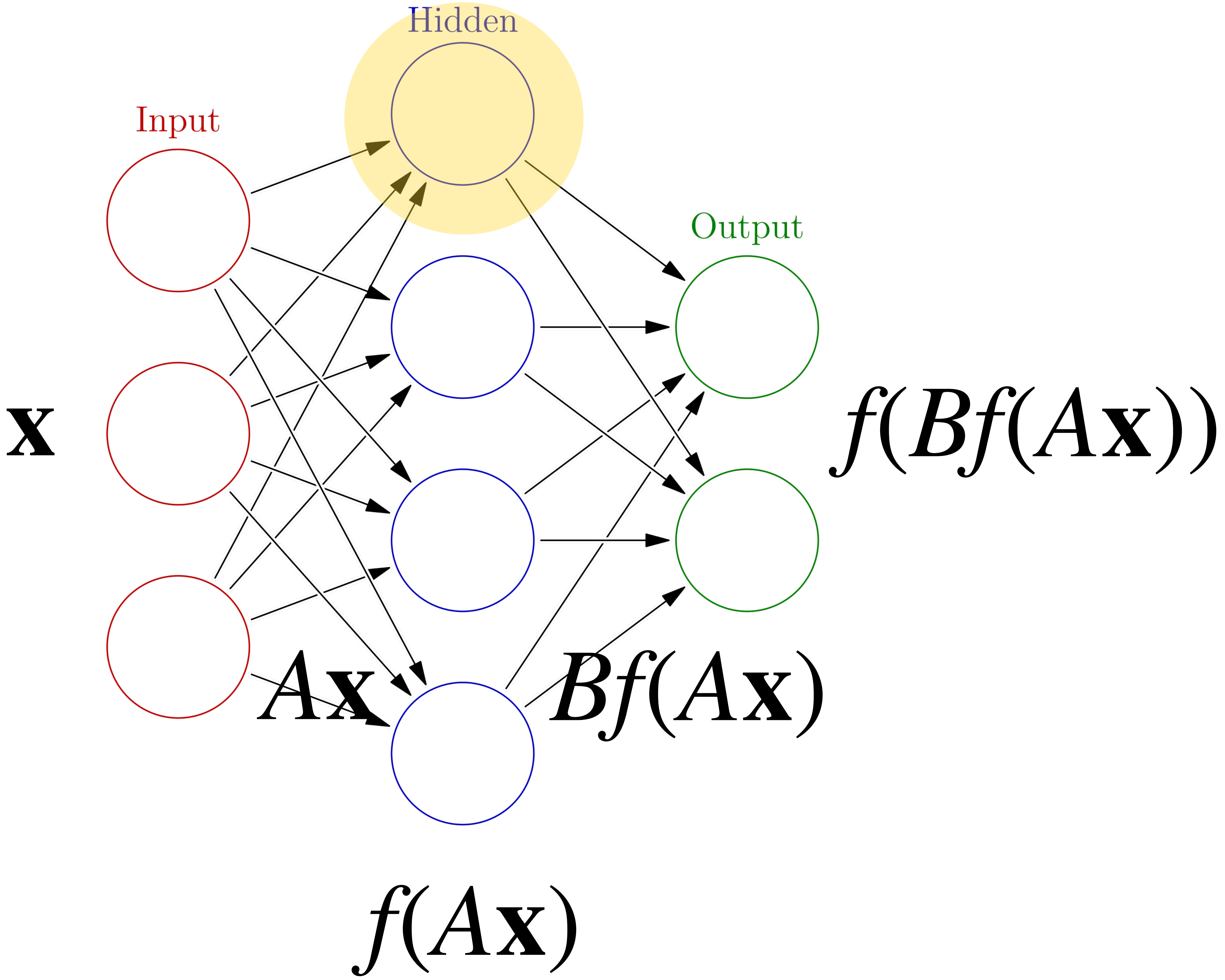
$$\mathrm{NN}(\mathbf{x}) = f(A_k(f(A_{k-1} \ldots f(A_1\mathbf{x}))$$



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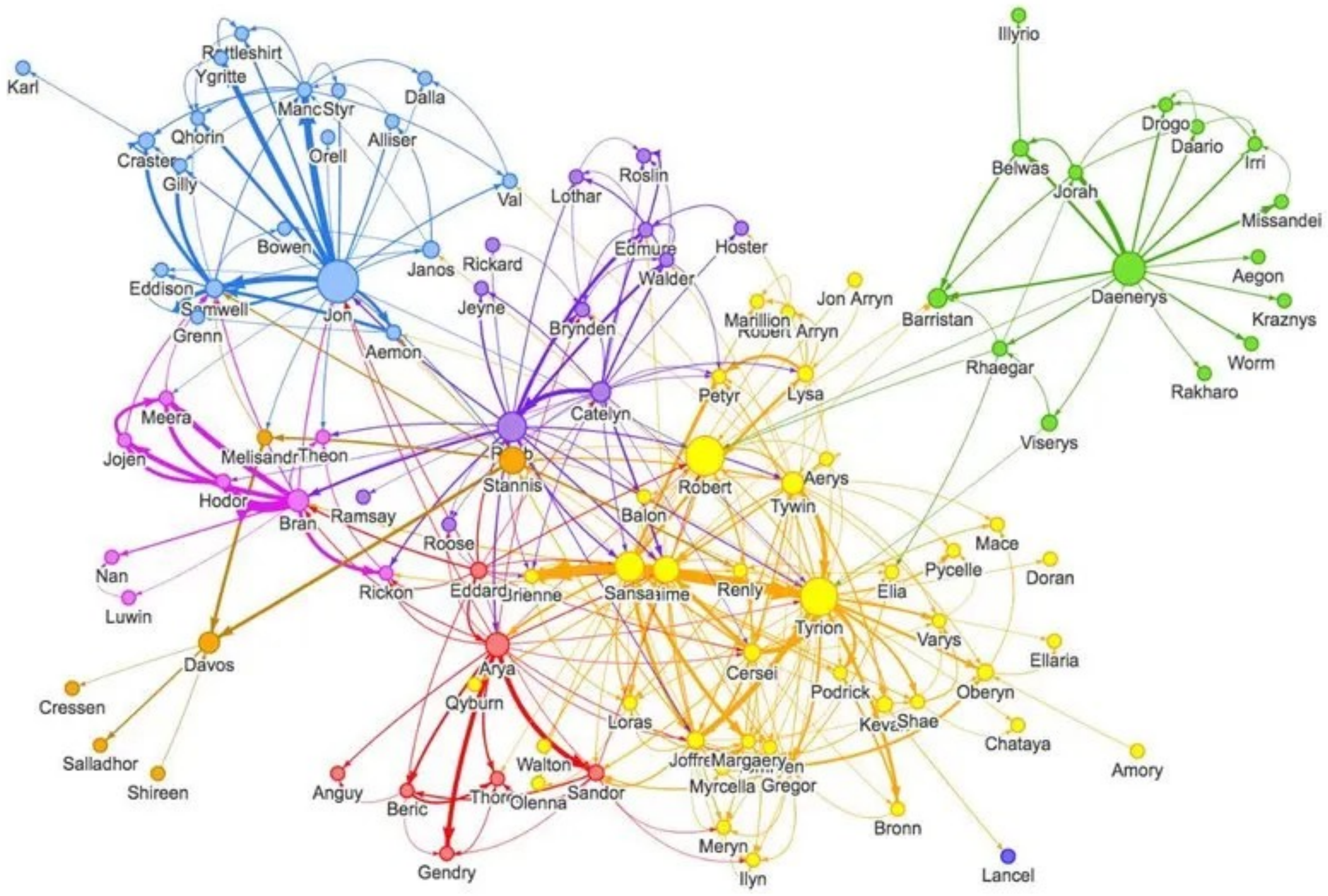
Spectral/Algebraic Graph Theory

Graphs can be viewed as matrices

The finding eigenvalues in graphs can gives use better clustering and cutting algorithms



https://medium.com/@n.rajadhyaksha/a-gentle-introduction-to-graph-spectral-filtering-df03ddc9d3f7

Abstract Algebra

$$\frac{U}{\mathsf{Nul}(f)} \cong \mathsf{Range}(f)$$

$$\begin{tikzcd}
U \arrow[r, "f"] \arrow[d] & V \\
U/\mathsf{Nul}(f) \arrow[ur] &
\end{tikzcd}$$

```
There's a lot of beautiful structure in the algebra
we've done in this course

And there are lots of directions to go from here
(infinite dimensional spaces, less restrictive
settings like groups and modules,...)
```

Course List

- CS 365 Foundations of Data Science
- CS 440 Intro to Artificial Intelligence
- CS 480 Intro to Computer Graphics
- CS 505 Intro to Natural Language Processing
- CS 506 Tools for Data Science
- CS 507 Intro to Optimization in ML
- CS 523 Deep Learning
- CS 530 Advanced Algorithms
- CS 531 Advanced Optimization Algorithms
- CS 542 Machine Learning
- CS 565 Algorithmic Data Mining
- CS 581 Computational Fabrication
- CS 583 Audio Computation

Some of these may not exist anymore...

Appreciations

The Course Staff

I'd like to thank:

Angelos Poulis, Gor Matcakian, Helen Zhou

If you see them around you should thank them as well

The CS Department Staff

If you're ever in the CS Department office, be kind to the people who work there. They work very hard to keep all our courses running

The Students of CS132

Thanks for sticking with it

Thanks for giving feedback

Thanks for participating

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